# Complementarity and Context: A Unified Framework for Economic Rationality

Gerhard Fehr
FehrAdvice & Partners AG, Zurich
Beatrix Research Group

January 2026 – Version 47

Main Text (without Appendices)

**Abstract**

Economics remains fragmented: classical theory explains market efficiency but not social preferences; behavioral economics explains social preferences but not institutional change; institutional economics explains institutional change but not financial crises. This paper provides a unified framework based on *complementarity*—the interdependence of choices across agents, domains, and time—and *context*—the technological, institutional, normative, and cultural environment that shapes payoff structures.

The framework is proposed as a *metatheory*: rather than replacing established theories, it aims to subsume them as limiting cases and specify *when* each applies. Arrow-Debreu emerges when complementarity is zero (see Appendix A for formal derivation) ($C_{ij} = 0$). Behavioral economics emerges when context is fixed ($\Psi = \bar{\Psi}$). Institutional economics emerges when complementarity is simplified. The framework's value-added lies in predicting which theoretical tradition applies to which situation—enabling conditional rather than unconditional predictions.

We introduce four core concepts: (1) the complementarity matrix $C$, capturing how choices reinforce or undermine each other; (2) the context vector $\Psi \in \mathbb{R}^8$, decomposed into eight primitive dimensions; (3) the reference structure $C^*(\Psi)$, derived as the self-consistent fixed point toward which systems gravitate; and (4) the coherence index $K$, measuring alignment between actual and reference structures.

**Key methodological contribution:** We operationalize context through eight independently measurable dimensions:

- $\Psi_I$ (Institutional): Rules, enforcement, property rights
- $\Psi_S$ (Social): Trust, norms, social capital
- $\Psi_C$ (Cognitive): Information processing, numeracy, biases

- $\Psi_K$ (Informational): Transparency, signal quality, asymmetries
- $\Psi_E$ (Economic): Development, market thickness, resources
- $\Psi_T$ (Temporal): Time horizons, uncertainty, stability
- $\Psi_M$ (Market Scope): Geographic reach, trade access, openness
- $\Psi_F$ (Factor Flexibility): Adjustment capacity, mobility

**Empirical validation:** Across six canonical economic relationships (education→earnings, management→productivity, democracy→growth, patience→growth, information→behavior, income→health), the eight $\Psi$ dimensions explain 70.1% of cross-country heterogeneity (Adjusted $R^2 = 55.2\%$). Different dimensions dominate for different effects: $\Psi_F$ (flexibility) matters most for management practices; $\Psi_K$ (information) matters most for education returns (Hanushek and Woessmann 2012); $\Psi_T$ (temporal stability) matters most for patience effects.

Welfare is measured through the FEPSDE framework: six dimensions (Financial, Emotional, Physical, Social, Digital, Ecological), three utility categories (Individual, Collective, Identity), four time horizons, and gains versus pains—yielding 144 components (detailed in Appendix C) that capture human flourishing beyond GDP.

Novel predictions include: (1) treatment effects are context-dependent functions, not universal constants—$\tau(X \to Y) = f(\Psi)$; (2) information interventions work when $\Psi_K$ is *low* (novel information) AND $\Psi_F$ is *high* (can act on it); (3) trade wars have persistent effects through context hysteresis; (4) financial crises cause political crises with 5-10 year lags; (5) recovery time is convex in crisis depth.

The framework's core falsifiable claim: *No economic effect is context-free.* If any intervention produces identical effects across all $\Psi$ configurations, the framework would be seriously challenged. This distinguishes our approach from theories that explain everything post hoc—we commit to context-dependence as a testable hypothesis.



**Keywords:** Complementarity, Context, Metatheory, Welfare, FEPSDE, Institutional Economics, Behavioral Economics, Supermodularity, Treatment Effect Heterogeneity

**JEL Classification:** B41 (Economic Methodology), C62 (Existence and Stability), D01 (Microeconomic Behavior), D02 (Institutions), E02 (Institutions and the Macroeconomy), O43 (Institutions and Growth)


# Contents


**1 Introduction** **9**

1.1 The Problem: Fragmentation . . . . . 9

1.2 The Puzzle: Why Integration Has Failed . . . . . 9

1.3 Our Solution: Metatheory Through Complementarity . . . . . 10

1.4 The Four Core Concepts . . . . . 11

1.5 Empirical Validation: The Eight Ψ Dimensions . . . 12
1.6 The Framework's Falsifiable Claim . . . 13
1.7 Roadmap . . . 14

**2 Rationality and Stability in Classical Economics 15**
2.1 The Classical Foundations: Smith to Mill . . . 15
2.2 The Marginalist Revolution: From Value to Utility . . . 16
2.3 General Equilibrium: The Arrow-Debreu Achievement . . . 16
2.4 Stability: The Missing Dimension . . . 17
2.5 Evolutionary Game Theory: Selection Without Rationality . . . 18
2.6 From Efficiency to Coherence . . . 20
2.7 Historical Examples of Coherence and Incoherence . . . 21
2.8 Summary: What Classical Economics Gets Right and Wrong . . . 22

**3 The Limits of Utility Maximization under Stable Preferences 24**
3.1 The Stable Preferences Assumption . . . 24
3.2 Six Domains Where Stable Preferences Fail . . . 25
3.2.1 Domain 1: Social Preferences . . . 25
3.2.2 Domain 2: Identity and Self-Image . . . 26
3.2.3 Domain 3: Reference Dependence and Loss Aversion . . . 27
3.2.4 Domain 4: Framing and Context Effects . . . 28
3.2.5 Domain 5: Preference Construction and Learning . . . 28
3.2.6 Domain 6: Temporal Preferences and Present Bias . . . 29
3.3 The Deeper Problem: Utility Interdependence . . . 31
3.4 What Utility Maximization Gets Wrong . . . 31
3.5 The Required Extension . . . 32

**4 Empirical Foundations: From Theory to Measurement 33**
4.1 The Experimental Revolution in Social Preferences . . . 33
4.1.1 The Ultimatum Game: Where It All Began . . . 33
4.1.2 The Dictator Game: Isolating Fairness . . . 33
4.1.3 Public Goods Games: Conditional Cooperation . . . 34
4.1.4 Trust Games: Reciprocity Quantified . . . 35
4.1.5 Formalizing Social Preferences . . . 35
4.2 Field Evidence: External Validity . . . 36
4.2.1 Does Lab Behavior Predict Field Behavior? . . . 36
4.2.2 Gift Exchange in Labor Markets . . . 36
4.2.3 Trust and Economic Development . . . 36
4.2.4 Cooperation in the Field . . . 37
4.3 The Replication Movement: What Survives? . . . 37
4.4 Complexity Economics and Aggregate Dynamics . . . 38

4.4.1 Increasing Returns and Path Dependence . . . 38
4.4.2 Network Effects . . . 38
4.4.3 The Santa Fe Perspective . . . 38
4.5 Neuroeconomic Substrates . . . 39
4.5.1 Neural Encoding of Social Value . . . 39
4.5.2 Oxytocin and Trust . . . 39
4.5.3 Punishment and Reward Circuits . . . 39
4.6 Cross-Cultural Evidence . . . 40
4.6.1 The WEIRD Problem . . . 40
4.6.2 Culture and Punishment . . . 40
4.7 Implications for the Framework . . . 40
4.8 Technological Enablement: Why Measurement Is Now Feasible . . . 41

**5 Complementarity as a Structural Principle 43**
5.1 The Intuition: Why Choices Interact . . . 43
5.2 Formal Definition: Supermodularity . . . 43
5.3 The Intellectual Heritage . . . 44
5.4 Types of Complementarity . . . 46
5.4.1 Strategic Complementarity . . . 46
5.4.2 Temporal Complementarity . . . 46
5.4.3 Cross-Domain Complementarity . . . 47
5.4.4 Cross-Agent Complementarity . . . 47
5.5 Why Complementarity Implies Coherence . . . 48
5.6 The Opposite Case: Substitutes . . . 48
5.7 Complementarity at Each Level . . . 49
5.7.1 Individual Level . . . 49
5.7.2 Household Level . . . 50
5.7.3 Organization Level . . . 50
5.7.4 Regional Level . . . 50
5.7.5 National Level . . . 51
5.7.6 Global Level . . . 51
5.8 The Coherence Index: Measuring Complementarity Alignment . . . 52
5.9 Complementarity and Classical Economics . . . 53
5.10 Summary: Complementarity as Foundation . . . 53
5.11 Why Complementarity Becomes Operational in 2026 . . . 54

**6 The Reference Structure: Three Derivations of $C^*$ (formal proofs in Appendix D) 56**
6.1 The Stigler-Becker Parallel . . . 56
6.2 Why $C^*$ Cannot Be Arbitrary . . . 57
6.3 Axiomatic Derivation: Self-Consistency . . . 58

6.4 Evolutionary Derivation: Stability under Selection . . . . . 60
6.5 Empirical Derivation: Long-Run Averages . . . . . 61
6.6 Convergence of the Three Approaches . . . . . 62
6.7 What $C^*$ Is and Is Not . . . . . 63
6.8 Implications for the Framework . . . . . 63

**7 Complementarity, Fit, and Non-Concavity 65**
7.1 Roberts' Insight: Organizations as Systems of Complements . . . . . 65
7.2 Non-Concavity: Why Complementarity Creates Peaks and Valleys . . . . . 66
7.3 Fit as Coherence . . . . . 67
7.4 Why Transitions Are So Difficult . . . . . 68
7.5 Multiple Peaks: Which Configuration Is Best? . . . . . 69
7.6 From Organizations to All Levels . . . . . 70
7.7 Implications for Strategy . . . . . 70

**8 Mathematical Formalization 71**
8.1 The Core Insight: Why Two Dimensions? . . . . . 71
8.2 The Two-Axis Model: Formal Definition . . . . . 72
8.3 The Payoff Function: Formalizing Complementarity . . . . . 72
8.4 Equilibria and Their Properties . . . . . 73
8.5 The Coherence Index K . . . . . 73
8.6 The Quality Index Q . . . . . 73
8.7 The 144-Component Structure . . . . . 73
8.8 The Complementarity Matrix . . . . . 74
8.9 Computational Tractability: Why This Is Now Feasible . . . . . 74
8.10 Summary: The Mathematical Architecture . . . . . 76

**9 Context as an Endogenous Variable 77**
9.1 What Is Context? . . . . . 77
9.2 The Feedback Loop . . . . . 78
9.3 A Simple Model of Context Dynamics . . . . . 78
9.4 Different Types of Context: Speed of Change . . . . . 79
9.5 Self-Reinforcing and Self-Correcting Dynamics . . . . . 80
9.6 Tipping Points and Regime Change . . . . . 81
9.7 Historical Examples . . . . . 82
9.8 Why Context Endogeneity Matters . . . . . 83
9.9 The Complete Dynamic System . . . . . 84
9.10 Theoretical Foundations: Connecting to Established Literature . . . . . 84
9.11 Spatial Context: Geography as Endogenous . . . . . 86
9.12 Complexity Economics: Emergence and Non-Equilibrium . . . . . 87
9.13 Formal Models for Each Context Type . . . . . 89

9.13.1 Technology Dynamics . . . 89
9.13.2 Norm Dynamics . . . 90
9.13.3 Institutional Dynamics . . . 90
9.13.4 Cultural Dynamics . . . 91
9.14 Cross-Level Interactions: How Context Types Influence Each Other . . . 92
9.15 Context Dynamics at Each Level . . . 93
9.15.1 Individual Level: Personal Context . . . 93
9.15.2 Household Level: Family Context . . . 94
9.15.3 Organizational Level: Corporate Context . . . 94
9.15.4 Regional Level: Local Context . . . 95
9.15.5 National Level: Institutional Context . . . 96
9.15.6 Global Level: International Context . . . 96
9.16 Policy Implications: Changing Context vs. Changing Behavior . . . 97
9.17 Limitations and Extensions . . . 99
9.18 Measuring Context: The Technological Breakthrough . . . 99

**10 Welfare and Equilibrium Quality: The FEPSDE Framework 102**
10.1 The Problem: $K$ Without $Q$ Is Incomplete . . . 102
10.2 The FEPSDE Dimensions: Why These Six? . . . 103
10.3 Each Dimension in Detail . . . 103
10.3.1 Financial Dimension ($U^F$) . . . 103
10.3.2 Emotional Dimension ($U^E$) . . . 104
10.3.3 Physical Dimension ($U^P$) . . . 105
10.3.4 Social Dimension ($U^S$) . . . 106
10.3.5 Digital Dimension ($U^D$) . . . 107
10.3.6 Ecological Dimension ($U^{Eco}$) . . . 108
10.4 The Three Utility Categories: INU, KNU, IDN . . . 109
10.4.1 Individual Utility (INU) . . . 109
10.4.2 Collective Utility (KNU) . . . 109
10.4.3 Identity Utility (IDN) . . . 110
10.5 The Time Structure: Four Horizons . . . 111
10.6 Gains and Pains: The Prospect Theory Connection . . . 111
10.7 The Complete 144-Component Structure . . . 112
10.8 The Aggregate Welfare Function . . . 113
10.9 FEPSDE vs. Alternative Welfare Measures . . . 113
10.10 Applying FEPSDE: Country Comparison . . . 114
10.11 The K-Q Relationship . . . 114
10.12 Summary: The Complete Welfare Framework . . . 115

**11 Novel Predictions** **116**
11.1 What Makes a Prediction "Novel"? . . . 116
11.2 Prediction 1: Trade War Dynamics—The Trump Tariff Analysis . . . 116
11.3 Prediction 2: Cross-Domain Coherence Spillovers . . . 118
11.4 Prediction 3: Asymmetric Recovery Dynamics . . . 119
11.5 Prediction 4: Innovation-Coherence Tradeoff . . . 120
11.6 Prediction 5: The Identity-Economics Feedback . . . 121
11.7 Prediction 6: The Coherence Trap . . . 122
11.8 Prediction 7: Multi-Level Coherence Conflicts . . . 123
11.9 Prediction 8: Mechanism Design for Context (Ψ-Design) . . . 123
11.10 Summary: The Prediction Matrix . . . 126

**12 Integration of Established Theories** **127**
12.1 The Unification Logic . . . 127
12.2 Classical Economics: The Separability Restriction . . . 127
12.3 Behavioral Economics: The Social Preference Extension . . . 128
12.4 Identity Economics: The Temporal Extension . . . 128
12.5 Growth Theory: The Endogenous Context Extension . . . 129
12.6 Institutional Economics: The Multiple Equilibria Extension . . . 130
12.7 Organizational Economics: The Supermodularity Foundation . . . 131
12.8 Financial Economics: The Market Coherence Extension . . . 131
12.9 International Trade Theory: The Factor Endowment Extension . . . 132
12.10 Agency Theory: The Alignment Extension . . . 135
12.11 Prospect Theory: The Loss Aversion Extension . . . 135
12.12 Summary: The Theory Space . . . 135

**13 Empirical and Policy Implications** **137**
13.1 General Principles . . . 137
13.2 Level 1: Individual — Coaching and Therapy . . . 137
13.3 Level 2: Household — Family Policy and Support . . . 139
13.4 Level 3: Organization — Management and Transformation . . . 140
13.5 Level 4: Region — Economic Development Policy . . . 141
13.6 Level 5: Nation — Institutional Design and Reform . . . 143
13.7 Level 6: Global — International Cooperation and Governance . . . 144
13.8 Cross-Level Interactions . . . 145
13.9 Summary: The Policy Matrix . . . 146
13.10 Implementation: The Technological Infrastructure . . . 146

**14 Limitations and Future Work** **150**
14.1 Theoretical Limitations . . . 150
14.1.1 The Contraction Assumption . . . 150

14.1.2 Linearity of Context Dynamics . . . 150
14.1.3 The Aggregation Problem . . . 151
14.1.4 No Theory of Equilibrium Selection . . . 152
14.2 Empirical Limitations . . . 152
14.2.1 Identification of Complementarities . . . 152
14.2.2 Parameter Proliferation . . . 153
14.2.3 No Empirical Validation Yet . . . 153
14.2.4 WEIRD Bias . . . 154
14.3 Conceptual Limitations . . . 155
14.3.1 Normative Neutrality . . . 155
14.3.2 Micro-Foundations . . . 155
14.3.3 Static vs. Dynamic Analysis . . . 156
14.4 What the Framework Cannot Do . . . 156
14.5 Epistemic Honesty: Integration vs. Prediction . . . 157
14.6 What the Framework Can Do . . . 160
14.7 Research Agenda . . . 161
14.8 Conclusion: The Value of Imperfect Frameworks . . . 162

**15 Conclusion** **163**
15.1 What We Have Done . . . 163
15.2 The Falsifiable Claim . . . 165
15.3 What This Changes . . . 165
15.4 Integration with Major Research Programs . . . 166
15.5 Limitations Acknowledged . . . 166
15.6 The Road Ahead . . . 167
15.7 Final Reflection . . . 167

# 1 Introduction

## 1.1 The Problem: Fragmentation

Economics appears fragmented. Classical theory explains market efficiency but struggles with social preferences. Behavioral economics explains social preferences but often holds institutions fixed. Institutional economics models institutional change but frequently ignores micro-level dynamics. Finance explains financial crises but rarely traces political consequences.

Each subdiscipline has its own assumptions, methods, and vocabulary. They coexist uneasily, sometimes contradicting each other, often talking past each other. Graduate students learn them as separate courses. Policy makers receive conflicting advice.

This fragmentation is not merely inconvenient—it is intellectually unsatisfying and practically costly. The great challenges of our time—climate change, inequality (Piketty 2014), institutional decay, technological disruption—do not respect disciplinary boundaries. They involve interactions between markets, behaviors, institutions, and identities. Understanding them requires a framework that integrates rather than fragments.

## 1.2 The Puzzle: Why Integration Has Failed

Why has economics remained fragmented despite decades of effort at integration? We suggest the answer lies in a hidden assumption that pervades most economic theory: *separability*.

**What Separability Assumes.**

- My utility depends on my consumption, not yours
- Your choices don't affect my optimal choice (except through prices)
- Context (technology, institutions, norms) is exogenous—given from outside
- Preferences are stable—they don't change with experience or social setting

Under separability, each agent's optimization problem is independent. Markets aggregate independent choices. Equilibrium is efficient. This is the Arrow-Debreu world (Arrow and Debreu 1954)—mathematically elegant, normatively powerful, empirically limited.

**What Happens When Separability Fails.** Empirical research has documented systematic violations:

- **Social preferences:** People reject unfair offers even at personal cost (Güth et al. 1982; Fehr and Schmidt 1999)

- **Identity:** People sacrifice material welfare to maintain self-image (Akerlof and Kranton 2000; Bénabou and Tirole 2006)
- **Reference dependence:** Choices depend on context, framing, and expectations (Kahneman and Tversky 1979)
- **Institutional effects:** The same people behave differently under different rules (North 1990; Acemoglu and Robinson 2012)

Each finding generated its own literature—behavioral economics (Camerer et al. 2004), identity economics, institutional economics. But these literatures developed independently, each relaxing separability in its own way, without a common framework to show how they fit together.

## 1.3 Our Solution: Metatheory Through Complementarity

This paper provides the missing framework. But we make a crucial distinction: our contribution is not another *theory* competing with existing ones, but a proposed *metatheory* that aims to organize them.

**What a Metatheory Does.** A metatheory does not replace component theories—it specifies when each applies. Consider an analogy: Newtonian mechanics, relativistic mechanics, and quantum mechanics are distinct theories. But physics also has organizing principles (symmetry, conservation, least action) that explain *when* each theory applies and *how* they relate.

Our framework provides this organizing structure for economics:

- **Arrow-Debreu** applies when complementarity is negligible ($C_{ij} \approx 0$) and context is stable ($\dot{\Psi} \approx 0$)
- **Behavioral economics** applies when individual biases dominate ($\Psi_C$ varies) but institutions are fixed
- **Institutional economics** applies when rules dominate ($\Psi_I$ varies) but individual behavior aggregates
- **Our full framework** applies when multiple $\Psi$ dimensions vary simultaneously with significant complementarity

The decisive advantage: existing theories tell you *that* certain effects occur. Our framework tells you *when* they occur—which contexts amplify, dampen, or reverse each effect.

## 1.4 The Four Core Concepts

**1. Complementarity Matrix $C$.** The second derivatives of utility functions:

$$C_{ij} = \frac{\partial^2 U_i}{\partial x_i \partial x_j} \tag{1}$$

This captures how my marginal utility changes when you change your behavior.

- $C_{ij} > 0$: Complements—your cooperation makes my cooperation more valuable
- $C_{ij} < 0$: Substitutes—your action crowds out mine
- $C_{ij} = 0$: Independent—the Arrow-Debreu case

**2. Context Vector $\Psi \in \mathbb{R}^8$.** Technology, institutions, norms, and culture—everything that shapes the payoff structure but is not under any single agent's control.

We operationalize $\Psi$ through eight primitive dimensions (Table 1), each derived from decision-theoretic requirements and independently measurable with established instruments.

Table 1: The Eight Context Dimensions

| Symbol | Dimension | What It Captures | Key Measures |
|---|---|---|---|
| $\Psi_I$ | Institutional | Formal rules and enforcement | Rule of Law, Polity IV |
| $\Psi_S$ | Social | Trust and social capital | WVS Trust, Civic Index |
| $\Psi_C$ | Cognitive | Information processing capacity | PISA, Financial Literacy |
| $\Psi_K$ | Informational | Transparency and signal quality | Press Freedom, CPI |
| $\Psi_E$ | Economic | Development and resources | GDP/capita, Credit Access |
| $\Psi_T$ | Temporal | Stability and time horizons | EPU Index, Inflation Vol. |
| $\Psi_M$ | Market Scope | Geographic reach and openness | Trade/GDP, LPI |
| $\Psi_F$ | Factor Flexibility | Adjustment capacity | EPL Index, Mobility |

**Empirical Grounding of Each Dimension.** Each $\Psi$ dimension is grounded in established research programs:

- $\Psi_C$ **(Cognitive):** Bounded rationality (Simon 1962), prospect theory (Kahneman and Tversky 1979), dual-process theory (Kahneman 2011). See Appendix U for Kahneman-Tversky foundations.
- $\Psi_I$ **(Institutional):** Institutional economics (North 1990), political institutions (Acemoglu and Robinson 2012). See Appendix L.
- $\Psi_S$ **(Social):** Social preferences including inequality aversion, reciprocity, and image concerns (Fehr and Charness 2024). See Appendix AJ.

- $\Psi_K$ **(Informational):** Information economics (Akerlof 1970; Spence 1973; Stiglitz 2000). See Appendix W.
- $\Psi_T$ **(Temporal):** Time preferences and present bias (Laibson 1997; Frederick et al. 2002). See Appendix AH.
- $\Psi_E$ **(Economic):** Resource constraints, budget effects, capital markets (Fama 1970). See Appendix Y.
- $\Psi_M$ / $\Psi_P$ **(Market/Physical):** Economic geography (Krugman 1991), urban economics (Glaeser 2008). See Appendix AI.
- $\Psi_F$ / $\Psi_{Cult}$ **(Flexibility/Cultural):** Cultural dimensions (Hofstede 1980; House et al. 2004), world values (Inglehart 2014). See Appendix V.

The Global Preference Survey (Falk et al. 2018) provides cross-country measures for patience ($\Psi_T$), risk tolerance, social preferences ($\Psi_S$), and trust across 76 countries, demonstrating substantial and systematic variation that predicts economic outcomes.

Crucially, $\Psi$ is *endogenous*:

$$\Psi_{t+1} = f(\Psi_t, a_t) \tag{2}$$

Actions today shape the context tomorrow. This feedback loop creates path dependence, institutional persistence, and the possibility of regime change.

**3. Reference Structure $C^*(\Psi)$.** Given context $\Psi$, there exists a *self-consistent* complementarity structure—the configuration where beliefs about others' behavior match actual behavior. This is the equilibrium toward which systems gravitate.

**4. Coherence Index $K$.** The alignment between actual structure $C$ and reference structure $C^*$:

$$K = 1 - \frac{\|C - C^*(\Psi)\|}{\|C^*(\Psi)\|} \tag{3}$$

High coherence ($K \approx 1$) means stability. Low coherence ($K \ll 1$) means pressure for change. Crucially, coherence is distinct from quality: a system can be coherent but bad (stable dystopia) or incoherent but improving (beneficial transition).

## 1.5 Empirical Validation: The Eight $\Psi$ Dimensions

The framework is not merely theoretical. We test whether the eight $\Psi$ dimensions actually explain treatment effect heterogeneity across contexts.

**The Core Prediction.** If context matters as we claim, then:

$$\tau(X \to Y|\Psi) = \beta_0 + \sum_{k=1}^{8} \beta_k \Psi_k + \text{interactions} \tag{4}$$

Treatment effects should be systematically related to context dimensions.

**Results.** Across six canonical economic relationships, the eight dimensions explain substantial heterogeneity:

Table 2: Empirical Validation: 8Ψ Model Across Six Outcomes

| **Relationship** | $R^2$ | **Adj.** $R^2$ | **Dominant** $\Psi$ |
|---|---|---|---|
| Education → Earnings | 84.5% | 76.8% | $\Psi_K$, $\Psi_M$ |
| Management → Productivity | 78.9% | 68.3% | $\Psi_F$ |
| Democracy → Growth | 39.8% | 9.7% | $\Psi_I$, $\Psi_E$ |
| Patience → Growth | 67.3% | 51.0% | $\Psi_T$, $\Psi_K$ |
| Information → Behavior | 82.4% | 73.7% | $\Psi_K$ (−), $\Psi_F$ |
| Income → Health | 67.7% | 51.6% | $\Psi_T$, $\Psi_I$ |
| **Average** | **70.1%** | **55.2%** | — |

Three findings emerge:

1. **Different dimensions dominate for different effects.** Factor flexibility ($\Psi_F$) matters most for management; information quality ($\Psi_K$) for education returns; temporal stability ($\Psi_T$) for patience effects.

2. **Micro effects are better explained than macro effects.** Individual decisions show more systematic context-dependence ($R^2 \approx 75$–$85\%$) than aggregate outcomes ($R^2 \approx 40$–$70\%$).

3. **No ninth dimension emerged.** We tested ethnic fractionalization, natural resources, and implementation capacity. None improved the model across all outcomes (see Appendix **??** for methodology).

## 1.6 The Framework's Falsifiable Claim

What would refute this framework? We commit to a specific, testable proposition:

> **Core Falsifiable Claim:** *No economic effect is context-free.*

If any intervention produces identical effects across all Ψ configurations—if education returns are the same in Switzerland and Nigeria, if information campaigns work equally in transparent and opaque environments, if management practices improve productivity identically with rigid and flexible labor markets—then our framework would face serious empirical challenges.

This is a strong hypothesis. Standard theory often assumes context-independence (the "average treatment effect" literature). We hypothesize that context-dependence is universal and systematic.

## 1.7 Roadmap

The paper proceeds as follows:

- **Section 2** reviews rationality concepts in classical economics.
- **Section 3** examines limits of utility maximization under stable preferences.
- **Section 4** establishes empirical foundations from theory to measurement.
- **Section 5** develops complementarity as a structural principle.
- **Section 6** derives the reference structure $C^*(\Psi)$ through three approaches.
- **Section 7** analyzes complementarity, fit, and non-concavity.
- **Section 8** provides mathematical formalization.
- **Section 9** develops context as an endogenous variable.
- **Section 10** presents the FEPSDE welfare framework.
- **Section 11** derives novel predictions.
- **Section 12** integrates established theories as limiting cases.
- **Section 13** discusses empirical and policy implications.
- **Section 14** addresses limitations and future work.
- **Section 15** concludes.

Appendices provide: (A) theoretical limiting cases with formal derivations; (B) complementarity conditions by level (**??**); (C) the complete 144-component FEPSDE matrix (**??**); (D) mathematical proofs (**??**); (E) empirical operationalization (**??**); (F) worked examples across levels; (G) glossary and notation (**??**); (H) computational history; (I) Nobel contributions 1969–2024; (J) prospective applications to recent papers (**??**); (K–O) detailed paper analyses: Fehr (**??**), Acemoglu (**??**), Shleifer (**??**), Heckman (**??**), Autor (**??**); (S) falsifiable predictions (**??**); (T) metatheory response to editorial concerns (**??**); (V) the eight $\Psi$ dimensions with empirical validation (**??**).

# 2 Rationality and Stability in Classical Economics

Economic rationality has been defined, refined, and contested for over two centuries. This section traces the evolution of the concept from classical political economy through neoclassical optimization to modern equilibrium theory, identifying both the achievements and the limitations that motivate our framework.

## 2.1 The Classical Foundations: Smith to Mill

**Adam Smith's Dual Insight.** The *Wealth of Nations* (Smith 1776) established two foundational ideas:

1. **Self-interest as mechanism:** "It is not from the benevolence of the butcher, the brewer, or the baker that we expect our dinner, but from their regard to their own interest."

2. **The invisible hand:** Individual self-interest, channeled through markets, produces socially beneficial outcomes.

What is often overlooked is Smith's complementary insight in the *Theory of Moral Sentiments* (Smith 1759): human beings are not merely self-interested but also possess "sympathy"— the capacity to share others' feelings and to care about their judgment. Smith's agent is already *interdependent*: utility depends on others' states and opinions.

**Example: The Butcher's Dilemma.** Consider Smith's butcher in a small village:

- Pure self-interest: Maximize profit by selling low-quality meat at high prices
- With reputation: Quality matters because repeat customers depend on trust
- With sympathy: The butcher cares about customers' welfare, not just their money

The classical tradition recognized that stable commerce requires more than self-interest—it requires *trust*, which emerges from repeated interaction and shared norms. In our framework: stable exchange requires $C_{ij} > 0$ (complementary interests) and context $\Psi$ that supports cooperation.

**John Stuart Mill's Refinement.** Mill (Mill 1848) distinguished between "higher" and "lower" pleasures, introducing a qualitative dimension to utility. This anticipates the FEPSDE distinction: financial gain (lower) vs. emotional fulfillment or identity satisfaction (higher).

Mill also emphasized that preferences are *formed* through education and experience—they are not exogenous but shaped by context. This is precisely the endogeneity of $\Psi$ in our framework.

## 2.2 The Marginalist Revolution: From Value to Utility

**The 1870s Breakthrough.** Jevons, Menger, and Walras independently developed marginal utility theory, shifting economics from objective labor value to subjective utility.

Key innovations:

- Utility is measurable (at least ordinally)
- Rationality means maximizing utility subject to constraints
- Equilibrium is achieved when marginal utilities equal price ratios

**Example: The Diamond-Water Paradox Resolved.** Why are diamonds expensive and water cheap, despite water being essential?

Classical answer (labor theory): Diamonds require more labor to produce.

Marginalist answer: Price reflects *marginal* utility, not total utility. Water is abundant; the marginal glass adds little. Diamonds are scarce; the marginal diamond adds much.

This resolution required the concept of diminishing marginal utility (Marshall 1890): $\partial^2 U/\partial x^2 < 0$. In our notation (see Appendix G for full glossary): the *own-complementarity* $C_{ii} < 0$ (concavity).

**What the Marginalists Assumed Away.** The marginalist framework achieved elegance by assuming:

- **Separability:** $\partial^2 U_i/\partial x_i \partial x_j = 0$ (no cross-effects)
- **Stability:** Preferences don't change
- **Independence:** My utility doesn't depend on your consumption

These assumptions enabled mathematical tractability but excluded precisely the interdependencies that drive real economic behavior.

## 2.3 General Equilibrium: The Arrow-Debreu Achievement

**The Existence Theorem.** Arrow and Debreu (1954) proved—and Debreu (1959) formalized axiomatically—that under certain conditions, a competitive equilibrium exists and is Pareto efficient.

Required conditions:

1. Convex preferences (diminishing marginal utility)
2. Convex production sets (diminishing returns)
3. Complete markets (all goods can be traded)
4. Perfect information (all agents know all prices)
5. No externalities (no unpriced interdependencies)

**Example: The Edgeworth Box.** Two agents, two goods. Each starts with an endowment. Trade moves them to a Pareto-efficient allocation on the contract curve. Competitive prices ensure market clearing.

This is the paradigm of classical rationality: given preferences and endowments, the invisible hand finds the efficient allocation.

**What Arrow-Debreu Achieved.**

- Rigorous proof that markets *can* work
- Clear conditions for efficiency
- Foundation for welfare economics
- Benchmark for policy analysis

**What Arrow-Debreu Assumed.** The fifth condition—no externalities—is crucial. It means:

$$C_{ij} = \frac{\partial^2 U_i}{\partial x_i \partial x_j} = 0 \quad \text{for all } i \neq j \tag{5}$$

Your consumption doesn't affect my utility. Your production doesn't affect my costs. We are connected only through prices.

This is the *separability assumption* that our framework relaxes.

## 2.4 Stability: The Missing Dimension

**Efficiency vs. Stability.** Arrow-Debreu proves that equilibrium is efficient. But is it stable? Will the economy return to equilibrium after a shock?

**The Sonnenschein-Mantel-Debreu Problem.** Sonnenschein (1972) and subsequent work showed that aggregate excess demand functions can have almost any shape consistent with Walras' law and homogeneity.

Implication: General equilibrium theory places almost no restrictions on dynamics. Multiple equilibria are possible. Stability is not guaranteed.

**Example: Coordination Failure.** Consider two workers choosing whether to specialize. If both specialize, they can trade and both gain. If one specializes and the other doesn't, the specialist loses.

| | Worker 2: Specialize | Worker 2: Autarky |
|---|---|---|
| Worker 1: Specialize | (3, 3) | (0, 2) |
| Worker 1: Autarky | (2, 0) | (2, 2) |

Two equilibria: (Specialize, Specialize) and (Autarky, Autarky). The first is Pareto-superior, but the second is also stable. Arrow-Debreu cannot predict which will emerge.

In our framework: Both have $K \approx 1$ (coherent), but different $Q$. The economy can be stably trapped in the inferior equilibrium.

**The Stability Gap.** Classical rationality defines an end-state property—a limitation Keynes (Keynes 1936) emphasized in challenging the assumption of automatic market clearing: optimal allocation given preferences and constraints. But it says nothing about:

- How equilibrium is reached
- Whether equilibrium persists under perturbation
- How the economy moves between multiple equilibria
- What happens when preferences themselves change

This is the gap our framework addresses.

## 2.5 Evolutionary Game Theory: Selection Without Rationality

**The Intuition.** Classical economics assumes people *calculate* the optimal choice. But nobody solves differential equations while shopping. Nobody knows all alternatives. Nobody has infinite computing power.

So where does "rational" behavior come from?

The evolutionary answer: **selection, not calculation**. You don't need to know what's optimal. You just need to survive. Those who play bad strategies disappear. Those who play good ones spread. Rationality emerges as a *result*, not as an assumption.

Think of it this way: A chess grandmaster doesn't calculate every possible move. She uses patterns learned over years of play—patterns that survived because they won games. The market works similarly: firms that make bad choices go bankrupt; strategies that work get imitated and spread.

**Why This Matters for the Framework.** This shift has profound implications:

- **Bounded rationality is not a bug:** Heuristics and biases evolved because they worked in ancestral environments. $\Psi_C$ reflects evolutionary calibration, not "irrationality."
- **Social preferences can survive:** Altruism seems costly, but evolution found ways to sustain it (kin selection, reciprocity, group selection). $\Psi_S$ is evolutionarily stable.
- **Multiple equilibria are natural:** Selection can stabilize different configurations. History determines which one emerges.

**The Technical Foundation.** The stability gap motivated a fundamentally different approach: rather than assuming agents calculate optimal strategies, evolutionary game theory asks which strategies *survive* competitive selection (Weibull 1995; Samuelson 1997; Gintis 2000).

**Maynard Smith: Evolutionary Stable Strategies.** Maynard Smith (1982) introduced the concept of an Evolutionarily Stable Strategy (ESS): a strategy that, once prevalent, cannot be invaded by any rare mutant strategy.

**Definition 2.1** (ESS). *Strategy $s^*$ is evolutionarily stable if for all $s \neq s^*$:*

$$\pi(s^*, s^*) > \pi(s, s^*) \quad \textit{or} \quad [\pi(s^*, s^*) = \pi(s, s^*) \textit{ and } \pi(s^*, s) > \pi(s, s)] \tag{6}$$

*where $\pi(s_i, s_j)$ is the payoff to strategy $s_i$ against $s_j$.*

**Key Insight.** ESS refines Nash equilibrium: every ESS is a Nash equilibrium, but not vice versa. ESS adds a *stability* requirement—the equilibrium must be robust to invasion.

**Replicator Dynamics.** The standard model of evolutionary dynamics (Taylor and Jonker 1978):

$$\dot{x}_i = x_i \left[\pi_i(x) - \bar{\pi}(x)\right] \tag{7}$$

where $x_i$ is the population share using strategy $i$, $\pi_i$ is its payoff, and $\bar{\pi}$ is the population average payoff.

Strategies that earn above-average payoffs grow; those below average shrink. This is selection without foresight—agents don't optimize, they are selected.

**Connection to Classical Equilibrium.** Weibull (1995) showed:

- Rest points of replicator dynamics are Nash equilibria
- Stable rest points are refinements of Nash (often ESS)
- Dynamics provide equilibrium selection among multiple Nash equilibria

**Bounded Rationality as Evolutionary Success.** Gintis (2009) argued that bounded rationality can be understood evolutionarily: heuristics and biases that seem "irrational" may be fitness-maximizing in ancestral environments. This connects $\Psi_C$ (cognitive context) to evolutionary foundations.

**Social Preferences and Evolution.** How can costly prosocial behavior survive selection? Several mechanisms:

- **Kin selection:** Altruism toward relatives (Hamilton 1964)

- **Reciprocity:** Conditional cooperation in repeated games (Axelrod 1984; Nowak 2006)
- **Group selection:** Groups with cooperators outcompete groups without (Boyd and Richerson 2005)
- **Indirect reciprocity:** Reputation sustains cooperation (Nowak and May 1998)

Fehr and Fischbacher (2004) showed that altruistic punishment—costly punishment of defectors—can stabilize cooperation even in one-shot anonymous interactions.

**Quantal Response Equilibrium.** McKelvey and Palfrey (1995) bridged rationality and bounded rationality with QRE:

$$P(s_i) = \frac{\exp(\lambda \cdot \pi_i)}{\sum_j \exp(\lambda \cdot \pi_j)} \tag{8}$$

where $\lambda$ measures rationality: $\lambda = 0$ is random, $\lambda \to \infty$ is fully rational.

QRE explains systematic deviations from Nash in experiments while maintaining equilibrium logic. It operationalizes $\Psi_C$: lower cognitive capacity means lower $\lambda$.

**Framework Integration.** Evolutionary game theory provides the dynamic foundation for our framework:

- $C^*$ is not calculated but *emerges* from evolutionary dynamics
- Multiple $C^*$ may exist; selection determines which is reached
- Bounded rationality ($\Psi_C < \infty$) is evolutionarily grounded
- Social preferences ($\Psi_S$) are evolutionarily stable under specific conditions

The equilibrium concept shifts from "what rational agents would choose" to "what survives selection"—a more robust foundation that doesn't require implausible computational abilities.

## 2.6 From Efficiency to Coherence

**The Limits of Optimization.** Classical rationality is agent-centric: each agent maximizes given the environment. But when agents' utilities are *interdependent*, optimization by each doesn't guarantee system-level stability.

**Example: The Tragedy of the Commons.** Each herder rationally adds cattle to the common pasture. Each optimizes individually. The pasture is destroyed—a collectively irrational outcome.

Arrow-Debreu would say: "This is an externality; define property rights." But property rights themselves require institutions, which require context $\Psi$.

**Example: Bank Runs.** Each depositor rationally withdraws funds when others do (Diamond and Dybvig 1983). Individual optimization leads to collective collapse. The efficient equilibrium (all deposit) is unstable.

In our framework: The "all deposit" equilibrium has high $Q$ but low $K$ when trust is fragile. A small shock triggers a cascade to the "all withdraw" equilibrium.

**The Coherence Criterion.** We propose that rationality should refer not only to individual optimization but to the *stability of the system of interdependent utilities*:

- How robust is the equilibrium to shocks?
- How well-aligned are agents' expectations with realized outcomes?
- How consistent are institutions, norms, and behavior?

This is what the coherence index $K$ measures.

## 2.7 Historical Examples of Coherence and Incoherence

**The Gold Standard (1870-1914): High K, Moderate Q.** The classical gold standard was remarkably coherent:

- Clear rules: currencies pegged to gold
- Automatic adjustment: gold flows balanced payments
- Shared expectations: credible commitment

$K \approx 1$: The system was internally consistent. $Q$ was moderate: growth was constrained by gold supply; crises were deflationary.

**Weimar Germany (1921-1923): Collapse of K.** Hyperinflation destroyed coherence:

- Monetary rules: broken by war finance
- Expectations: deanchored, self-fulfilling collapse
- Institutions: lost credibility

$K \to 0$: Complete breakdown of the institutional framework. $Q$ collapsed: economic and social devastation.

**Post-War Reconstruction (1945-1970): High K, High Q.** The Bretton Woods system and welfare state created:

- Clear rules: fixed exchange rates, capital controls
- Shared expectations: growth, stability, inclusion
- Complementary institutions: monetary policy + fiscal policy + social insurance

$K \approx 1$: Coherent configuration of all elements. $Q$ high: "Les Trente Glorieuses," unprecedented prosperity.

**The 2008 Financial Crisis: K Collapse.** Before the crisis:

- Apparent coherence: Rising asset prices, low volatility
- Hidden incoherence: Risk models assumed stable correlations

During the crisis:

- Correlation breakdown: "All assets fell together"
- Expectation collapse: Counterparty trust evaporated
- Institutional failure: Models proved wrong

$K_{market}$: Dropped from $\approx 0.95$ to $\approx 0.3$. Recovery required massive intervention to restore coherence.

## 2.8 Summary: What Classical Economics Gets Right and Wrong

**Achievements of Classical Rationality.**

- Rigorous framework for optimization
- Proof that markets can achieve efficiency
- Clear welfare criteria
- Foundation for policy analysis

**Limitations of Classical Rationality.**

- Assumes away interdependence ($C_{ij} = 0$)
- Treats preferences as exogenous (no $\Psi$ dynamics)
- Describes efficiency but not stability
- Cannot explain multiple equilibria selection
- Silent on institutional design

**The Extension Required.** A complete theory of rationality must include:

1. Utility interdependence: $C_{ij} \neq 0$
2. Endogenous context: $\Psi_{t+1} = f(\Psi_t, a_t)$
3. Stability criterion: $K$ as measure of coherence
4. Quality criterion: $Q$ as measure of welfare
5. Multi-level analysis: Individual to global

This is what the $(C, \Psi)$ framework provides.

# 3 The Limits of Utility Maximization under Stable Preferences

The classical framework assumes that agents maximize well-defined utility functions with stable preferences. This section examines the empirical and theoretical challenges to this assumption, demonstrating that the stability assumption is not merely a simplification but a substantive restriction that excludes central economic phenomena. Formal derivations showing how each behavioral phenomenon emerges as a limiting case appear in Appendix A.

## 3.1 The Stable Preferences Assumption

**What the Assumption Claims.** Stable preferences means:

1. Utility functions don't change over time: $U_i(x, t) = U_i(x)$
2. Preferences are independent of others: $U_i(x_i, x_j) = U_i(x_i)$
3. Context doesn't affect valuation: $U_i(x; \Psi) = U_i(x)$

**Why Economists Made This Assumption.** The assumption is not arbitrary—it serves crucial theoretical functions:

- **Tractability:** Fixed preferences enable optimization
- **Prediction:** Stable preferences allow demand estimation
- **Welfare:** Unchanging preferences define a welfare criterion
- **Identification:** Without stability, any behavior is "rational"

**Stigler-Becker Defense.** Stigler and Becker (1977) argued that apparent preference changes can always be reinterpreted as responses to changing constraints. "De gustibus non est disputandum"—tastes don't change; only opportunities do.

**Example: Addiction.** Is addiction a changing preference or a stable preference with habit formation?

Stigler-Becker interpretation:

$$U(c_t, S_t) \quad \text{where } S_t = \sum_{\tau < t} c_\tau \text{ is consumption stock} \tag{9}$$

The utility function is stable; past consumption enters as a state variable.

**The Problem.** This reinterpretation saves stability but at a cost: it makes the theory unfalsifiable. Any behavior pattern can be rationalized with a sufficiently complex constraint structure.

## 3.2 Six Domains Where Stable Preferences Fail

### 3.2.1 Domain 1: Social Preferences

**The Evidence.** Thousands of experiments show that people care about others' outcomes:

- **Ultimatum games:** Responders reject unfair offers, sacrificing money (Güth et al. 1982)
- **Dictator games:** Dictators share even when anonymous
- **Public goods games:** Many contribute despite dominant strategy to free-ride (Andreoni 1988)
- **Trust games:** Investors trust and trustees reciprocate (Berg et al. 1995)

**Example: The Ultimatum Game.** Proposer divides \$10. Responder accepts or rejects. If reject, both get zero.

**Classical prediction:** Proposer offers \$0.01; responder accepts.

**Actual behavior:** Modal offer is \$4-5; offers below \$2 often rejected.

**Implication.** Utility is not $U_i(x_i)$ but $U_i(x_i, x_j)$. The cross-derivative is non-zero:

$$\frac{\partial^2 U_i}{\partial x_i \partial x_j} \neq 0 \tag{10}$$

This is precisely the complementarity that classical theory assumes away.

**Models of Social Preferences.** Several models formalize other-regarding preferences:

- **Inequality aversion** (Fehr and Schmidt 1999): $U_i = \pi_i - \alpha \max\{\pi_j - \pi_i, 0\} - \beta \max\{\pi_i - \pi_j, 0\}$
- **ERC model** (Bolton and Ockenfels 2000): Utility depends on own payoff and relative share
- **Social welfare** (Charness and Rabin 2002): Combines efficiency and maximin concerns
- **Reciprocity** (Rabin 1993): Kindness begets kindness; unkindness begets punishment

**The State of the Art.** The comprehensive survey by Fehr and Charness (2024) synthesizes three decades of research on social preferences, documenting:

1. **Distributional preferences:** Inequality aversion and altruism
2. **Belief-dependent preferences:** Reciprocity and guilt aversion (Charness and Dufwenberg 2006)
3. **Image concerns:** Self-image and social image (Bénabou and Tirole 2006)

**Distribution of Types.** A key finding: the majority of individuals have social preferences; purely selfish individuals are a minority:

| Type | Students | General Population |
|---|---|---|
| Selfish | 40–60% | 20–30% |
| Inequality averse | 10–15% | 15–20% |
| Altruistic | 25–45% | 45–55% |

**Merit and Desert.** Social preferences are context-dependent. Tolerance for inequality depends on its source (Cappelen et al. 2007; Almas et al. 2020):

- Inequality from effort/merit: More accepted
- Inequality from luck: Less accepted, more redistribution demanded
- Cultural variation: Americans more meritocratic, Scandinavians more egalitarian

**Context Dependence.** The magnitude of social preferences varies with context:

- Higher in face-to-face interactions
- Higher with in-group members
- Lower in competitive frames
- Lower when "deserving" vs. "undeserving" recipients

Thus: $C_{ij} = C_{ij}(\Psi)$—complementarity depends on context.

### 3.2.2 Domain 2: Identity and Self-Image

**The Evidence.** People sacrifice material payoffs to maintain consistent self-image:

- **Professional identity:** Doctors resist cost-cutting that conflicts with medical ethics
- **Group identity:** Fans support losing teams; patriots sacrifice for country
- **Moral identity:** People avoid situations that would reveal their selfishness

**Example: The Moral Wiggle Room Experiment.** Dana et al. (2007) offered subjects a choice:

- Option A: Get \$6, other gets \$1
- Option B: Get \$5, other gets \$5

Many chose B (fair). But when they could choose "not to know" what the other gets, many chose ignorance and then A. They preferred not to know they were being unfair.

**Implication.** Utility includes self-image:

$$U = U^{material} + U^{IDN} \tag{11}$$

where $U^{IDN}$ is identity utility—the value of being (or seeing oneself as) a good person.

**Temporal Complementarity.** Identity creates intertemporal linkage:

$$\frac{\partial^2 U}{\partial a_t \partial a_{t+1}} > 0 \tag{12}$$

Behaving as type $X$ today makes it easier to see oneself as type $X$ tomorrow. This is the mechanism behind habit, character, and culture.

### 3.2.3 Domain 3: Reference Dependence and Loss Aversion

**The Evidence.** Kahneman and Tversky (1979) demonstrated that outcomes are evaluated relative to a reference point, with losses weighted more than gains:

$$v(x) = \begin{cases} x^\alpha & \text{if } x \geq 0 \\ -\lambda(-x)^\beta & \text{if } x < 0 \end{cases} \tag{13}$$

with $\lambda \approx 2$ (loss aversion coefficient).

**Example: The Endowment Effect.** Knetsch et al. (1990) gave subjects mugs or chocolate bars. Willingness to trade was far below 50%—people valued what they had more than equivalent items they didn't have.

Classical prediction: Indifferent subjects should trade 50% of the time.

**Implication.** The reference point $r$ is part of context:

$$U = U(x - r(\Psi)) \tag{14}$$

Reference points depend on:

- Status quo (what I have)
- Expectations (what I thought I'd get) (Kőszegi and Rabin 2006)
- Social comparison (what others have)
- Aspirations (what I want to have)

All are context-dependent: $r = r(\Psi)$.

#### 3.2.4 Domain 4: Framing and Context Effects

**The Evidence.** Identical choice problems elicit different responses depending on framing:

- **Asian Disease Problem:** "200 saved" vs. "400 die" reverses preferences (Tversky and Kahneman 1981)
- **Default effects:** Organ donation rates vary 10x based on opt-in vs. opt-out
- **Anchoring:** Arbitrary numbers influence willingness to pay

**Example: Organ Donation Defaults.** Johnson et al. (2003) showed:

- Germany (opt-in): 12% donation rate
- Austria (opt-out): 99% donation rate

Same populations, similar cultures, radically different behavior— entirely due to default setting.

**Implication.** Choices depend on how options are presented:

$$\text{Choice}(A, B; \Psi_1) \neq \text{Choice}(A, B; \Psi_2) \tag{15}$$

even when $A$ and $B$ are objectively identical.

Stable preferences would require: same options $\Rightarrow$ same choice. This is empirically false.

#### 3.2.5 Domain 5: Preference Construction and Learning

**The Evidence.** For many choices, preferences are not "retrieved" but "constructed" at decision time:

- **Unfamiliar domains:** How much is a clean environment worth?
- **Complex tradeoffs:** How to weigh career vs. family?
- **New products:** Did you "want" a smartphone before it existed?

**Example: Contingent Valuation.** Willingness to pay for environmental goods varies wildly depending on:

- Order of questions
- Starting point for bidding
- Whether WTP or WTA is elicited
- What comparison goods are mentioned

**Implication.** Preferences for complex goods are not stable attributes but emerge from deliberation in context:

$$U(x; \Psi) = f(\text{construction process}(\Psi)) \tag{16}$$

This is especially true for the "higher" FEPSDE dimensions: emotional, social, identity, and ecological utility are constructed through social processes, not simply discovered.

#### 3.2.6 Domain 6: Temporal Preferences and Present Bias

**The Evidence.** A vast literature documents systematic deviations from exponential discounting (Frederick et al. 2002):

- **Present bias:** People overweight immediate rewards relative to future rewards, even controlling for pure time preference
- **Preference reversals:** The same person prefers \$110 in 31 days over \$100 in 30 days, but \$100 today over \$110 tomorrow
- **Commitment demand:** People pay to restrict their future choices (Ashraf et al. 2006; Bryan et al. 2010)
- **Procrastination:** Costly tasks are perpetually postponed

**The Time Inconsistency Problem.** Strotz (1956) first identified that non-exponential discounting leads to dynamically inconsistent preferences: optimal plans made today are not followed tomorrow.

**The $\beta$-$\delta$ Model.** Laibson (1997) formalized quasi-hyperbolic discounting:

$$U_t = u(c_t) + \beta \sum_{\tau=1}^{T} \delta^\tau u(c_{t+\tau}) \quad \text{where } \beta < 1 \tag{17}$$

The parameter $\beta$ captures present bias—the additional weight on immediate consumption beyond standard exponential discounting $\delta$.

**Empirical Estimates.** Studies consistently find $\beta \approx 0.7$ and $\delta \approx 0.99$ (Frederick et al. 2002; Cohen et al. 2020). This implies:

- Substantial present bias (30% premium on immediate rewards)
- Modest long-run impatience
- Time-inconsistent behavior: plans made today are not followed tomorrow

**Sophistication vs. Naïveté.** O'Donoghue and Rabin (1999, 2001) distinguished:

- **Sophisticates:** Know their $\beta < 1$, may seek commitment
- **Naïfs:** Believe $\hat{\beta} = 1$, perpetually surprised by own behavior
- **Partial naïfs:** $\beta < \hat{\beta} < 1$—underestimate but don't ignore bias

**Framework Integration.** Temporal preferences constitute the $\Psi_T$ dimension of context:

$$\Psi_T = (\beta, \delta, \hat{\beta}) \tag{18}$$

Present bias implies that $C^*$ depends on the timing of decisions:

$$C^*_{immediate} \neq C^*_{planned} \tag{19}$$

This creates demand for commitment devices—institutional arrangements ($\Psi_I$) that constrain future choices. Examples include:

- Illiquid savings accounts (401k, Christmas clubs)
- Deadlines and penalties
- Default enrollment in beneficial programs (Madrian and Shea 2001)

Chetty et al. (2014) found that 85% of savers are "passive" (follow defaults) while only 15% are "active" (respond to incentives)—demonstrating the power of institutional design for present-biased agents.

**Implications.** Temporal context interacts with other $\Psi$ dimensions:

- $\Psi_T \times \Psi_C$: Cognitive load worsens self-control
- $\Psi_T \times \Psi_I$: Institutions can provide commitment
- $\Psi_T \times \Psi_E$: Poverty may intensify present bias

## 3.3 The Deeper Problem: Utility Interdependence

**Beyond Individual Violations.** Each domain above could be treated as a "behavioral anomaly"— a deviation from rationality requiring a patch.

But the deeper issue is *structural*: utilities are interdependent in ways that classical theory cannot accommodate.

**Four Types of Interdependence.** **Cross-agent interdependence:**

$$\frac{\partial U_i}{\partial x_j} \neq 0 \tag{20}$$

My utility depends on your outcome (social preferences).

**Temporal interdependence:**

$$\frac{\partial U_t}{\partial a_{t-1}} \neq 0 \tag{21}$$

My current utility depends on my past actions (habits, identity).

**Intertemporal inconsistency:**

$$U_t(c_t, c_{t+1}) \neq \delta \cdot U_{t+1}(c_{t+1}, c_{t+2}) \tag{22}$$

My time preferences are not constant (present bias).

**Context interdependence:**

$$\frac{\partial U}{\partial \Psi} \neq 0 \tag{23}$$

My utility depends on the institutional environment (framing, norms).

**The Complementarity Implication.** All four types generate non-zero second derivatives:

$$C_{ij} = \frac{\partial^2 U_i}{\partial x_i \partial x_j} \neq 0 \tag{24}$$

This is not a minor technical detail—it is the structural feature that creates coordination problems, multiple equilibria, and path dependence.

## 3.4 What Utility Maximization Gets Wrong

**Not Wrong, But Incomplete.** The problem is not that utility maximization is false. Given a utility function and constraints, maximization is a reasonable model of choice.

The problem is that the framework presupposes:

1. The utility function is known and stable
2. The constraints are exogenous
3. Agents optimize independently

When utilities are interdependent and context is endogenous, "maximize utility" doesn't define behavior— it merely redescribes whatever happens as "utility-maximizing."

**The Identification Problem.** If utility can include anything (other-regarding preferences, identity, reference points, framing effects), then any behavior is "rational." The theory becomes unfalsifiable.

**The Equilibrium Selection Problem.** With interdependent utilities, multiple equilibria exist. Utility maximization says: at equilibrium, everyone is maximizing. It doesn't say which equilibrium will emerge.

**The Stability Problem.** Utility maximization describes optimal response to given conditions. It doesn't describe whether those conditions will persist— whether the equilibrium is stable or fragile.

## 3.5 The Required Extension

**From Optimization to Coherence.** The limits of utility maximization suggest a different criterion: not "is each agent optimizing?" but "is the system of utilities coherent?"

Coherence means:

- Expectations match outcomes
- Institutions support behavior
- Norms align with incentives
- Identity is consistent with action

**The Framework Extension.**

1. Allow $C_{ij} \neq 0$: Interdependent utilities
2. Allow $\Psi_{t+1} \neq \Psi_t$: Endogenous context
3. Define $K$: Coherence as structural stability
4. Define $Q$: Quality as welfare at equilibrium
5. Analyze stability: When does $K \approx 1$ persist?

This is what the following sections develop.

# 4 Empirical Foundations: From Theory to Measurement

The theoretical apparatus developed in previous sections—complementarity matrices, context dynamics, coherence indices—would be empty without empirical grounding. This section establishes that the framework's core concepts are not merely logically possible but empirically real, measurable, and policy-relevant.

We review evidence from five research programs: experimental economics, field studies, complexity economics (Arrow et al. 1997), neuroeconomics, and cross-cultural research. Each contributes distinct evidence that complementarities are pervasive, context-dependent, and consequential.

## 4.1 The Experimental Revolution in Social Preferences

### 4.1.1 The Ultimatum Game: Where It All Began

**The Setup.** Güth et al. (1982) introduced what became the most influential experiment in behavioral economics. Two players divide a sum (say, $10):

- Proposer offers a split (e.g., "$7 for me, $3 for you")
- Responder accepts (both get proposed amounts) or rejects (both get $0)

**Standard Prediction.** Under separable preferences ($C_{ij} = 0$), the Responder should accept any positive offer. The Proposer, anticipating this, should offer the minimum ($0.01). Subgame perfect equilibrium: ($9.99, $0.01).

**Actual Results.** Across hundreds of replications in dozens of countries:

- Modal offer: 40-50% of the pie ($4-5)
- Mean offer: 30-40%
- Rejection rate for offers below 20%: 50%+
- Rejection rate for offers below 10%: 80%+

**Interpretation.** People reject "unfair" offers even at personal cost. This requires $\partial^2 U_i/\partial x_i \partial x_j \neq 0$: Responder's utility depends not just on own payoff but on the split.

### 4.1.2 The Dictator Game: Isolating Fairness

**The Setup.** Same as Ultimatum, but Responder has no veto. Proposer simply decides the split.

**Standard Prediction.** With no strategic reason to share, Proposer should keep everything: ($10, $0).

**Actual Results.**

- Mean transfer: 20-30% of the pie
- Fraction giving $0: 30-40%
- Fraction giving 50%: 10-20%

**Interpretation.** Even without strategic incentives, many people share. This reflects pure social preferences: $U_i$ depends on $x_j$.

### 4.1.3 Public Goods Games: Conditional Cooperation

**The Setup.** $n$ players each receive endowment $e$. Each chooses contribution $c_i \in [0, e]$ to a public good. The public good is multiplied by factor $m > 1$ and shared equally:

$$\pi_i = e - c_i + \frac{m \sum_j c_j}{n} \tag{25}$$

If $m < n$, the private return to contributing is negative. Nash equilibrium: $c_i = 0$ for all $i$.

**Actual Results.** Fehr and Gächter (2000) and many replications find:

- Initial contributions: 40-60% of endowment
- With repetition: Contributions decline toward 10-20%
- With punishment option: Contributions rise to 80-90%
- Strong heterogeneity: "Conditional cooperators" (50%), "Free riders" (30%), "Unconditional cooperators" (10%)

**Interpretation.** Most people are *conditional cooperators*: they contribute if others contribute. This is strategic complementarity:

$$\frac{\partial c_i}{\partial c_{-i}} > 0 \tag{26}$$

The complementarity structure depends on context (punishment, communication, group identity).

### 4.1.4 Trust Games: Reciprocity Quantified

**The Setup.** Berg, Dickhaut, and McCabe (1995): Sender receives \$10, can send any amount $s$ to Receiver. Amount triples. Receiver can return any amount $r \in [0, 3s]$.

**Standard Prediction.** Receiver returns \$0 (no incentive). Sender anticipates this, sends \$0.

**Actual Results.**

- Mean sent: \$5 (50%)
- Mean returned: 90-95% of amount sent
- Strong positive correlation: More sent $\Rightarrow$ more returned

**Interpretation.** Reciprocity is quantifiable: $\partial r/\partial s > 0$. Trust and trustworthiness are complements. The complementarity $C_{trust}$ can be estimated from experimental data.

### 4.1.5 Formalizing Social Preferences

**Fehr-Schmidt Inequity Aversion.** Fehr and Schmidt (1999) proposed:

$$U_i(x) = x_i - \alpha_i \cdot \frac{1}{n-1} \sum_{j \neq i} \max(x_j - x_i, 0) - \beta_i \cdot \frac{1}{n-1} \sum_{j \neq i} \max(x_i - x_j, 0) \tag{27}$$

where $\alpha_i$ captures disadvantageous inequity aversion (envy) and $\beta_i$ captures advantageous inequity aversion (guilt).

**Estimated Parameters.** From experimental data:

- $\alpha$: Mean $\approx 0.85$, range $[0, 4+]$
- $\beta$: Mean $\approx 0.315$, range $[0, 0.6]$
- Constraint: $\beta_i \leq \alpha_i$ (guilt $\leq$ envy)
- High variance across individuals

**What This Means for $C$.** The second derivative:

$$C_{ij} = \frac{\partial^2 U_i}{\partial x_i \partial x_j} \neq 0 \tag{28}$$

The magnitude depends on $(\alpha_i, \beta_i)$, which vary across people and contexts.

**Bolton-Ockenfels ERC.** Bolton and Ockenfels (2000) proposed an alternative model where utility depends on own payoff and *share*:

$$U_i = U_i\left(x_i, \frac{x_i}{\sum_j x_j}\right) \tag{29}$$

Also generates $C_{ij} \neq 0$ through the share term.

## 4.2 Field Evidence: External Validity

### 4.2.1 Does Lab Behavior Predict Field Behavior?

**The Critique.** Laboratory experiments use small stakes, student subjects, and artificial settings. Do results generalize?

**The Evidence.** Levitt and List (2007) and subsequent research show:

- Stakes effects: Moderate. Doubling stakes increases selfishness slightly.
- Subject pool: Some differences. Non-students often more prosocial.
- Scrutiny effects: Significant. Anonymity reduces prosociality.
- Core findings: Robust. Social preferences exist outside the lab.

### 4.2.2 Gift Exchange in Labor Markets

**The Phenomenon.** Gneezy and Rustichini (2000) showed that workers respond to wage increases with higher effort—but only when framed as gifts. Explicit incentive contracts can *crowd out* intrinsic motivation.

**Field Evidence.** Fehr et al. (2009) and List and Millimet (2008) conducted field experiments:

- Tree planters paid unexpectedly high wages increased output 10-20%
- Effect disappeared when framed as piece rate adjustment
- Complementarity between perceived fairness and effort: $C_{wage,effort}(\Psi_{framing})$

### 4.2.3 Trust and Economic Development

**Cross-Country Correlations.** Knack and Zak (2003) established that generalized trust correlates with GDP per capita ($r \approx 0.6$).

**Causal Evidence.** Algan and Cahuc (2010) exploited variation in inherited trust among US immigrants from different countries:

- Trust levels persist across generations
- Inherited trust predicts economic behavior in the US
- Effect is causal (immigrants face same institutions)

**Interpretation.** Trust is a form of complementarity: "I cooperate because I expect you to cooperate." Trust structures are context-dependent ($\Psi$) and persistent.

#### 4.2.4 Cooperation in the Field

**Common Pool Resources.** Ostrom (1990) documented how communities manage common resources without privatization or government regulation. Key finding: Communities develop norms and institutions that sustain cooperation (Axelrod 1984)— *endogenous context* that shapes complementarity structures.

**Tax Compliance.** Slemrod (1998) shows that tax compliance depends heavily on perceived fairness of the tax system and beliefs about others' compliance. This is conditional cooperation at scale: $C_{compliance} = C_{compliance}(\Psi_{legitimacy})$.

### 4.3 The Replication Movement: What Survives?

**The Crisis.** Psychology's replication crisis (Camerer et al. 2016) raised questions about behavioral economics findings. Camerer (2018) systematically replicated 21 experimental economics studies.

**The Results.**

- 62% replicated (same direction and significant)
- 77% replicated (same direction, allowing for smaller effects)
- Effect sizes: Average 75% of original

**What Replicates Reliably.**

- Ultimatum game rejections: Yes
- Dictator game giving: Yes (smaller than originally thought)
- Public goods conditional cooperation: Yes
- Trust game reciprocity: Yes
- Context effects on prosociality: Yes (effect sizes vary)

**What This Means.** Core findings on social preferences are robust. Complementarities are real. Effect sizes are context-dependent— exactly what the $(C, \Psi)$ framework predicts.

## 4.4 Complexity Economics and Aggregate Dynamics

### 4.4.1 Increasing Returns and Path Dependence

**Arthur's Insight.** Arthur (1989) showed that positive feedbacks—complementarity at the aggregate level— produce phenomena impossible under diminishing returns:

- Path dependence: History matters
- Lock-in: Inferior technologies can dominate
- Multiple equilibria: Same fundamentals, different outcomes

**Example: QWERTY.** The keyboard layout persists not because it's optimal but because complementarities (training, software, expectations) create lock-in. Switching requires coordinated change across the complementarity structure.

### 4.4.2 Network Effects

**Formal Structure.** Katz and Shapiro (1985): If utility from joining network is

$$U_i = v(n) - p \tag{30}$$

where $v'(n) > 0$ (value increases with network size), then adoption decisions are strategic complements.

**Dynamics.** Network effects generate:

- Tipping points: Small advantages become large
- Winner-take-all: One network dominates
- Hysteresis: Hard to reverse once established

These are precisely the dynamics that emerge when $C_{ij} > 0$ at population scale.

### 4.4.3 The Santa Fe Perspective

**Economy as Adaptive System.** Arthur (2015) and the Santa Fe school reframe economics:

- Agents have bounded rationality
- Expectations and behaviors co-evolve

- Equilibrium is temporary coherence, not permanent rest

This aligns with our framework: stability is coherence ($K \approx 1$), not absence of change. The reference structure $C^*$ is what expectations converge toward, not a fixed point forever.

## 4.5 Neuroeconomic Substrates

### 4.5.1 Neural Encoding of Social Value

**Key Finding.** Rangel et al. (2008) and Fehr and Camerer (2007) show:

- Social rewards (fairness, reciprocity) activate same brain regions as material rewards
- Ventromedial prefrontal cortex encodes "common currency" of value
- Social and material rewards are substitutable in the brain

**Interpretation.** The FEPSDE dimensions are not arbitrary—they reflect distinct but neurally integrated value systems. $U^F$, $U^E$, $U^S$ are processed by overlapping neural circuits.

### 4.5.2 Oxytocin and Trust

**The Experiment.** Kosfeld et al. (2005): Subjects given oxytocin (nasal spray) vs. placebo played trust games.

**Results.**

- Oxytocin increased trust (amount sent) by 17%
- No effect on risk-taking in non-social contexts
- Effect is specifically social

**Interpretation.** Neuroendocrine states modulate $C_{ij}$. This is biological evidence for context-dependent complementarity: $C_{trust} = C_{trust}(\Psi_{oxytocin})$.

### 4.5.3 Punishment and Reward Circuits

**Key Finding.** de Quervain et al. (2004): Punishing free-riders activates reward circuits (striatum). People experience satisfaction from enforcing norms.

**Interpretation.** Third-party punishment—crucial for sustaining cooperation at scale—is not just strategic but intrinsically rewarding. This explains why complementarity structures can be self-enforcing: people *want* to punish defectors.

## 4.6 Cross-Cultural Evidence

### 4.6.1 The WEIRD Problem

**The Critique.** Henrich et al. (2010) showed that most behavioral research uses Western, Educated, Industrialized, Rich, Democratic (WEIRD) subjects. WEIRD populations are often outliers, not representative of humanity.

**Cross-Cultural Ultimatum Game.** Henrich et al. (2000) played ultimatum games in 15 small-scale societies:

- Mean offers ranged from 26% (Machiguenga) to 58% (Lamelara)
- Rejection rates varied from 0% to 40%+
- Variation correlated with market integration and cooperation norms

**Interpretation.** The *structure* of social preferences is universal: $C_{ij} \neq 0$ everywhere. But the *parameters* vary with context: $C_{ij} = C_{ij}(\Psi_{culture})$.

### 4.6.2 Culture and Punishment

**The Finding.** Herrmann et al. (2008): In some societies, not only free-riders but also *high contributors* get punished ("antisocial punishment").

**Interpretation.** Complementarity structures can be negative: in low-trust societies, cooperation can be seen as threatening (showing off, making others look bad). This is $C_{ij} < 0$—choices are substitutes, not complements.

## 4.7 Implications for the Framework

The empirical evidence establishes five facts that the framework takes as axiomatic:

1. **Complementarities are real.** Hundreds of experiments across decades confirm that $C_{ij} \neq 0$ is the rule, not the exception.
2. **Complementarities are measurable.** Structural estimation methods can recover $(\alpha, \beta)$ parameters and thus estimate the complementarity matrix $C$.
3. **Complementarities are context-dependent.** The same individuals exhibit different $C_{ij}$ under different institutions, framings, and social settings. This motivates $C = C(\Psi)$.
4. **Complementarities have neural substrates.** Social preferences are not metaphors but are encoded in brain circuits, supporting the biological reality of FEPSDE dimensions.

5. **Complementarities vary cross-culturally.** The structure is universal; the parameters are culturally specific. This supports the context-dependence of $C^*(\Psi)$.

**Methodological Implications.**

- Experimental methods can estimate components of $C$
- Field experiments can test external validity
- Cross-cultural comparisons can identify how $\Psi$ shapes $C$
- Neuroscience can validate the dimensional structure of utility

The framework is not purely theoretical—it has empirical foundations and generates empirically testable predictions.

## 4.8 Technological Enablement: Why Measurement Is Now Feasible

The empirical foundations described above have existed for decades. What has changed is our ability to *scale* and *integrate* measurement.

**Traditional Limitations.**

- Laboratory experiments: Small samples, limited contexts
- Surveys: Self-report bias, costly administration
- Field studies: Case-by-case, hard to aggregate
- Cross-cultural research: Expensive, slow, sparse coverage

**New Capabilities (2024-2026). Digital Behavioral Data.** Every online interaction generates data about revealed preferences:

- Purchasing patterns reveal $C^{FF}$ (financial complementarities)
- Social media behavior reveals $C^{SS}$ (social complementarities)
- Health app data reveals $C^{PP}$ (physical utility patterns)
- Platform usage reveals $C^{DD}$ (digital complementarities)

The scale is unprecedented: billions of observations across diverse contexts, enabling estimation of context-dependent complementarities $C(\Psi)$.

**NLP for Soft Variables.** Context $\Psi$ includes norms, trust, and institutional quality—variables that resist direct measurement. NLP and text-as-data methods (Gentzkow et al. 2019) now enable:

- Sentiment analysis of institutional legitimacy
- Topic modeling of policy discourse
- Embedding-based measures of cultural similarity
- Automated coding of legal and regulatory texts

**LLM-Assisted Survey Design.** Traditional surveys are expensive and slow to adapt. LLM-powered systems (Brown et al. 2020; Vaswani et al. 2017) can:

- Generate context-appropriate survey instruments
- Conduct adaptive questioning in natural language
- Translate across languages while preserving meaning
- Analyze open-ended responses at scale

**Computational Structural Estimation.** Estimating the Fehr-Schmidt parameters $(\alpha, \beta)$ from experimental data required bespoke econometric procedures. Modern tools enable:

- Automatic likelihood construction from model specification
- Bayesian estimation with flexible priors
- Model comparison and selection
- Propagation of uncertainty through downstream analysis

**From Parameters to Matrices.** The leap from estimating two parameters $(\alpha, \beta)$ to estimating a $144 \times 144$ complementarity matrix seemed impossible. But with modern computational infrastructure:

- Sparse matrix representations reduce dimensionality
- Hierarchical models pool information across contexts
- Transfer learning leverages patterns across domains
- Active learning prioritizes informative observations

The empirical program outlined above is now *tractable*— challenging but achievable with current technology.

For a detailed history of how statistical and computational capabilities evolved from the 1960s to today, see Appendix **??**.

# 5 Complementarity as a Structural Principle

This section develops the central concept of the framework: complementarity. We show that complementarity is not merely a technical property of utility functions but a structural principle that explains stability, coordination, and coherence across all levels of economic organization.

## 5.1 The Intuition: Why Choices Interact

**A Simple Example.** Consider a firm choosing two policies: investment in employee training ($x_1$) and adoption of advanced technology ($x_2$).

**Independent choices** ($C_{12} = 0$): The return to training doesn't depend on technology level. The return to technology doesn't depend on training. Each choice can be optimized separately.

**Complementary choices** ($C_{12} > 0$): Trained employees get more out of advanced technology. Advanced technology makes training more valuable. The *whole* exceeds the sum of *parts*.

**Substitute choices** ($C_{12} < 0$): If we invest in training, we don't need technology (people can improvise). If we invest in technology, we don't need training (machines do the work). One investment crowds out the other.

**The Fundamental Question.** Classical economics assumes independence: $C_{ij} = 0$. This makes optimization easy—each choice can be made separately. But real choices interact. The question is: *How do we analyze systems where choices are interdependent?*

Complementarity theory provides the answer.

## 5.2 Formal Definition: Supermodularity

**Definition 5.1** (Supermodularity)**.** *A function $f : \mathbb{R}^n \to \mathbb{R}$ is* supermodular *if for all $i \neq j$:*

$$\frac{\partial^2 f}{\partial x_i \partial x_j} \geq 0 \tag{31}$$

*Equivalently, for all $x, y \in \mathbb{R}^n$:*

$$f(x \vee y) + f(x \wedge y) \geq f(x) + f(y) \tag{32}$$

*where $x \vee y$ is the component-wise maximum and $x \wedge y$ the component-wise minimum.*

**Interpretation.** Supermodularity means: *Doing more of one thing increases the return to doing more of another.* The cross-partial derivative is non-negative. Actions are complements, not substitutes.

**The Lattice Formulation.** The $\vee/\wedge$ formulation shows supermodularity is about *order*: moving "up" in all dimensions together is better than moving up in some and down in others. This connects complementarity to coordination: everyone moving together beats mixed movement.

## 5.3 The Intellectual Heritage

**Edgeworth (1881): Complementarity in Consumption.** Edgeworth (1881) first noted that goods can be complements: the marginal utility of coffee increases with sugar. This is consumer-level complementarity: $\partial^2 U/\partial c_{coffee}\partial c_{sugar} > 0$.

**Milgrom and Roberts (1990): The Foundational Contributions.** Milgrom and Roberts (1990a) established the mathematical foundations for analyzing complementarity in economic systems. Their key results:

1. Pure strategy Nash equilibria exist in supermodular games (no convexity needed)
2. Equilibria form a complete lattice (smallest and largest equilibria exist)
3. Comparative statics are monotone (higher parameters $\rightarrow$ higher equilibrium)
4. Best responses are isotone (if others do more, optimal response is more)

These results revolutionized the analysis of strategic interaction by providing powerful tools that do not require the restrictive assumptions of classical game theory.

**Modern Manufacturing.** In a companion paper, Milgrom and Roberts (1990b) applied complementarity theory to explain the transformation of manufacturing. They identified a cluster of complementary practices:

- Shorter production runs
- Reduced inventories (JIT)
- Improved quality control (TQM)
- Flexible manufacturing systems
- Increased worker training
- Flatter organizational hierarchies

The key insight: these practices are *complements*, not independent choices. Adopting one raises the return to adopting others. In our notation:

$$C_{JIT,TQM} > 0, \quad C_{JIT,flex} > 0, \quad C_{TQM,training} > 0 \tag{33}$$

This explains why firms adopt coherent "bundles" of practices rather than mixing traditional and modern approaches—piecemeal reform breaks complementarities without creating new ones.

**Strategic Fit.** Milgrom and Roberts (1995) extended this to corporate strategy. Successful firms achieve "fit" across activities. Southwest Airlines succeeds through complementary choices:

- Point-to-point routes (no hub)
- No meals, no assigned seats
- Fast turnaround (25 minutes)
- One aircraft type (Boeing 737)
- High employee motivation

Imitating one element without the others fails. The system is coherent—$K \approx 1$— and that coherence is the source of competitive advantage.

**Topkis (1998): The Mathematics.** Topkis (1998) provided the rigorous mathematical foundations for supermodularity and complementarity theory, establishing:

- Supermodular games have pure-strategy Nash equilibria
- Equilibria form a complete lattice (highest and lowest equilibria exist)
- Best responses are monotone (if others do more, you should do more)
- Comparative statics are well-behaved

**Empirical Evidence: HRM Bundles.** Ichniowski et al. (1997) tested complementarity in human resource management. Studying 36 steel production lines, they found:

- Individual HRM practices had small effects
- "Bundles" of complementary practices (incentive pay + teams + training + job security) raised productivity by 7% and reduced defects
- Piecemeal adoption was ineffective

This empirical finding confirms the theoretical prediction: with strong complementarities, coherent bundles dominate mixed strategies.

**Vives (2005): Complementarity and Competition.** Vives (2005) showed that complementarity shapes market structure:

- Complementary products tend toward integration or coordination
- Substitute products tend toward competition
- Industry structure reflects underlying complementarity patterns

**Bulow, Geanakoplos, and Klemperer (1985): Strategic Substitutes and Complements.** Bulow et al. (1985) introduced the influential distinction between strategic complements and strategic substitutes in oligopoly theory, showing how the sign of $\partial^2 \pi_i / \partial a_i \partial a_j$ determines whether competition leads to matching or differentiation.

## 5.4 Types of Complementarity

Complementarity manifests in several distinct forms:

### 5.4.1 Strategic Complementarity

**Definition.** My best response increases when you do more:

$$\frac{\partial^2 \Pi_i}{\partial a_i \partial a_j} > 0 \quad \Rightarrow \quad \frac{\partial a_i^*(a_j)}{\partial a_j} > 0 \tag{34}$$

**Examples.**

- **Bank runs:** If others withdraw, I should withdraw (Diamond-Dybvig)
- **Technology adoption:** If others adopt, adoption is more valuable for me
- **Cooperation:** If others cooperate, cooperation is less risky for me

**Consequence: Multiple Equilibria.** Strategic complementarity generates multiple equilibria:

- High equilibrium: Everyone cooperates / adopts / trusts
- Low equilibrium: No one cooperates / adopts / trusts
- Both are self-fulfilling

### 5.4.2 Temporal Complementarity

**Definition.** My action today increases the return to my action tomorrow:

$$\frac{\partial^2 U}{\partial a_t \partial a_{t+1}} > 0 \tag{35}$$

**Examples.**

- **Habit formation:** Exercise today makes exercise tomorrow easier
- **Human capital:** Learning today increases return to learning tomorrow
- **Identity:** Honest behavior today reinforces honest self-image tomorrow

**Consequence: Path Dependence.** Temporal complementarity creates momentum: early choices shape later choices, creating lock-in effects.

### 5.4.3 Cross-Domain Complementarity

**Definition.** My action in domain $A$ increases the return to action in domain $B$:

$$\frac{\partial^2 U}{\partial a^A \partial a^B} > 0 \tag{36}$$

**Examples.**

- **Work-family:** Success at work can enhance (or undermine) family life
- **Health-wealth:** Health enables earning; wealth enables health investment
- **Trust-trade:** Social trust enables economic exchange; trade builds trust

**Consequence: Spillovers.** Cross-domain complementarity means interventions in one domain affect others. This is why narrow policy analysis often fails: it ignores spillovers.

### 5.4.4 Cross-Agent Complementarity

**Definition.** My utility increases when you do more:

$$\frac{\partial^2 U_i}{\partial x_i \partial x_j} > 0 \tag{37}$$

**Examples.**

- **Network effects:** Your participation makes the platform more valuable to me
- **Herd immunity:** Your vaccination protects me
- **Social norms:** Your compliance makes compliance easier for me

**Consequence: Coordination Problems.** Cross-agent complementarity creates coordination problems: we could all be better off if we all moved together, but individual incentives may not support the move.

## 5.5 Why Complementarity Implies Coherence

**The Logic.** When choices are complementary:

1. High levels reinforce high levels
2. Low levels reinforce low levels
3. Mixed configurations are unstable

Therefore, systems with strong complementarity tend toward *coherent configurations*—states where all elements are aligned, either all high or all low.

**Formal Statement.**

**Proposition 5.2.** *If the payoff function $f$ is supermodular and $\Phi$ is the best-response mapping, then:*

1. *$\Phi$ is monotone increasing*
2. *The set of fixed points forms a complete lattice*
3. *Highest and lowest Nash equilibria exist*
4. *Starting from any point, best-response dynamics converge to a fixed point*

**Interpretation.** Complementarity creates *natural attractors*—coherent configurations that the system tends toward. These are the $C^*$ of our framework.

## 5.6 The Opposite Case: Substitutes

**Definition.** Actions are *substitutes* when:

$$\frac{\partial^2 f}{\partial x_i \partial x_j} < 0 \tag{38}$$

Doing more of one reduces the return to the other.

**Examples.**

- **Cournot competition:** If you produce more, my marginal revenue falls
- **Arms races:** If you arm more, I must arm more (but arms are costly)
- **Status goods:** Your status gain is my status loss

**Consequences.** With substitutes:

- Best responses are decreasing (you do more $\Rightarrow$ I do less)
- Equilibria may be unique (not multiple)
- Coordination is less important (we naturally differentiate)

**Mixed Systems.** Real systems have both complements and substitutes. The complementarity matrix $C$ captures this structure:

$$C_{ij} = \frac{\partial^2 U}{\partial x_i \partial x_j} \begin{cases} > 0 & \text{complements} \\ = 0 & \text{independent} \\ < 0 & \text{substitutes} \end{cases} \tag{39}$$

The pattern of signs in $C$ determines system dynamics.

## 5.7 Complementarity at Each Level

Complementarity operates at every level of the framework:

### 5.7.1 Individual Level

**Internal Coherence.** An individual's choices across life domains are complementary:

- Values and behavior: Living according to values reinforces identity
- Health and productivity: Exercise improves work performance and mood
- Learning and earning: Skills enable income; income enables learning

**The Coherent Individual.** A person with $K_{individual} \approx 1$ has aligned:

- Goals and habits
- Values and actions
- Short-term and long-term interests

Incoherence ($K_{individual} < 1$) manifests as: inner conflict, procrastination, regret, identity confusion.

### 5.7.2 Household Level

**Family Complementarity.** Household members' choices interact:

- Specialization: One earns, one cares (traditional complementarity)
- Coordination: Shared schedules, resources, goals
- Conflict: Zero-sum decisions, incompatible preferences

**The Coherent Household.** A family with $K_{household} \approx 1$ has:

- Clear role expectations
- Aligned priorities
- Effective communication

Incoherence manifests as: conflict, misunderstanding, divorce.

### 5.7.3 Organization Level

**Organizational Complementarity.** This is Milgrom and Roberts (1990b)'s core domain:

- Strategy and structure must fit
- Processes and culture must align
- Incentives and goals must match

**The Coherent Organization.** A firm with $K_{org} \approx 1$:

- All elements reinforce the strategy
- Employees understand and support direction
- Processes enable rather than obstruct goals

Incoherence manifests as: bureaucratic dysfunction, strategic failure, employee disengagement.

### 5.7.4 Regional Level

**Regional Complementarity.** A region's elements interact:

- Industry clusters: Firms benefit from proximity to related firms
- Infrastructure and economy: Roads enable trade enable investment in roads
- Education and employment: Skills attract firms attract investment in skills

**The Coherent Region.** Silicon Valley, Wall Street, Hollywood:

- Specialized ecosystem
- Self-reinforcing advantages
- High barriers to replication

Incoherence manifests as: declining regions, brain drain, economic stagnation.

### 5.7.5 National Level

**Institutional Complementarity.** Hall and Soskice (2001) documented varieties of capitalism:

- **Liberal market economies (US, UK):** Flexible labor, dispersed ownership, general education
- **Coordinated market economies (Germany, Japan):** Rigid labor, concentrated ownership, vocational training

Each is coherent ($K \approx 1$) but different. Mixing elements from different models reduces coherence.

**The Coherent Nation.** A country with $K_{national} \approx 1$:

- Institutions support each other
- Culture aligns with economic structure
- Political system reflects social bargains

Incoherence manifests as: institutional dysfunction, reform failure, social conflict.

### 5.7.6 Global Level

**International Complementarity.** The global order involves complementarities:

- Trade and peace: Economic interdependence raises cost of war
- Institutions and cooperation: WTO, IMF, UN enable coordination
- Norms and behavior: Shared expectations stabilize interactions

**The Coherent Global Order.** Post-WWII Bretton Woods system:

- US hegemony provided public goods
- Institutions coordinated policy
- Norms supported cooperation

Current global *incoherence*:

- Hegemonic transition (US $\rightarrow$ multipolar)
- Institutional erosion (WTO, WHO, climate)
- Norm conflict (sovereignty vs. intervention, free trade vs. industrial policy)

## 5.8 The Coherence Index: Measuring Complementarity Alignment

**Definition Recap.** The coherence index measures alignment between actual and reference structure:

$$K = 1 - \frac{\|C - C^*\|}{\|C^*\|} \tag{40}$$

where $C$ is the observed complementarity matrix and $C^*$ is the self-consistent reference.

**What $K$ Captures.**

- $K = 1$: Perfect alignment. Expectations match behavior. Stable equilibrium.
- $K \approx 1$: Near alignment. Small deviations. System is stable.
- $K \ll 1$: Misalignment. Expectations diverge from behavior. Instability.
- $K \approx 0$: Chaos. No stable pattern. Crisis.

**Why $K$ Matters.** Coherence ($K$) is distinct from quality ($Q$). A system can be:

- High $K$, high $Q$: Stable and good (Nordic countries)
- High $K$, low $Q$: Stable but bad (authoritarian equilibria)
- Low $K$, high $Q$ potential: Unstable, transitioning to good (reform periods)
- Low $K$, low $Q$: Unstable and bad (failed states)

## 5.9 Complementarity and Classical Economics

**The Hidden Assumption.** Arrow-Debreu general equilibrium (Arrow and Debreu 1954) assumes $C_{ij} = 0$:

$$U_i(x_i, x_j) = U_i(x_i) \quad \Rightarrow \quad \frac{\partial^2 U_i}{\partial x_i \partial x_j} = 0 \tag{41}$$

Under this assumption:

- Each agent optimizes independently
- Market prices coordinate without explicit coordination
- Equilibrium is unique and efficient

**What Happens When $C_{ij} \neq 0$.** With interdependence:

- Independent optimization is insufficient
- Prices may not coordinate effectively
- Multiple equilibria can exist
- Coordination becomes essential

**The Framework's Contribution.** We don't reject classical economics—we show it as a *special case*. When $C_{ij} \approx 0$ (atomistic markets, weak interdependence), classical results hold. When $C_{ij} \neq 0$ (social preferences, network effects (Granovetter 1985), institutions), the fuller framework is needed.

## 5.10 Summary: Complementarity as Foundation

Complementarity is the structural foundation of the framework because:

1. **It captures interdependence.** Real economic choices interact. Complementarity formalizes how.
2. **It explains coordination.** When choices are complementary, coordination matters. Coherence emerges from aligned complementarities.
3. **It generates multiple equilibria.** Complementarity creates multiple stable configurations ($C^*$). History matters for which one is selected.
4. **It connects levels.** The same logic applies from individuals to global systems. Complementarity is scale-invariant.
5. **It unifies theories.** Classical economics assumes $C_{ij} = 0$. Behavioral economics studies $C_{ij} \neq 0$. The framework embraces both.

## 5.11 Why Complementarity Becomes Operational in 2026

**The Historical Barrier.** Milgrom and Roberts wrote about complementarity in 1990. Topkis formalized supermodularity in 1998. Why did it take 25+ years to build a comprehensive framework?

The barrier was *computational*. Consider what analyzing complementarity requires:

**Matrix Operations.** The complementarity matrix $C \in \mathbb{R}^{n \times n}$ for $n$ choice variables has $n^2$ entries. For our 144-component FEPSDE structure:

- Full matrix: $144^2 = 20,736$ parameters
- Symmetric matrix: $144 \times 145/2 = 10,440$ unique entries
- With block structure: Still thousands of parameters

**Equilibrium Computation.** Finding $C^*$ requires solving fixed-point equations in high-dimensional space. Traditional methods (Newton-Raphson, iterative best response) scale poorly.

**Comparative Statics.** Understanding how $C^*(\Psi)$ changes with context requires differentiating through the fixed-point mapping—computationally intensive.

**What Modern Infrastructure Provides. 1. GPU-Accelerated Linear Algebra.** Matrix operations that took hours now take seconds:

- Eigendecomposition of $C$ for stability analysis
- Matrix inversion for comparative statics
- Singular value decomposition for dimensionality reduction

**2. Automatic Differentiation.** Computing $\partial C^*/\partial \Psi$ no longer requires hand-derived analytical expressions. Autodiff libraries (JAX, PyTorch) compute gradients through arbitrary computational graphs.

**3. Fixed-Point Solvers.** Modern numerical libraries include robust fixed-point algorithms:

- Anderson acceleration for contraction mappings
- Continuation methods for tracking equilibria as $\Psi$ changes
- Homotopy methods for finding multiple equilibria

**4. Sparse and Structured Matrices.** Real complementarity matrices are not dense—most entries are near zero. Sparse matrix representations and low-rank approximations make high-dimensional analysis tractable.

**5. LLM-Assisted Implementation.** Perhaps most importantly, researchers can now describe a model in natural language and have code generated automatically:

> "Build a model with 6 FEPSDE dimensions, 3 agent types, and complementarity structure where financial and social utilities interact positively. Compute the coherence index and simulate context dynamics."

This dramatically lowers the barrier to entry for using the framework.

**Implication.** Complementarity theory was "ahead of its time" in 1990. The theoretical insights were valid, but implementation was impractical. Today, the computational infrastructure has caught up with the theory.

The next section derives $C^*$—a computation that is now routine but would have been prohibitive a decade ago. See Appendix **??** for a detailed account of how computational capabilities evolved from chalk-and-blackboard correlation analysis in the 1960s to today's LLM-assisted frameworks.

# 6 The Reference Structure: Three Derivations of $C^*$ (formal proofs in Appendix D)

The coherence index $K = 1 - \|C - C^*\| / \|C^*\|$ measures deviation from a reference complementarity structure $C^*$. For this measure to be scientifically meaningful and non-arbitrary, $C^*$ must be derived from first principles, not assumed.

This section provides three independent derivations—axiomatic, evolutionary, and empirical—that converge on the same object. The convergence establishes that $C^*$ is not a normative ideal but a structural attractor: the configuration toward which systems gravitate.

## 6.1 The Stigler-Becker Parallel

**The Classical Defense of Stable Preferences.** Stigler and Becker (1977) famously argued that economists should not explain behavior by invoking changing tastes. "De gustibus non est disputandum"—tastes are stable; only constraints change.

This methodological commitment served crucial purposes:

- It made behavior predictable (same preferences $\Rightarrow$ same response to same constraints)
- It enabled welfare analysis (stable preferences define welfare)
- It prevented ad hoc explanations ("preferences changed" explains everything, hence nothing)

**The Problem with Stigler-Becker.** As Section 3 documented, the stable preferences assumption is empirically untenable. Preferences demonstrably depend on:

- Others' outcomes (social preferences)
- Past behavior (identity, habits)
- Framing and context (reference dependence)
- Institutional environment (norms)

**Our Resolution: Stable $C^*$, Variable $C$.** We preserve the Stigler-Becker insight at a higher level of abstraction:

| | **Stigler-Becker** | **Our Framework** |
|---|---|---|
| Stable element | Preferences $U$ | Reference structure $C^*(\Psi)$ |
| Variable element | Constraints | Actual structure $C$ |
| Behavior explained by | Optimization given $U$ | Convergence toward $C^*$ |
| Anchor | Exogenous tastes | Endogenous fixed point |

The key advance: $C^*$ is not exogenously assumed but *derived* as the unique self-consistent configuration. And $C^*$ depends on context $\Psi$—so when institutions change, the reference structure changes, but it remains determinate.

**"De Complementaritate Non Est Disputandum."** Just as Stigler-Becker said "don't argue about tastes," we say "don't argue about $C^*$—derive it."

$C^*$ is not what we think the world *should* look like. It is what the world *must* look like if beliefs and behavior are mutually consistent. Any other configuration contains internal contradictions that generate pressure toward $C^*$.

## 6.2 Why $C^*$ Cannot Be Arbitrary

**The Arbitrariness Objection.** A skeptic might ask: "Why is *this* $C^*$ the reference point? Couldn't you choose any structure and measure deviation from it?"

**The Answer: Self-Consistency.** $C^*$ is not chosen—it is the unique solution to a fixed-point equation. Consider what happens with any other structure $C \neq C^*$:

1. Agents believe the complementarity structure is $C$
2. They optimize given this belief
3. Their behavior generates a *different* structure $\Phi(C)$
4. The belief $C$ contradicts the reality $\Phi(C)$
5. Agents update beliefs toward $\Phi(C)$
6. The process continues until $C = \Phi(C)$

Only $C^*$ satisfies $C^* = \Phi(C^*)$. It is the unique structure where beliefs and reality are mutually consistent.

**Example: Trust in a Trading Community.** Consider a community of merchants. The complementarity structure captures how much each merchant's cooperation depends on others' cooperation.

**Scenario 1: Beliefs match reality.**

- Merchants believe others are trustworthy ($C_{ij} > 0$: cooperation complements cooperation)
- They extend credit, honor contracts, build relationships
- This behavior confirms the belief
- $C = C^*$: Stable high-trust equilibrium

**Scenario 2: Beliefs too optimistic.**

- Merchants believe trust is higher than it is
- They extend too much credit
- Some get cheated
- Beliefs adjust downward toward reality

**Scenario 3: Beliefs too pessimistic.**

- Merchants believe trust is lower than it could be
- They demand cash payment, avoid relationships
- Opportunities for mutually beneficial trade are missed
- Some merchants experiment with trust, find it works
- Beliefs adjust upward

In all cases, beliefs gravitate toward $C^*$—the structure that reproduces itself.

## 6.3 Axiomatic Derivation: Self-Consistency

We now formalize the intuition.

**Definition 6.1** (Complementarity Structure)**.** *Let $\mathcal{A} = \{1, \ldots, n\}$ be a set of agents and $\mathcal{D} = \{1, \ldots, m\}$ a set of utility dimensions. A* complementarity structure *is a matrix $C \in \mathbb{R}^{(n\times m)\times(n\times m)}$ with elements*

$$C_{id,jd'} = \frac{\partial^2 U_i}{\partial x_{id}\, \partial x_{jd'}}$$

*representing the cross-effect of agent $j$'s action in dimension $d'$ on agent $i$'s marginal utility in dimension $d$.*

**Definition 6.2** (Behavioral Response Operator)**.** *Given a complementarity structure $C$ and context $\Psi$, let $a^*(C, \Psi)$ denote the profile of optimal actions when agents maximize utility taking $C$ as given. The* behavioral response operator $\Phi$ *maps $C$ to the complementarity structure implied by these optimal actions:*

$$\Phi(C, \Psi) = \left[ \frac{\partial^2 U_i(a^*(C, \Psi))}{\partial x_{id} \, \partial x_{jd'}} \right].$$

**Intuition.** $\Phi$ answers the question: "If everyone believes the world has structure $C$ and acts optimally, what structure does their behavior actually create?"

**Definition 6.3** (Self-Consistent Structure)**.** *A complementarity structure $C^*$ is* self-consistent *at context $\Psi$ if*

$$C^* = \Phi(C^*, \Psi). \tag{42}$$

**Intuition.** $C^*$ is a *rational expectations equilibrium* for complementarity beliefs. If everyone believes $C^*$, their behavior confirms $C^*$. No one has reason to revise their beliefs.

**Axiom 6.4** (Regularity)**.** *The utility functions $U_i$ are twice continuously differentiable, strictly concave in own actions, and satisfy $\|\partial^2 U_i / \partial x_{id} \partial x_{jd'}\| < B$ for some bound $B > 0$.*

**Meaning.** Utility functions are smooth and bounded—no infinite complementarities. Concavity in own actions ensures optimal responses are well-defined.

**Axiom 6.5** (Reciprocity)**.** *Cross-effects are symmetric in expectation:*

$$\mathbb{E}[C_{id,jd'}] = \mathbb{E}[C_{jd',id}] \quad \textit{for all } i, j, d, d'.$$

**Meaning.** On average, if my action affects your marginal utility, your action affects mine symmetrically. This rules out pure exploitation structures.

**Axiom 6.6** (Contraction)**.** *The operator $\Phi$ is a contraction in the Frobenius norm:*

$$\|\Phi(C_1, \Psi) - \Phi(C_2, \Psi)\|_F \leq \gamma \|C_1 - C_2\|_F$$

*for some $\gamma \in (0, 1)$.*

**Meaning.** Small differences in beliefs lead to even smaller differences in implied structure. The system is self-correcting: errors in beliefs dampen rather than amplify.

**When Contraction Fails.** The contraction axiom fails when complementarities are very strong— when small changes in beliefs cause large changes in behavior. This happens during:

- Financial panics (beliefs about others' beliefs spiral)
- Bank runs (withdrawal beliefs self-fulfill)
- Revolutions (coordination on rebellion)
- Hyperinflation (price expectations self-fulfill)

In these cases, the system can have multiple $C^*$ or no stable $C^*$. This is precisely when $K$ becomes undefined or the system jumps between equilibria.

**Theorem 6.7** (Existence and Uniqueness)**.** *Under Axioms 6.4–6.6, for each context $\Psi$ there exists a unique self-consistent complementarity structure $C^*(\Psi)$ satisfying (42).*

*Proof.* By Axiom 6.4, $\Phi$ maps the bounded set $\mathcal{C} = \{C : \|C\|_F \leq B\}$ into itself. By Axiom 6.6, $\Phi$ is a contraction. The Banach fixed-point theorem guarantees existence and uniqueness of $C^* \in \mathcal{C}$ satisfying $C^* = \Phi(C^*, \Psi)$. □

**Constructive Interpretation.** The Banach theorem is constructive: starting from any $C_0$, the sequence $C_{n+1} = \Phi(C_n, \Psi)$ converges to $C^*$. This describes how a society "finds" $C^*$ through adaptive learning.

## 6.4 Evolutionary Derivation: Stability under Selection

An alternative derivation (Bowles 2004) treats $C^*$ as the evolutionarily stable complementarity structure in a population of heterogeneous agents.

**Setup.** Imagine a population where each agent has beliefs $C_i$ about complementarity. Agents interact, and their success (fitness) depends on how well their beliefs match reality.

**Definition 6.8** (Fitness)**.** *Let agent $i$ have complementarity beliefs $C_i$. Fitness is defined as realized utility when interacting with a population whose average beliefs are $\bar{C}$:*

$$\pi_i(C_i, \bar{C}) = U_i(a^*(C_i), a^*_{-i}(\bar{C})).$$

**Intuition.** An agent with wrong beliefs $C_i \neq C^*$ will:

- Mispredict others' responses
- Choose suboptimal actions
- Achieve lower utility
- Be outcompeted by agents with correct beliefs

**Definition 6.9** (Evolutionarily Stable Structure)**.** *A structure $C^*$ is* evolutionarily stable *if no mutant $C' \neq C^*$ can invade a population at $C^*$:*

$$\pi(C^*, C^*) > \pi(C', C^*) \quad \textit{for all } C' \neq C^*.$$

**Proposition 6.10.** *Under Axioms 6.4–6.6, the self-consistent structure $C^*$ is evolutionarily stable.*

**Proof Sketch.** At $C^*$, beliefs match reality. Any mutant $C' \neq C^*$ has incorrect beliefs. When interacting with a population at $C^*$, the mutant mispredicts responses and achieves lower fitness. Hence $C^*$ cannot be invaded.

**Example: Cultural Evolution.** Consider norms about cooperation.

- A society at $C^*$ has stable norms: people cooperate when expected, defect when expected
- A "naive cooperator" ($C' > C^*$) gets exploited
- A "paranoid defector" ($C' < C^*$) misses cooperative gains
- Both are outcompeted by agents with accurate beliefs $C^*$

This connects to the cultural evolution literature (Boyd and Richerson 1985; Gintis 2007): $C^*$ is the culturally stable configuration of social norms.

## 6.5 Empirical Derivation: Long-Run Averages

A third approach identifies $C^*$ from observed behavior without imposing theoretical structure.

**Definition 6.11** (Empirical Reference Structure)**.** *Let $C_t$ denote the complementarity structure estimated at time $t$. The* empirical reference structure *is the long-run average:*

$$C^*_{emp} = \lim_{T\to\infty} \frac{1}{T} \sum_{t=1}^{T} C_t. \tag{43}$$

**Intuition.** If we observe a system long enough, transient deviations average out. What remains is the structural attractor—the configuration around which the system fluctuates.

**Proposition 6.12.** *If the system is ergodic and $C_t$ evolves according to adaptive dynamics with unique stationary distribution, then $C^*_{emp} = C^*$ almost surely.*

**Practical Implication.** We can estimate $C^*$ empirically:

1. Collect data on complementarities over time (behavioral elasticities, covariances)
2. Compute the long-run average
3. This estimates the theoretical $C^*$

**Example: Stock Market Correlations.**

- Daily correlations fluctuate wildly
- During crises, correlations spike ("all assets fall together")
- During calm periods, correlations return to normal
- The long-run average correlation structure is $\Sigma^*$
- Market coherence $K = 1 - \|\Sigma_t - \Sigma^*\|/\|\Sigma^*\|$

## 6.6 Convergence of the Three Approaches

The remarkable result is that all three derivations yield the same $C^*$ (proofs in Appendix D).

**Theorem 6.13** (Equivalence)**.** *Under Axioms 6.4–6.6 and ergodicity of the adaptive dynamics, the axiomatic, evolutionary, and empirical definitions of $C^*$ coincide:*

$$C^*_{axiomatic} = C^*_{evolutionary} = C^*_{empirical}.$$

**Why This Matters. 1. Scientific robustness.** Three different methodologies— mathematical, biological, statistical— converge on the same answer. This is strong evidence that $C^*$ is a real structural feature, not an artifact of our assumptions.

**2. Theoretical unity.** The theorem connects:

- Fixed-point theory (mathematics)
- Evolutionary game theory (biology)
- Ergodic theory (statistics)

**3. Empirical testability.** We can estimate $C^*_{empirical}$ from data and compare it to theoretically derived $C^*_{axiomatic}$. If they match, the theory is confirmed.

**4. Policy relevance.** Since $C^* = C^*(\Psi)$, policy that changes context $\Psi$ changes the reference structure. This is how institutional reform works: it shifts the attractor, and behavior follows.

## 6.7 What $C^*$ Is and Is Not

**$C^*$ Is:**

- A fixed point of the belief-behavior system
- The structure that reproduces itself under rational behavior
- The evolutionarily stable configuration
- The long-run average of observed structures
- Context-dependent: $C^* = C^*(\Psi)$
- Empirically estimable

**$C^*$ Is Not:**

- A normative ideal ("how things should be")
- An arbitrary benchmark
- Fixed forever (it changes when $\Psi$ changes)
- Necessarily optimal (a bad equilibrium can be stable)
- Observable directly (only its effects are observable)

**The Crucial Distinction: $K$ vs. $Q$.** $C^*$ determines where the system gravitates (coherence $K \to 1$). But $C^*$ can be a bad equilibrium (low $Q$).

A society of mutual exploitation can have $K \approx 1$ (everyone correctly believes others will exploit, and acts accordingly) but $Q$ is low (everyone is worse off than in a cooperative equilibrium).

This is why we need both:

- $K$: Are we at *a* stable configuration?
- $Q$: Is it a *good* configuration?

## 6.8 Implications for the Framework

**The Coherence Index Is Well-Defined.** With $C^*$ rigorously derived, the coherence index

$$K = 1 - \frac{\|C - C^*(\Psi)\|_F}{\|C^*(\Psi)\|_F} \tag{44}$$

is a meaningful measure of structural stability, not an arbitrary metric.

**Dynamics Are Predictable.** Under the contraction axiom, deviations from $C^*$ shrink:

$$\|C_{t+1} - C^*\| \leq \gamma \|C_t - C^*\| \tag{45}$$

The system returns to coherence after shocks.

**Context Changes Are Fundamental.** Since $C^* = C^*(\Psi)$, changing context changes the attractor. Institutional reform doesn't just change constraints— it changes what configuration the system gravitates toward.

**Multiple Equilibria Are Possible.** When the contraction axiom fails (strong complementarities), multiple $C^*$ may exist. The system can be trapped in a bad equilibrium. Policy then requires "big push" to shift between equilibria, not marginal adjustment within one.

# 7 Complementarity, Fit, and Non-Concavity

Section 6 established that $C^*$ is the structural attractor— the configuration toward which systems gravitate. But why do systems gravitate anywhere? Why can't they wander freely across configuration space?

The answer lies in *complementarity*: when elements reinforce each other, payoff functions become non-concave, creating peaks and valleys. Systems climb toward peaks and resist leaving them.

This section develops the connection to organizational economics (see Appendix A for the Milgrom-Roberts limiting case), showing how Roberts (2004)'s concept of "fit" is precisely our coherence $K$, and why the landscape metaphor is more than a metaphor.

## 7.1 Roberts' Insight: Organizations as Systems of Complements

**The Problem Roberts Addressed.** In the 1990s, organizational economics faced a puzzle: why do successful firms look so different from each other, yet each successful firm's elements fit together so well?

- **Toyota:** Lean production, worker empowerment, continuous improvement, lifetime employment
- **Goldman Sachs:** Partnership culture, high risk tolerance, aggressive incentives, up-or-out
- **Southwest Airlines:** Point-to-point routes, single aircraft type, no frills, fun culture

Each is internally coherent. Each would fail if you mixed their elements. Lean production with aggressive individual incentives? Disaster. Partnership culture with standardized processes? Contradiction.

**Roberts' Answer: Complementarity.** Roberts (2004) explained this with *complementarity*: organizational elements are supermodular complements— doing more of one increases the return to doing more of others.

$$\frac{\partial^2 \Pi}{\partial x_i \partial x_j} > 0 \quad \text{for all } i \neq j \tag{46}$$

where $x_i$ are organizational choices: strategy, structure, processes, people, culture, incentives.

**Example: Strategy and Structure.** Consider two strategic choices:

- **Innovation strategy:** Develop new products, enter new markets

- **Efficiency strategy:** Cut costs, standardize, optimize

And two structural choices:

- **Flat structure:** Few hierarchy levels, autonomous teams
- **Hierarchical structure:** Many levels, clear command chains

The cross-derivatives are positive:

$$\frac{\partial^2 \Pi}{\partial \text{Innovation}\, \partial \text{Flat}} > 0 \quad \text{(Innovation needs autonomy)} \tag{47}$$

$$\frac{\partial^2 \Pi}{\partial \text{Efficiency}\, \partial \text{Hierarchy}} > 0 \quad \text{(Efficiency needs control)} \tag{48}$$

But the cross-configurations are penalized:

$$\frac{\partial^2 \Pi}{\partial \text{Innovation}\, \partial \text{Hierarchy}} < 0 \quad \text{(Innovation stifled by bureaucracy)} \tag{49}$$

$$\frac{\partial^2 \Pi}{\partial \text{Efficiency}\, \partial \text{Flat}} < 0 \quad \text{(Efficiency lost to coordination problems)} \tag{50}$$

## 7.2 Non-Concavity: Why Complementarity Creates Peaks and Valleys

**The Mathematical Structure.** Under complementarity, the payoff function $\Pi(x_1, \ldots, x_n)$ is *supermodular*: its Hessian has positive off-diagonal elements.

This has a crucial implication: the function is *not globally concave.*

**Proposition 7.1** (Non-Concavity from Complementarity)**.** *If $\partial^2 \Pi / \partial x_i \partial x_j > 0$ for all $i \neq j$, then $\Pi$ is not globally concave. The payoff landscape has multiple local maxima.*

*Proof.* The Hessian matrix $H$ of $\Pi$ has:

- Diagonal elements $H_{ii} = \partial^2 \Pi / \partial x_i^2 < 0$ (concavity in own choice)
- Off-diagonal elements $H_{ij} = \partial^2 \Pi / \partial x_i \partial x_j > 0$ (complementarity)

For global concavity, $H$ must be negative semi-definite: all eigenvalues $\leq 0$. But with positive off-diagonal elements, the matrix has at least one positive eigenvalue. Hence $\Pi$ is not concave. □

**The Landscape Metaphor.** Think of $\Pi$ as a landscape over configuration space:

- **Peaks:** Coherent configurations where all elements fit together
- **Valleys:** Incoherent configurations where elements clash
- **Ridges:** Partial coherence—some elements fit, others don't

**Example: The Two-Peak Landscape.** With two choices (strategy $s$, structure $\sigma$), each in $[0, 1]$:

| | $\sigma = 0$ (Hierarchy) | $\sigma = 1$ (Flat) |
|---|---|---|
| $s = 0$ (Efficiency) | **Peak A**: $\Pi = \beta$ | Valley: $\Pi < 0$ |
| $s = 1$ (Innovation) | Valley: $\Pi < 0$ | **Peak B**: $\Pi = \alpha$ |

Two peaks (A and B), with a valley between them. An organization at (0.5, 0.5)—half innovative, half efficient—is in the valley.

## 7.3 Fit as Coherence

**The Roberts-Framework Connection.** Roberts (2004) defines **fit** as the degree to which organizational elements reinforce each other. This maps directly to our framework:

| **Roberts (2004)** | **Present Framework** | **Meaning** |
|---|---|---|
| Fit | $K \approx 1$ | At a local maximum |
| Misfit | $K < 1$ | In the valley between peaks |
| Configuration quality | $Q(C^*)$ | Height of the peak |
| Multiple configurations | Multiple $C^*$ | Multiple local maxima |
| Strategy-structure alignment | $s \approx \sigma$ | On the diagonal |

**Example: Southwest Airlines.** Southwest's configuration:

- Strategy: Low cost, high frequency
- Structure: Decentralized, empowered employees
- Processes: Fast turnaround, no assigned seats
- People: Hired for attitude, trained for skill
- Culture: Fun, team-oriented
- Incentives: Profit sharing, no layoffs

Every element reinforces every other. Change one, and the system breaks. This is $K \approx 1$: high coherence.

**Example: Failed Transformation at Continental.** In the 1990s, Continental Airlines tried to copy Southwest:

- Adopted low-cost strategy (changed $s$)
- Kept hierarchical structure (didn't change $\sigma$)
- Kept old processes, people, culture, incentives

Result: Chaos. The strategy required different structure, but structure didn't change. $K \ll 1$: Low coherence. Employees didn't know whether to innovate or follow rules. Continental retreated and eventually found success with a *different* coherent configuration.

## 7.4 Why Transitions Are So Difficult

**The Valley Problem.** To move from Peak A to Peak B, an organization must:

1. Leave Peak A (coherence falls)
2. Cross the valley (incoherence, pain)
3. Climb to Peak B (coherence restored)

$$\text{Transition path: } C_A^* \xrightarrow{K\downarrow} \text{Valley} \xrightarrow{K\uparrow} C_B^* \tag{51}$$

**Why Partial Change Fails.** The complementarity structure means you can't change elements one at a time. If you change strategy but not structure, you move *sideways*—not toward the new peak, but into the valley.

| **Change** | $\Delta s$ | $\Delta\sigma$ | Direction | Result |
|---|---|---|---|---|
| Full transformation | +0.8 | +0.8 | Diagonal | Reach Peak B |
| Strategy only | +0.8 | 0 | Horizontal | Stuck in valley |
| Structure only | 0 | +0.8 | Vertical | Stuck in valley |
| Gradual (sequential) | +0.8, then +0.8 | | Two valleys | Worse than staying |

**Example: Digital Transformation.** A traditional manufacturing firm wants to become a digital innovator.

**The wrong way (sequential):**

1. Year 1: Announce digital strategy
2. Year 2: Start changing processes
3. Year 3: Realize culture doesn't fit

4. Year 4: Hire new people
5. Year 5: Change incentives
6. Years 6-10: Still in the valley, key people have left

**The right way (simultaneous):**

1. Year 0: Plan complete transformation
2. Year 1: Change strategy, structure, processes, hiring, culture, incentives *together*
3. Years 2-3: Accept low $K$, support the transition
4. Year 4: New coherent configuration emerges

The second path is harder in the short run but reaches the new peak.

## 7.5 Multiple Peaks: Which Configuration Is Best?

**K vs. Q Again.** Both Peak A (Efficiency) and Peak B (Innovation) have $K \approx 1$. Both are coherent. But they differ in $Q$—the height of the peak.

Which peak is higher depends on context $\Psi$:

- Stable industry, commodity product: Peak A (Efficiency) is higher
- Turbulent industry, differentiated product: Peak B (Innovation) is higher

**Example: Kodak's Coherent Decline.** Kodak in 2000 was highly coherent ($K \approx 1$):

- Strategy: Dominate film photography
- Structure: Vertically integrated
- Processes: Optimized for chemical film
- Culture: Engineering excellence in film

But context changed: digital photography emerged. Kodak's peak (film excellence) became a valley in the new landscape. High $K$, but $Q$ collapsed because $\Psi$ shifted.

**The Trap of Coherent Obsolescence.** A firm can be perfectly coherent and still fail— if it's coherent in a configuration that the market no longer rewards.

This is the $K$ vs. $Q$ distinction in action:

- $K$ measures fit with self: Are our elements aligned?
- $Q$ measures fit with world: Is our configuration valued?

## 7.6 From Organizations to All Levels

**Roberts' Insight Generalizes.** The complementarity-fit-non-concavity logic applies beyond firms:

| Level | Elements | Fit Means | Example |
|---|---|---|---|
| Individual | Values, goals, behavior, habits | Integrity | Live your values |
| Household | Roles, expectations, resources | Family harmony | Shared parenting |
| Organization | Strategy, structure, culture | Competitive fit | Southwest |
| Region | Economy, education, infrastructure | Economic coherence | Silicon Valley |
| Nation | Institutions, norms, policies | Institutional fit | Nordic model |
| Global | Trade, climate, security regimes | International order | Bretton Woods |

At every level, complementarity creates peaks and valleys. At every level, partial change lands you in the valley. At every level, $K$ and $Q$ are distinct.

## 7.7 Implications for Strategy

**Diagnostic Questions.** Before any change, ask:

1. Where are we? (Which peak, or are we in a valley?)
2. Where do we want to be? (Which peak is best given $\Psi$?)
3. How deep is the valley? (Transition cost)
4. Can we change everything together? (Transformation capacity)

**Strategic Options.**

- **Optimize:** Stay on current peak, improve $Q$ marginally
- **Transform:** Full-stack change to different peak
- **Hybrid:** Rarely works—usually lands in valley
- **Exit:** If valley too deep and transformation impossible

**The Key Insight.** Complementarity is both blessing and curse:

- Blessing: Coherent configurations are stable (hard to dislodge)
- Curse: Coherent configurations are traps (hard to leave)

Understanding this structure is the first step toward navigating it.

# 8 Mathematical Formalization

The preceding sections developed intuition through examples. This section provides the mathematical machinery (for standard microeconomic foundations, see Mas-Colell et al. 1995) that makes the framework precise, testable, and applicable. We build up from simple two-dimensional models to the full 144-component structure, always keeping the intuition visible.

## 8.1 The Core Insight: Why Two Dimensions?

**The Roberts Reduction.** Roberts (2004) showed that organizational choices—however numerous— cluster along two fundamental dimensions:

- **What we want** (goals, strategy, aspirations)
- **How we organize** (structure, processes, constraints)

This insight generalizes beyond organizations. At every level, agents face a version of the same choice:

| **Level** | **"What we want"** | **"How we organize"** |
|---|---|---|
| Individual | Life goals, values | Lifestyle, habits, routines |
| Household | Family priorities | Role allocation, rules |
| Organization | Business strategy | Structure, processes |
| Region | Economic positioning | Institutions, infrastructure |
| Nation | Societal model | State organization, laws |
| Global | World order vision | Governance architecture |

**The Two Poles.** Each dimension spans between two poles:

- **Strategy axis:** Stability/security ($s = 0$) vs. Dynamism/growth ($s = 1$)
- **Structure axis:** Hierarchy/control ($\sigma = 0$) vs. Flexibility/autonomy ($\sigma = 1$)

**Example: National Models.**

- Germany ($s \approx 0.3$, $\sigma \approx 0.3$): Stability-oriented, rule-based, coordinated
- USA ($s \approx 0.7$, $\sigma \approx 0.7$): Growth-oriented, flexible, market-based
- Both coherent ($s \approx \sigma$), different configurations

## 8.2 The Two-Axis Model: Formal Definition

**Definition 8.1** (Configuration Space)**.** *For each level $\ell \in \mathcal{L} = \{Ind, HH, Org, Reg, Nat, Glob\}$, the configuration space is the unit square $\mathcal{S} = [0,1] \times [0,1]$ with coordinates $(s, \sigma)$ representing strategy and structure.*

**Definition 8.2** (Strategy and Structure Axes)**.**

$$s \in [0,1] \quad \textit{(Strategy: stability} \leftrightarrow \textit{dynamism)} \tag{52}$$

$$\sigma \in [0,1] \quad \textit{(Structure: hierarchy} \leftrightarrow \textit{flexibility)} \tag{53}$$

**Interpretation.** A point $(s, \sigma) = (0.7, 0.3)$ represents: strategy leaning toward dynamism (70%) but structure leaning toward hierarchy (30%). This is a *misfit*: innovative goals with rigid structure.

## 8.3 The Payoff Function: Formalizing Complementarity

The payoff function captures three forces:

1. **Misfit penalty:** Deviation between strategy and structure is costly
2. **Innovation synergy:** High $s$ and high $\sigma$ together create value
3. **Stability synergy:** Low $s$ and low $\sigma$ together create value

**Definition 8.3** (Payoff Function)**.**

$$\Pi(s, \sigma; \Psi) = \underbrace{-\gamma(s-\sigma)^2}_{\textit{Misfit penalty}} + \underbrace{\alpha(\Psi) \cdot s \cdot \sigma}_{\textit{Innovation synergy}} + \underbrace{\beta(\Psi) \cdot (1-s) \cdot (1-\sigma)}_{\textit{Stability synergy}} \tag{54}$$

*where $\gamma > 0$ measures misalignment cost, $\alpha(\Psi) > 0$ the return to innovation configuration, and $\beta(\Psi) > 0$ the return to stability configuration.*

**Why This Functional Form?** The misfit term $-\gamma(s-\sigma)^2$ is zero when $s = \sigma$ and increasingly negative as $|s - \sigma|$ grows. The innovation term $\alpha \cdot s \cdot \sigma$ is highest at $(1,1)$ and captures complementarity: high $s$ needs high $\sigma$ to pay off. The stability term $\beta \cdot (1-s) \cdot (1-\sigma)$ is highest at $(0,0)$ with similar logic.

**Example: Evaluating Configurations.** With $\gamma = 1$, $\alpha = 2$, $\beta = 2$:

| Configuration | $(s, \sigma)$ | $\Pi$ | Interpretation |
|---|---|---|---|
| Stability Peak (A) | $(0, 0)$ | $0 + 0 + 2 = 2$ | Coherent, stable |
| Innovation Peak (B) | $(1, 1)$ | $0 + 2 + 0 = 2$ | Coherent, innovative |
| Valley (M) | $(0.5, 0.5)$ | $0 + 0.5 + 0.5 = 1$ | Coherent but low synergy |
| Misfit 1 | $(1, 0)$ | $-1 + 0 + 0 = -1$ | Innovation + hierarchy |
| Misfit 2 | $(0, 1)$ | $-1 + 0 + 0 = -1$ | Stability + flexibility |

## 8.4 Equilibria and Their Properties

**Proposition 8.4** (Equilibrium Configurations)**.** *The payoff function (54) has two local maxima at $A = (0, 0)$ with $\Pi_A = \beta$ and $B = (1, 1)$ with $\Pi_B = \alpha$, one saddle point at $M = (0.5, 0.5)$, and two local minima at $(0, 1)$ and $(1, 0)$ with $\Pi = -\gamma$.*

**Which Peak Is Higher?** $\Pi_A = \beta$ vs. $\Pi_B = \alpha$ depends on context $\Psi$: if $\alpha(\Psi) > \beta(\Psi)$, the innovation peak B is higher; if $\beta(\Psi) > \alpha(\Psi)$, the stability peak A is higher.

## 8.5 The Coherence Index K

**Definition 8.5** (Coherence Index)**.**

$$K(s, \sigma) = 1 - \frac{\min\{d((s, \sigma), A), d((s, \sigma), B)\}}{d_{max}} \tag{55}$$

*where $d$ is Euclidean distance and $d_{max} = 1/\sqrt{2}$.*

$K = 1$ at peaks A and B (perfect coherence); $K \to 0$ at the point equidistant from both peaks.

## 8.6 The Quality Index Q

**Definition 8.6** (Quality Index)**.** $Q(C^*) = \Pi(C^*; \Psi)$ *where $C^*$ is the equilibrium configuration.*

Realized welfare depends on both: $W = Q(C^*) \cdot g(K)$ where $g(1) = 1$, $g(0) = 0$.

## 8.7 The 144-Component Structure

The two-axis model captures essentials but lacks richness. Real decisions involve multiple utility types, dimensions, and time horizons.

**Definition 8.7** (Utility Components)**.** *The full structure has 144 components $U = \{u_{i,v,d,t}\}$ indexed by:*

- $i \in \{INU, KNU, IDN\}$*: Utility category (Individual, Collective, Identity) — 3*

- $v \in \{G, P\}$: *Valence (Gain or Pain) — 2*
- $d \in \{F, E, P, S, D, Eco\}$: *FEPSDE dimension — 6*
- $t \in \{0, 1, 2, 3\}$: *Time horizon (instant, short, medium, long) — 4*

*Total:* $3 \times 2 \times 6 \times 4 = 144$ *components.*

**Definition 8.8** (Aggregate Welfare Function)**.**

$$W = \sum_{i \in \{INU, KNU, IDN\}} \sum_{d \in FEPSDE} \sum_{t=0}^{3} \delta_d(\Psi)^t \left[G_{i,d,t} - \lambda_d(\Psi) \cdot P_{i,d,t}\right] \tag{56}$$

*with discount factor* $\delta_d(\Psi) \in (0, 1)$ *and loss aversion* $\lambda_d(\Psi) > 1$.

**Key Features.** Discounting weights future less than present. Loss aversion ($\lambda > 1$) weights pains more than gains, following Prospect Theory. Dimension-specific parameters allow health to be discounted differently than money.

## 8.8 The Complementarity Matrix

With 144 components, the full complementarity matrix is $C \in \mathbb{R}^{144 \times 144}$ with 20,736 entries (10,440 unique by symmetry).

The matrix has block structure reflecting interactions between utility categories:

$$C = \begin{pmatrix} C_{INU,INU} & C_{INU,KNU} & C_{INU,IDN} \\ C_{KNU,INU} & C_{KNU,KNU} & C_{KNU,IDN} \\ C_{IDN,INU} & C_{IDN,KNU} & C_{IDN,IDN} \end{pmatrix} \tag{57}$$

## 8.9 Computational Tractability: Why This Is Now Feasible

**The Apparent Problem.** A $144 \times 144$ matrix with 10,440 unique parameters seems intractable:

- How do we estimate so many parameters?
- How do we compute equilibria?
- How do we perform comparative statics?

This concern explains why comprehensive complementarity frameworks were not developed earlier—the computational burden seemed prohibitive.

**The Solution: Structure and Technology.** **1. Exploiting Block Structure.** The matrix is not arbitrary. It has hierarchical block structure:

- Within-category blocks ($C_{INU,INU}$, etc.): Dense, strong interactions
- Cross-category blocks ($C_{INU,KNU}$, etc.): Sparser, weaker interactions
- Within-dimension blocks: Strongest complementarities

This structure reduces effective dimensionality dramatically. A sparse representation with ∼500 non-negligible entries captures most variation.

**2. Low-Rank Approximation.** Empirically, complementarity matrices are approximately low-rank:

$$C \approx U \Sigma V^T \tag{58}$$

where $U, V \in \mathbb{R}^{144 \times k}$ and $k \ll 144$.

With $k = 10 - 20$ factors, we capture 90%+ of the variance. This reduces estimation from 10,440 parameters to ∼3,000.

**3. Hierarchical Bayesian Estimation.** Parameters are not estimated independently but hierarchically:

$$C_{ij} \sim \mathcal{N}(\mu_{block(i,j)}, \sigma^2_{block}) \tag{59}$$

$$\mu_{block} \sim \mathcal{N}(\mu_{global}, \tau^2) \tag{60}$$

Information is pooled across similar entries, regularizing estimates and enabling learning from limited data.

**4. Modern Computational Infrastructure.**

- **GPU acceleration:** Matrix operations parallelize efficiently
- **Automatic differentiation:** Gradients computed without hand derivation
- **Probabilistic programming:** Bayesian inference automated (Stan, PyMC, NumPyro)
- **Cloud computing:** Massive parallel estimation feasible

**5. LLM-Assisted Implementation.** Researchers can specify models in natural language:

> "Estimate a hierarchical complementarity model with block structure for INU/KNU/IDN categories, sparse cross-category effects, and context-dependent parameters varying by institutional quality."

The code is generated automatically, errors are debugged interactively, and results are explained in accessible language.

**Concrete Example: Computing $K$.** Suppose we have estimated $\hat{C}$ from data and derived $C^*(\Psi)$ from theory. Computing the coherence index:

$$K = 1 - \frac{\|\hat{C} - C^*\|_F}{\|C^*\|_F} \tag{61}$$

**Before (2010):**

- Derive $C^*$ analytically (intractable for complex specifications)
- Implement Frobenius norm calculation (error-prone)
- Debug numerical issues (weeks of work)
- Document and validate (more weeks)

**Now (2026):**

```
K = 1 - np.linalg.norm(C_hat - C_star, 'fro') / np.linalg.norm(C_star, 'fro')
```

Or simply: "Calculate the coherence index for this estimated complementarity matrix."

**The Transformation.** What would have been a multi-year computational project in 2010 is now an afternoon's work in 2026. This is not a minor convenience—it is what makes the framework *usable*.

The historical evolution from manual matrix computation to AI-assisted analysis is documented in Appendix **??**, tracing the 60-year journey that made this framework operationally feasible.

## 8.10 Summary: The Mathematical Architecture

| **Object** | **Definition** | **Meaning** |
|---|---|---|
| $(s, \sigma)$ | Configuration in $[0,1]^2$ | What we want + how we organize |
| $\Pi(s, \sigma; \Psi)$ | Payoff function | Value of configuration in context |
| $C^*$ | Fixed point of $\Phi$ | Self-consistent structure |
| $K$ | Distance to nearest peak | Coherence/stability |
| $Q$ | Payoff at equilibrium | Quality of configuration |
| $W$ | 144-component aggregation | Full welfare |
| $\Psi$ | Context variable | Technology, institutions, norms |

# 9 Context as an Endogenous Variable

The preceding sections treated context $\Psi$ as a parameter that shapes the reference structure $C^*(\Psi)$ and the payoff landscape $\Pi(s, \sigma; \Psi)$. But where does context come from? How does it change?

This section develops the crucial insight that context is not exogenous but *endogenous*—shaped by the very behavior it influences. This feedback loop is what creates path dependence (Arthur 1994), institutional persistence, and the possibility of regime change.

## 9.1 What Is Context?

**The Components of $\Psi$.** Context encompasses everything that shapes the payoff structure but is not under any single agent's immediate control:

| Component | Examples | Time Scale |
|---|---|---|
| Technology | Production methods, digital infrastructure | Years to decades |
| Institutions | Laws, property rights, constitutions | Decades to centuries |
| Norms | Social expectations, fairness standards | Years to generations |
| Culture | Values, worldviews, identity categories | Generations |
| Physical environment | Climate, geography, infrastructure | Decades to centuries |

**Why Classical Economics Treats $\Psi$ as Exogenous.** Arrow-Debreu and most microeconomic models assume:

$$\Psi_{t+1} = \Psi_t = \bar{\Psi} \tag{62}$$

This simplification enables equilibrium analysis but rules out:

- Institutional change
- Technological progress
- Cultural evolution
- Path dependence

**Our Approach: Endogenous Context.** We model context as evolving through aggregate behavior:

$$\Psi_{t+1} = f(\Psi_t, a_t) \tag{63}$$

Actions today shape the context tomorrow. This closes the feedback loop that makes economic systems truly dynamic.

## 9.2 The Feedback Loop

**The Circular Causation.**

$$\Psi_t \text{ (Context)}$$
$$\downarrow$$
$$C^*(\Psi_t) \text{ (Reference Structure)}$$
$$\downarrow$$
$$a_t^*(C^*) \text{ (Optimal Actions)}$$
$$\downarrow$$
$$\Psi_{t+1} = f(\Psi_t, a_t) \text{ (New Context)}$$
$$\downarrow \text{ cycle repeats¿cycle repeats}$$

**Example: Trust and Institutions.**

1. High trust ($\Psi_t$: trust = high) $\Rightarrow$
2. Cooperation pays ($C^*$: $C_{ij} > 0$) $\Rightarrow$
3. People cooperate ($a_t$: cooperative behavior) $\Rightarrow$
4. Trust reinforced ($\Psi_{t+1}$: trust = high)

But also:

1. Low trust ($\Psi_t$: trust = low) $\Rightarrow$
2. Defection pays ($C^*$: $C_{ij} < 0$) $\Rightarrow$
3. People defect ($a_t$: defensive behavior) $\Rightarrow$
4. Trust eroded ($\Psi_{t+1}$: trust = lower)

Both are stable equilibria. Which one emerges depends on history.

## 9.3 A Simple Model of Context Dynamics

**Linear Approximation.** Near equilibrium, context dynamics can be approximated as:

$$\Psi_{t+1} = \Psi_t + \eta \cdot (a_t - a^*(\Psi_t)) \tag{64}$$

where:

- $\eta > 0$: Adjustment speed (how fast context responds to behavior)
- $a_t$: Actual aggregate behavior at time $t$
- $a^*(\Psi_t)$: Equilibrium behavior given context $\Psi_t$

**Interpretation.** If people behave *as expected* ($a_t = a^*$), context doesn't change. If people behave *differently* ($a_t \neq a^*$), context shifts.

**Example: Norm Erosion.** Consider a norm against littering.

- If everyone follows the norm ($a_t = a^*$): Norm persists
- If some people litter ($a_t > a^*$): Norm weakens
- Weaker norm $\Rightarrow$ more littering acceptable $\Rightarrow$ more littering
- Feedback can lead to norm collapse

## 9.4 Different Types of Context: Speed of Change

Not all context changes at the same speed.

**Fast-Moving Context: Technology.**

- Can change in years (smartphones: 2007-2015)
- Driven by innovation, investment, adoption
- Individuals can influence through entrepreneurship
- $\eta_{tech}$ is large

**Medium-Moving Context: Norms.**

- Changes over years to decades (smoking norms: 1970-2000)
- Driven by social learning, signaling, coordination
- Requires critical mass to shift
- $\eta_{norm}$ is medium

**Slow-Moving Context: Institutions.**

- Changes over decades to centuries (property rights, democracy)
- Driven by political conflict, crises, external pressure
- Highly persistent—"sticky"
- $\eta_{inst}$ is small

**Very Slow Context: Culture.**

- Changes over generations (religious values, family structures)
- Driven by socialization, education, demographic change
- Extremely persistent
- $\eta_{culture}$ is very small

**Implication: Layered Dynamics.**

$$\Psi = (\Psi_{tech}, \Psi_{norm}, \Psi_{inst}, \Psi_{culture}) \tag{65}$$

Each component has its own dynamics:

$$\Psi_{tech,t+1} = \Psi_{tech,t} + \eta_{tech} \cdot \Delta a_t \tag{66}$$
$$\Psi_{norm,t+1} = \Psi_{norm,t} + \eta_{norm} \cdot \Delta a_t \tag{67}$$
$$\Psi_{inst,t+1} = \Psi_{inst,t} + \eta_{inst} \cdot \Delta a_t \tag{68}$$
$$\Psi_{culture,t+1} = \Psi_{culture,t} + \eta_{culture} \cdot \Delta a_t \tag{69}$$

Technology moves fast; culture moves slow. This creates the phenomenon of "cultural lag" (Ogburn 1922)—technology changes faster than norms can adapt.

## 9.5 Self-Reinforcing and Self-Correcting Dynamics

**Self-Reinforcing (Positive Feedback).** Some context changes amplify themselves:

$$\frac{\partial^2 \Psi_{t+1}}{\partial \Psi_t \partial a_t} > 0 \tag{70}$$

**Example: Network Effects.**

- More people use a platform ⇒ platform becomes more valuable
- More valuable platform ⇒ more people join
- Winner-take-all dynamics

**Example: Institutional Decay.**

- Corruption increases ⇒ honest people exit public service
- Lower quality officials ⇒ more corruption
- Downward spiral

**Self-Correcting (Negative Feedback).** Some context changes dampen themselves:

$$\frac{\partial^2 \Psi_{t+1}}{\partial \Psi_t \partial a_t} < 0 \tag{71}$$

**Example: Price Mechanisms.**

- High demand $\Rightarrow$ high prices
- High prices $\Rightarrow$ reduced demand
- Equilibrating dynamics

**Example: Social Backlash.**

- Norm violation increases $\Rightarrow$ moral outrage increases
- Moral outrage $\Rightarrow$ punishment of violators
- Norm restored

**Which Dominates?** The stability of context depends on whether self-reinforcing or self-correcting dynamics dominate:

- Self-correcting dominant: Single stable equilibrium
- Self-reinforcing dominant: Multiple equilibria, path dependence
- Balance: Tipping points, regime shifts

## 9.6 Tipping Points and Regime Change

**When Linear Dynamics Fail.** The linear model (64) assumes smooth adjustment. But real context change often exhibits:

- **Thresholds:** Nothing happens until a critical point, then sudden change
- **Hysteresis:** The path back is different from the path forward
- **Irreversibility:** Some changes cannot be undone

**A Nonlinear Model.**

$$\Psi_{t+1} = \Psi_t + \eta \cdot g(\Psi_t) \cdot (a_t - a^*(\Psi_t)) \tag{72}$$

where $g(\Psi)$ varies with context:

- $g(\Psi) \approx 0$ when context is "locked in"
- $g(\Psi)$ large when context is "fluid"

**Example: Revolution.**

- Normal times: $g(\Psi) \approx 0$, institutions resist change
- Crisis: $g(\Psi)$ suddenly large, small actions have big effects
- New equilibrium: $g(\Psi) \approx 0$ again, new institutions lock in

**The S-Curve of Norm Change.** Many norm changes follow an S-curve:

$$\Psi_{t+1} = \Psi_t + \eta \cdot \Psi_t(1 - \Psi_t) \cdot \Delta a_t \tag{73}$$

- Early stage ($\Psi \approx 0$): Change is slow (few adopters)
- Middle stage ($\Psi \approx 0.5$): Change is fast (critical mass)
- Late stage ($\Psi \approx 1$): Change slows (saturation)

## 9.7 Historical Examples

**Example 1: The Fall of Communism (1989).**

- For decades: $g(\Psi) \approx 0$, system appeared stable
- Gorbachev reforms: $g(\Psi)$ increased, system became fluid
- Small protests in Leipzig $\Rightarrow$ massive cascade
- Within months: Complete regime change across Eastern Europe
- New equilibrium: $g(\Psi) \approx 0$, but at different $\Psi$

**Framework interpretation:** The old regime was a coherent but fragile equilibrium ($K \approx 1$ but $g(\Psi)$ latently high). A small perturbation triggered a cascade to a new equilibrium.

**Example 2: The Smartphone Revolution (2007-2015).**

- 2006: Mobile phones are for calls and SMS
- 2007: iPhone launches—$\Psi_{tech}$ begins shifting
- 2010: Critical mass reached—app ecosystem explodes
- 2015: Smartphones ubiquitous—new equilibrium

**Framework interpretation:** Technology ($\eta_{tech}$ high) moved fast. Norms ($\eta_{norm}$ medium) followed: always-on connectivity became expected. Institutions ($\eta_{inst}$ low) lagged: privacy law still catching up.

**Example 3: The Decline of Smoking (1965-2020).**

- 1965: 42% of US adults smoke; smoking is normal
- 1970s-80s: Scientific evidence accumulates; norms begin shifting
- 1990s: Critical mass—smoking becomes stigmatized
- 2020: 14% smoke; smoking is deviant

**Framework interpretation:** Norm change followed S-curve over 55 years. Institutions (smoking bans, taxes) followed and reinforced norm change. Self-reinforcing: as fewer smoke, smoking becomes more stigmatized.

**Example 4: Climate Change (Ongoing).**

- Scientific consensus since 1990s
- Norms shifting slowly (awareness increasing)
- Institutions lagging (Paris Agreement weak)
- Technology accelerating (renewables cost decline)

**Framework interpretation:** Different context components moving at different speeds. Technology may force institutional change (if renewables become cheaper than fossil fuels, the political economy shifts). Possible tipping point ahead.

## 9.8 Why Context Endogeneity Matters

**For Theory.** Endogenous context explains phenomena that exogenous models cannot:

- **Path dependence:** History matters because past actions shaped current $\Psi$
- **Multiple equilibria:** Same fundamentals, different outcomes depending on $\Psi$
- **Institutional persistence:** Low $\eta_{inst}$ explains why bad institutions persist
- **Sudden change:** Tipping points explain revolutions and crises

**For Policy.** Policy can target context directly, not just behavior:

- **Nudges:** Change the choice architecture ($\Psi_{norm}$)
- **Constitutional reform:** Change the rules ($\Psi_{inst}$)
- **R&D investment:** Change technology ($\Psi_{tech}$)
- **Education:** Change culture ($\Psi_{culture}$)

**For Strategy.** Organizations and individuals can:

- **Anticipate context change:** Position for where $\Psi$ is going, not where it is
- **Shape context:** Large actors can influence $\Psi$ (platform companies, governments)
- **Exploit tipping points** (Schelling 1978): Small actions at critical moments have large effects

## 9.9 The Complete Dynamic System

Combining context dynamics with coherence dynamics:

$$C_{t+1} = \Phi(C_t, \Psi_t) \quad \text{(Behavior adapts to context)} \tag{74}$$

$$\Psi_{t+1} = f(\Psi_t, a_t(C_t)) \quad \text{(Context adapts to behavior)} \tag{75}$$

$$K_t = 1 - \|C_t - C^*(\Psi_t)\| / \|C^*(\Psi_t)\| \quad \text{(Coherence measure)} \tag{76}$$

$$Q_t = \Pi(C_t; \Psi_t) \quad \text{(Quality measure)} \tag{77}$$

**Steady State.** At steady state:

- $C = C^*(\Psi)$: Behavior matches beliefs
- $\Psi_{t+1} = \Psi_t$: Context is stable
- $K = 1$: Full coherence
- $Q = Q(C^*; \Psi)$: Quality determined by which equilibrium

**Transitions.** During transitions:

- $C \neq C^*$: Behavior and beliefs diverge
- $\Psi$ changing: Context in flux
- $K < 1$: Incoherence
- $Q$ may rise or fall depending on direction

## 9.10 Theoretical Foundations: Connecting to Established Literature

The endogeneity of context is not our invention—it is implicit in several major research programs. Our contribution is to integrate these insights into a unified framework.

**Aghion-Howitt: Technology as Endogenous Context.** Aghion and Howitt (1998) modeled technological progress as the outcome of purposeful R&D investment. In our framework:

$$\Psi_{tech,t+1} = \Psi_{tech,t} + \phi(R\&D_t) \cdot \epsilon_t \tag{78}$$

where $\phi(\cdot)$ is the innovation production function and $\epsilon_t$ captures stochastic breakthrough.

**Key insight from Aghion-Howitt:** Innovation causes "creative destruction"— new technology renders old configurations obsolete. In our terms: $\Delta\Psi_{tech} > 0$ shifts the optimal $C^*$, forcing systems to transform or decline.

**Acemoglu-Robinson: Institutions as Endogenous Context.** For detailed paper-by-paper analysis, see Appendix **??**. Acemoglu and Robinson (2012) showed that institutions evolve through political conflict between elites and citizens. In our framework:

$$\Psi_{inst,t+1} = \Psi_{inst,t} + g(power_t, conflict_t, crisis_t) \tag{79}$$

**Key insight from Acemoglu-Robinson:** Institutions are persistent because they create constituencies that defend them. In our terms: $\eta_{inst}$ is small because institutions shape the distribution of power that determines institutional change.

**Critical junctures:** Moments when $g(\cdot)$ becomes large— wars, revolutions, economic crises—allow rapid institutional change. Between critical junctures, institutions are nearly fixed.

**North: Institutional Change and Path Dependence.** North (1990) emphasized that institutions evolve incrementally and exhibit strong path dependence. In our framework:

$$\Psi_{inst,t+1} = (1-\delta) \cdot \Psi_{inst,t} + \delta \cdot \Psi_{inst}^{target}(a_t) \tag{80}$$

where $\delta \ll 1$ captures the slow adjustment speed.

**Key insight from North:** History matters because past institutions constrain current choices, which shape future institutions. Small initial differences can lead to vastly different long-run outcomes.

**Boyd-Richerson: Culture as Endogenous Context.** Boyd and Richerson (1985) modeled cultural evolution through social learning, biased transmission, and group selection. In our framework:

$$\Psi_{culture,t+1} = \sum_j w_j(success_j, prestige_j) \cdot \Psi_{culture,j,t} \tag{81}$$

where agents copy cultural traits from successful or prestigious models.

**Key insight from Boyd-Richerson:** Culture is inherited but not genetic. It evolves

faster than genes but slower than individual learning. Cultural group selection can maintain cooperation even when individually costly.

**Synthesis.** Our framework integrates these insights:

| Theory | Context Type | Mechanism | Speed |
|---|---|---|---|
| Aghion-Howitt | Technology | R&D, innovation | Fast |
| Social learning lit. | Norms | Imitation, signaling | Medium |
| Acemoglu-Robinson | Institutions | Political conflict | Slow |
| Boyd-Richerson | Culture | Biased transmission | Very slow |

## 9.11 Spatial Context: Geography as Endogenous

Physical location constitutes a fundamental dimension of context ($\Psi_P$). The New Economic Geography (Krugman 1991) showed how spatial equilibria emerge from the interaction of increasing returns, transport costs, and factor mobility.

**Agglomeration as Endogenous Context.** Marshall (1890) identified three sources of agglomeration benefits:

1. **Labor pooling:** Thick markets reduce matching frictions
2. **Input sharing:** Specialized suppliers viable at scale
3. **Knowledge spillovers:** Ideas flow locally—"the mysteries of the trade"

In our notation, these are spatial complementarities:

$$C_{ij}^{spatial} > 0 \quad \text{(co-location is complementary)} \tag{82}$$

**Core-Periphery Dynamics.** Krugman (1991) showed that spatial equilibria exhibit:

- Multiple equilibria: Both dispersed and agglomerated configurations are stable
- Path dependence: Small initial advantages can lock in spatial structure
- Discontinuous transitions: Gradual parameter change can trigger sudden concentration

In our framework:

$$C^*_{spatial} \in \{C^*_{dispersed}, C^*_{agglomerated}\} \tag{83}$$

The selected equilibrium depends on history and the sequence of $\Psi$ changes.

**Urban Wage Premium.** Glaeser (2008); Glaeser et al. (2009) document that urban workers earn 25–35% more than rural workers with similar characteristics. Part is selection (skilled workers move to cities), but substantial productivity gains from agglomeration remain after controlling for sorting.

Rosenthal (2004); Combes and Gobillon (2015) estimate agglomeration elasticities:

$$\frac{\partial \ln(\text{productivity})}{\partial \ln(\text{density})} \approx 0.03\text{–}0.08 \tag{84}$$

Doubling density raises productivity 3–8%.

**Spatial Path Dependence.** Davis and Holt (1993) studied Japanese cities after WWII bombing. Despite massive destruction, cities returned to pre-war size within 15–20 years. Geographic fundamentals determine city location; history selects among multiple equilibria.

Bleakley (2012) found that historical portage sites (where goods were carried around waterfalls) became cities that persist long after portage became obsolete. Spatial $C^*$ exhibits strong path dependence:

$$C^*_{spatial,t} = f(C^*_{spatial,t-1}, \Psi_P^{historical}) \tag{85}$$

**Spatial Misallocation.** Hsieh and Moretti (2019) estimate that housing constraints in productive cities (San Francisco, New York) reduce US GDP by 36%. Workers cannot move to where they would be most productive because housing supply is restricted.

This is spatial misallocation:

$$Q^{actual} < Q^{optimal} \quad \text{due to spatial frictions} \tag{86}$$

Policy implication: place-based policies (Kline and Moretti 2014) can be efficient when multiple spatial equilibria exist and agglomeration externalities are not internalized.

## 9.12 Complexity Economics: Emergence and Non-Equilibrium

Standard economics assumes equilibrium exists, is unique, and is reached quickly. Complexity economics (Arthur 2015) challenges all three assumptions.

**Economy as Complex Adaptive System.** The Santa Fe approach (Arthur 1997) characterizes the economy as:

- **Heterogeneous agents:** Not identical, not representative
- **Local interaction:** Agents interact with neighbors, not all
- **Bounded rationality:** $\Psi_C < \infty$

- **Adaptation:** Agents learn, strategies evolve
- **Emergence:** Macro patterns not derivable from micro rules

**Increasing Returns and Lock-In.** Arthur (1989, 1994) showed that with increasing returns, small early events can have large long-run consequences:

$$C^*_{selected} \in \{C^*_1, C^*_2, \ldots\} \quad \text{history selects which} \tag{87}$$

Multiple equilibria exist; path determines which one is reached. The selected equilibrium may be inefficient—lock-in to inferior technology or institution.

**Emergence and Segregation.** Schelling (1971, 1978) demonstrated that macro patterns can emerge from micro behavior in unexpected ways. His segregation model shows that mild individual preferences for similar neighbors (e.g., wanting 30% similar) produce complete segregation—an outcome far exceeding individual preferences:

$$C^*_{macro} \neq \sum_i C^*_i \quad \text{(emergent properties)} \tag{88}$$

**Self-Organized Criticality.** Bak (1996) showed that complex systems naturally evolve to critical states where small perturbations can trigger avalanches of any size:

$$P(\text{event size } s) \sim s^{-\alpha} \quad \text{(power law)} \tag{89}$$

This explains "fat tails" in economic outcomes—the frequency of large crises, crashes, and disruptions that standard models underpredict.

**Framework Integration.** Complexity economics provides the dynamic foundation for our framework:

- $C^*$ is not calculated but *emerges* from agent interaction
- Multiple $C^*$ may exist; history and $\Psi$ select among them
- Heterogeneity in $\Psi$ across agents is fundamental, not a nuisance
- Transitions between $C^*$ may be discontinuous (tipping points)
- Fat-tailed outcomes require non-normal $Q$ distributions

**Implications for the Framework.** Complexity perspective enriches our analysis:

1. **Equilibrium as attractor:** $C^*$ is not calculated by a social planner but emerges from decentralized interaction

2. **Coherence as emergent:** $K \approx 1$ emerges when agents' adaptive behavior aligns, not from explicit coordination
3. **Context as enabling:** $\Psi$ shapes the "fitness landscape" on which agents adapt
4. **Transitions as phase changes:** Moving from one $C^*$ to another resembles phase transitions in physics

## 9.13 Formal Models for Each Context Type

### 9.13.1 Technology Dynamics

**The Innovation-Diffusion Model.** Technology evolves in two phases:

1. **Innovation:** New technology is created ($A_{new}$)
2. **Diffusion:** Technology spreads through adoption ($A_{adopted}$)

$$\Psi_{tech,t+1} = \underbrace{A_t + \phi(R\&D_t) \cdot \epsilon_t}_{\text{Innovation}} + \underbrace{\lambda(A_{frontier} - A_t)}_{\text{Diffusion}} \tag{90}$$

where:

- $\phi(R\&D)$: Innovation production function (diminishing returns)
- $\epsilon_t$: Stochastic breakthrough
- $\lambda$: Diffusion speed
- $A_{frontier}$: Best available technology

**Implication: Technology Leaders vs. Followers.**

- Leaders ($A_t \approx A_{frontier}$): Depend on innovation, high variance
- Followers ($A_t \ll A_{frontier}$): Can grow through adoption, lower variance

**Example: China's Technological Catch-Up.**

- 1980-2010: High $\lambda$ (adoption), rapid growth without much innovation
- 2010-present: Approaching frontier, must shift to innovation
- Challenge: Innovation requires different institutions than adoption

### 9.13.2 Norm Dynamics

**The Social Learning Model.** Norms evolve through observation and imitation:

$$\Psi_{norm,t+1} = \Psi_{norm,t} + \eta_{norm} \cdot (\bar{a}_t - \Psi_{norm,t}) \tag{91}$$

where $\bar{a}_t$ is average observed behavior.

**Interpretation:** Norms track behavior with a lag. If people start behaving differently, norms eventually follow.

**The Conformity-Threshold Model.** More realistically, norm adoption depends on how many others have adopted:

$$Prob(\text{adopt norm}) = \begin{cases} 0 & \text{if } \bar{a}_t < \theta_i \\ 1 & \text{if } \bar{a}_t \geq \theta_i \end{cases} \tag{92}$$

where $\theta_i$ is individual $i$'s conformity threshold.

**Implication:** S-curve dynamics. If thresholds are distributed, a small initial group can trigger a cascade.

**Example: Same-Sex Marriage Acceptance.**

- 2004: 30% support in US
- 2008: 40% support—early adopters
- 2012: 50% support—critical mass reached
- 2015: 60% support—cascade, Supreme Court ruling
- 2023: 71% support—new norm established

The S-curve pattern: slow start, rapid middle, slow finish.

### 9.13.3 Institutional Dynamics

**The Political Economy Model.** Institutions change when the political coalition supporting them weakens:

$$\Psi_{inst,t+1} = \begin{cases} \Psi_{inst,t} & \text{if } power(\text{defenders}) > power(\text{challengers}) \\ \Psi_{inst}^{new} & \text{if } power(\text{challengers}) > power(\text{defenders}) \end{cases} \tag{93}$$

**Implication:** Institutions are sticky because they create their own defenders. Change requires either gradual power shift or sudden crisis.

**The Punctuated Equilibrium Model.** Combining stability and sudden change:

$$\Psi_{inst,t+1} = \Psi_{inst,t} + \eta_{inst}(crisis_t) \cdot \Delta a_t \tag{94}$$

where $\eta_{inst}(crisis) \gg \eta_{inst}(normal)$.

**Interpretation:** Most of the time, institutions barely change. During crises, change is rapid. This matches historical observation.

**Example: The New Deal.**

- Pre-1929: Limited federal government, laissez-faire institutions
- 1929-1933: Crisis—$\eta_{inst}$ becomes large
- 1933-1938: Rapid institutional change (Social Security, SEC, NLRA, etc.)
- Post-1938: New institutions lock in, $\eta_{inst}$ returns to small

### 9.13.4 Cultural Dynamics

**The Intergenerational Transmission Model.** Culture evolves through what parents teach children:

$$\Psi_{culture,t+1} = (1 - \mu) \cdot \Psi_{culture,t} + \mu \cdot \bar{a}_t \tag{95}$$

where $\mu$ is the weight on observed behavior vs. inherited values.

**Implication:** $\mu$ is small, so culture changes slowly. One generation's behavior slightly shifts the next generation's values.

**The Dual Inheritance Model.** Combining vertical (parent-child) and horizontal (peer) transmission:

$$\Psi_{culture,t+1} = \alpha \cdot \Psi_{parent} + (1 - \alpha) \cdot \Psi_{peers} \tag{96}$$

**Implication:** When $\alpha$ is high (traditional societies), culture is very stable. When $\alpha$ is low (modern societies), culture changes faster but remains slower than norms.

**Example: Secularization in Western Europe.**

- 1900: 90%+ religious affiliation
- 1950: 80%+ religious affiliation (war had little effect)
- 2000: 50-60% religious affiliation
- 2020: 30-40% religious affiliation

120 years for a major cultural shift—roughly 4-5 generations. Each generation slightly less religious than parents.

## 9.14 Cross-Level Interactions: How Context Types Influence Each Other

Context types don't evolve independently. Technology affects norms, norms affect institutions, institutions affect culture.

**Technology → Norms.** New technology creates new behaviors, which eventually become normalized.

- Smartphone → Always-on connectivity becomes expected
- Social media → Public sharing of personal life becomes normal
- Contraception → Sexual norms change

**Lag:** Technology changes in years; norms follow in decades.

**Norms → Institutions.** When norms shift sufficiently, institutional change becomes possible.

- Anti-smoking norms → Smoking bans become politically feasible
- Environmental awareness → Climate legislation possible
- Civil rights norms → Legal discrimination becomes untenable

**Lag:** Norms must reach critical mass before institutions follow.

**Institutions → Culture.** Institutions shape what behaviors are rewarded, which shapes values over generations.

- Democratic institutions → Democratic values strengthen over generations
- Market institutions → Individualistic values strengthen
- Welfare state → Egalitarian values strengthen

**Lag:** Institutional effects on culture take generations to manifest.

**The Cascade Model.**

$$\Delta\Psi_{tech} \xrightarrow{\tau_1} \Delta\Psi_{norm} \xrightarrow{\tau_2} \Delta\Psi_{inst} \xrightarrow{\tau_3} \Delta\Psi_{culture} \tag{97}$$

where $\tau_1 < \tau_2 < \tau_3$ are the lag times.

**Example: The Internet Revolution.**

- 1990s: Technology arrives (Internet, email, web)
- 2000s: Norms shift (online commerce normal, email expected)
- 2010s: Institutions adapt (GDPR, platform regulation, digital taxation)
- 2020s+: Culture shifts (digital natives, attention economy, privacy expectations)

**Cultural Lag (Ogburn, 1922).** When technology changes faster than norms, and norms faster than institutions, the system is perpetually "out of sync."

$$K < 1 \text{ because } \eta_{tech} > \eta_{norm} > \eta_{inst} > \eta_{culture} \tag{98}$$

This explains why modern societies often feel incoherent: technology has changed, but institutions still reflect older norms and values.

## 9.15 Context Dynamics at Each Level

Context operates differently at each analytical level.

### 9.15.1 Individual Level: Personal Context

**Components.**

- Habits and routines
- Physical environment (home, workspace)
- Social network
- Skills and knowledge
- Health status

**Dynamics.**

$$\Psi_{personal,t+1} = \Psi_{personal,t} + \eta_{habit} \cdot (a_t - a^*) \tag{99}$$

Habits form slowly but persist strongly (small $\eta_{habit}$ but high autocorrelation).

**Example: Career Change.** Changing careers requires changing personal context:

- New skills (education, training)
- New network (professional connections)
- New identity (how you see yourself)

- New habits (daily routines)

The difficulty of career change reflects the stickiness of personal $\Psi$.

### 9.15.2 Household Level: Family Context

**Components.**

- Role expectations (who does what)
- Communication patterns
- Resource allocation rules
- Parenting norms
- Extended family relationships

**Dynamics.** Family context is negotiated between members:

$$\Psi_{family,t+1} = \Psi_{family,t} + \sum_i w_i \cdot (a_{i,t} - a_i^*) \tag{100}$$

where $w_i$ is member $i$'s influence weight.

**Example: Division of Housework.**

- 1970: Traditional division (women do housework)
- 2020: More equal, but still unequal
- Change is slow because it requires renegotiating family $\Psi$
- Each generation's behavior slightly shifts norms for next generation

### 9.15.3 Organizational Level: Corporate Context

**Components.**

- Corporate culture
- Formal structure and processes
- Informal power networks
- Shared mental models
- Organizational routines

**Dynamics.**

$$\Psi_{org,t+1} = \Psi_{org,t} + \eta_{org}(leadership_t) \cdot \Delta strategy_t \tag{101}$$

where $\eta_{org}$ depends heavily on leadership commitment and tenure.

**Example: Cultural Transformation at Microsoft.**

- Pre-2014: Competitive, "stack ranking" culture
- 2014: New CEO (Nadella), explicit cultural change program
- 2014-2020: Gradual shift to "growth mindset" culture
- 6+ years for substantial cultural change, with strong leadership

#### 9.15.4 Regional Level: Local Context

**Components.**

- Local industry structure
- Educational institutions
- Physical infrastructure
- Local norms and trust
- Political governance

**Dynamics.** Regional context is shaped by investment, migration, and policy:

$$\Psi_{region,t+1} = \Psi_{region,t} + \eta_{infra} \cdot I_t + \eta_{migration} \cdot M_t \tag{102}$$

**Example: Silicon Valley.**

- 1950s: Stanford + defense contracts create initial tech cluster
- 1960s-70s: Semiconductor industry develops
- 1980s-90s: Software and internet boom
- 2000s+: Self-reinforcing ecosystem

The regional context (VC, talent, culture) took 50+ years to develop and is now extremely difficult to replicate elsewhere.

### 9.15.5 National Level: Institutional Context

**Components.**

- Constitutional framework
- Legal system
- Economic institutions
- Welfare state
- National identity

**Dynamics.** National institutions change through politics, rarely smoothly:

$$\Psi_{national,t+1} = \begin{cases} \Psi_{national,t} & \text{normal times} \\ \Psi_{national,t} + \Delta_{crisis} & \text{critical juncture} \end{cases} \tag{103}$$

**Example: Germany's Economic Model.**

- 1949: Post-war, new constitution, "Soziale Marktwirtschaft"
- 1950-2000: Remarkable stability of institutional framework
- 2003-2005: Hartz reforms (labor market liberalization)—rare change
- 2010s: Minor adjustments but basic model intact

70+ years of institutional continuity with only incremental change.

### 9.15.6 Global Level: International Context

**Components.**

- International organizations (UN, WTO, IMF)
- Trade regimes
- Security alliances
- Global norms (human rights, sovereignty)
- Climate agreements

**Dynamics.** Global context is the hardest to change—no central authority:

$$\Psi_{global,t+1} = \Psi_{global,t} + \eta_{hegemony} \cdot a_{hegemon} + \eta_{crisis} \cdot crisis_t \tag{104}$$

**Example: Post-WWII International Order.**

- 1944-1948: Bretton Woods, UN, GATT created (crisis → change)
- 1948-1989: Cold War framework, remarkably stable
- 1989-2000: "End of history," liberal order expands
- 2000s-present: Erosion, but no new framework yet

Global $\Psi$ changes only after major crises or hegemonic shifts.

## 9.16 Policy Implications: Changing Context vs. Changing Behavior

**The Traditional Policy Approach: Change Behavior.** Most policies target behavior directly:

- Taxes and subsidies (change incentives)
- Regulations (mandate or prohibit behavior)
- Information campaigns (change beliefs)

**Problem:** If context ($\Psi$) is unchanged, behavior reverts when policy is removed.

**The Context Approach: Change $\Psi$ Directly.** Alternative: target context itself:

- **Technology policy:** R&D funding, infrastructure investment
- **Norm entrepreneurship:** Role models, media campaigns, social movements
- **Institutional reform:** Constitutional change, legal reform
- **Cultural policy:** Education, national narratives

**Advantage:** Changed context sustains changed behavior even without ongoing intervention.

**When to Target Context vs. Behavior.**

| Situation | Target | Example |
|---|---|---|
| Quick fix needed | Behavior | Tax on plastic bags |
| Sustainable change needed | Context | Norm against single-use plastic |
| Individual problem | Behavior | Fine for littering |
| Systemic problem | Context | Redesign waste system |
| Context already favorable | Behavior | Enforce existing norms |
| Context unfavorable | Context | Change the norms first |

**The Sequencing Problem.** Context change is slow but durable. Behavior change is fast but fragile. Optimal policy often sequences both:

1. Start behavior change (regulations, incentives)
2. Behavior change shifts norms (people adapt)
3. Norm change enables institutional reform
4. New institutions sustain new behavior
5. Remove original policy—new equilibrium is self-sustaining

**Example: Seat Belt Laws.**

1. 1968: Federal mandate for seat belts in cars (technology)
2. 1984+: States pass mandatory wearing laws (institution)
3. 1985-2000: Compliance increases, wearing becomes normal (norm)
4. Today: 90%+ wear seat belts, law rarely enforced—norm self-sustains

**Example: Failed Policy—Prohibition.**

1. 1920: Constitutional amendment bans alcohol (institution)
2. But: Drinking norms unchanged, technology (speakeasies) adapts
3. Institution-behavior gap too large—massive non-compliance
4. 1933: Amendment repealed

**Lesson:** Institutional change without norm change is unsustainable.

**The "Big Push" Question.** Sometimes context change requires simultaneous movement on multiple fronts. Partial change fails because the system returns to old equilibrium.

- **Successful big push:** Singapore 1960s-80s (simultaneous change in institutions, education, housing, economic policy)
- **Failed partial push:** Many developing country reforms that changed laws but not enforcement, or policy but not norms

## 9.17 Limitations and Extensions

**Limitations of the Linear Model.** The linear dynamics (64) cannot capture:

- Threshold effects (change only after critical mass)
- Hysteresis (asymmetric adjustment)
- Irreversibility (some changes cannot be undone)
- Strategic context manipulation (large actors shaping $\Psi$)

**Future Extensions.**

- Regime-switching models for tipping points
- Agent-based models for micro-to-macro aggregation
- Network models for contagion dynamics
- Political economy models for institutional change

The framework is compatible with these extensions. The key insight—that $\Psi$ is endogenous and co-evolves with behavior—remains regardless of the specific functional form.

## 9.18 Measuring Context: The Technological Breakthrough

Context $\Psi$ includes norms, trust, culture, and institutional quality— variables that have traditionally resisted quantification. This is why most economic models treat $\Psi$ as exogenous or ignore it entirely.

**The Traditional Problem.**

- **Norms:** How do you measure what people think is appropriate?
- **Trust:** Survey responses are noisy and may not predict behavior
- **Culture:** Anthropological methods don't scale
- **Institutions:** Legal texts don't capture enforcement

**The NLP Revolution (2020-2026).** Natural Language Processing has transformed context measurement:

**1. Norm Extraction from Text.** Large corpora (social media, news, policy documents) contain implicit norms. Modern NLP can extract:

- What behaviors are praised or condemned

- How justifications shift over time
- Which violations trigger outrage
- When norm change reaches tipping points

**2. Trust from Communication Patterns.** Trust manifests in language:

- Cooperative vs. defensive framing
- Future-oriented vs. present-oriented discourse
- Inclusive vs. exclusive pronouns
- Conditional vs. unconditional commitments

LLM-based analysis can score these patterns at scale.

**3. Institutional Quality from Legal Text.** Laws and regulations are text. Modern NLP can:

- Compare legal frameworks across jurisdictions
- Track regulatory change over time
- Measure consistency and contradictions
- Identify gaps between law and enforcement

**4. Cultural Values from Large-Scale Data.**

- Word embeddings reveal implicit associations
- Topic models identify cultural concerns
- Sentiment analysis tracks attitude change
- Cross-lingual analysis enables comparison

**LLM-Powered Context Analysis.** The most powerful recent development is using LLMs directly for context assessment:

> "Analyze these 1000 policy documents from Germany (2000-2025). Extract: (1) implicit assumptions about labor market flexibility, (2) attitudes toward risk-sharing between firms and workers, (3) changes in emphasis on competition vs. coordination. Score each dimension 0-10 and track trends."

This produces quantified context measures that would have required years of expert coding—in hours.

**From Qualitative to Quantitative $\Psi$.** Context is no longer a residual category or a qualitative description. It is a measurable, trackable, comparable variable. This transforms the $(C, \Psi)$ framework from theory to empirical tool.

The evolution of context measurement capabilities—from expert judgment in the 1970s to LLM-powered analysis today—is detailed in Appendix **??**.

# 10 Welfare and Equilibrium Quality: The FEPSDE Framework

The coherence index $K$ measures whether a system is in equilibrium. It does not measure whether the equilibrium is *good*. A society of mutual exploitation can be perfectly coherent ($K = 1$) yet deeply impoverished in welfare.

This section introduces the FEPSDE framework—a multidimensional welfare criterion (the complete 144-component matrix appears in Appendix C) that ranks equilibria and answers the question: What makes one coherent configuration better than another?

## 10.1 The Problem: $K$ Without $Q$ Is Incomplete

**Two Coherent Societies.** Consider two societies, both with $K \approx 1$:

**Society A: Extractive Coherence**

- Elites extract resources from masses
- Masses expect extraction and don't invest
- Institutions enforce extraction
- Everyone's beliefs match reality—coherent
- But welfare is low for most people

**Society B: Inclusive Coherence**

- Property rights protect investment
- People expect returns and invest
- Institutions enforce contracts
- Everyone's beliefs match reality—coherent
- Welfare is high for most people

Both have $K \approx 1$. But B is clearly better than A. We need a criterion to capture this: the quality index $Q$.

**Why Not Just Use GDP?** Gross Domestic Product measures market transactions. It misses:

- Non-market production (household work, volunteering)
- Distribution (GDP can rise while most people suffer)

- Health and longevity
- Environmental degradation
- Social cohesion and trust
- Subjective wellbeing
- Sustainability (GDP today vs. GDP tomorrow)

A framework that claims to capture "rationality" must have a richer welfare concept.

## 10.2 The FEPSDE Dimensions: Why These Six?

**The Selection Criteria.** The dimensions were selected based on:

1. **Empirical distinctness:** Each captures variance not captured by others
2. **Universal relevance:** Applies across cultures, levels, time periods
3. **Measurability:** Can be operationalized for empirical work
4. **Policy relevance:** Corresponds to distinct policy domains
5. **Theoretical grounding:** Connects to established research programs

**The Six Dimensions.**

**Definition 10.1** (FEPSDE Utility Dimensions)**.**

$$
\begin{aligned}
U^F &: \textbf{\textit{Financial}} \text{ — } \textit{material wealth, income, economic security} \\
U^E &: \textbf{\textit{Emotional}} \text{ — } \textit{psychological wellbeing, life satisfaction, meaning} \\
U^P &: \textbf{\textit{Physical}} \text{ — } \textit{health, energy, bodily integrity, longevity} \\
U^S &: \textbf{\textit{Social}} \text{ — } \textit{relationships, belonging, trust, status, community} \\
U^D &: \textbf{\textit{Digital}} \text{ — } \textit{connectivity, access, digital participation, data rights} \\
U^{Eco} &: \textbf{\textit{Ecological}} \text{ — } \textit{environmental quality, sustainability, nature access}
\end{aligned}
$$

## 10.3 Each Dimension in Detail

### 10.3.1 Financial Dimension ($U^F$)

**What It Captures.** Material resources and economic security:

- Income and wealth
- Consumption possibilities

- Economic security (protection against shocks)
- Property and assets
- Access to credit and financial services

**Why It's Not Sufficient Alone.** The "Easterlin Paradox": above a threshold, more income doesn't increase happiness. Cross-country: GDP per capita correlates with well-being only up to $20,000. Within countries: relative position matters more than absolute level.

**Measurement.**

- Income (individual, household, equivalized)
- Wealth (assets minus liabilities)
- Consumption expenditure
- Economic security indices
- Gini coefficient (distribution)

**Example: High $U^F$, Low Other Dimensions.** A wealthy executive: high income, long hours, no time for family, chronic stress, sedentary lifestyle, no community involvement. $U^F$ high; $U^E$, $U^P$, $U^S$ low.

### 10.3.2 Emotional Dimension ($U^E$)

**What It Captures.** Psychological wellbeing and subjective experience:

- Life satisfaction
- Positive and negative affect
- Meaning and purpose
- Autonomy and self-determination
- Mental health

**Theoretical Grounding.** Connects to:

- Subjective wellbeing research (Diener et al. 1999)
- Self-determination theory (autonomy, competence, relatedness)
- Positive psychology (PERMA: Positive emotion, Engagement, Relationships, Meaning, Achievement)

**Measurement.**

- Life satisfaction scales (Cantril ladder, SWLS)
- Positive/negative affect schedules (PANAS)
- Depression and anxiety inventories
- Meaning in life questionnaires
- Experience sampling (momentary wellbeing)

**Example: The Nordic Paradox.** Nordic countries: moderate $U^F$ (high taxes reduce disposable income) but very high $U^E$ (life satisfaction, low anxiety, high trust). Shows $U^E$ is distinct from $U^F$.

### 10.3.3 Physical Dimension ($U^P$)

**What It Captures.** Bodily health and functioning:

- Health status and absence of disease
- Physical functioning and mobility
- Energy and vitality
- Longevity and life expectancy
- Freedom from pain
- Sleep quality

**Why It's Fundamental.** Health is a precondition for enjoying other dimensions. Severe illness reduces capacity to work ($U^F$), enjoy relationships ($U^S$), and experience positive emotions ($U^E$).

**Measurement.**

- Life expectancy and healthy life years
- Self-reported health status
- Disability-adjusted life years (DALYs)
- Quality-adjusted life years (QALYs)
- Biomarkers (blood pressure, BMI, etc.)
- Functional capacity assessments

**Example: The American Health Paradox.** USA: Highest healthcare spending per capita. Yet: Lower life expectancy than peer countries, high rates of chronic disease. High $U^F$ investment in health doesn't automatically produce high $U^P$.

### 10.3.4 Social Dimension ($U^S$)

**What It Captures.** Relationships and social integration:

- Close relationships (family, friends, partners)
- Social support networks
- Community belonging
- Social status and recognition
- Trust (interpersonal and institutional)
- Civic participation

**Theoretical Grounding.** Connects to:

- Social capital literature (Putnam 2000)
- Attachment theory
- Belongingness as fundamental need
- Status and relative position research

**Why Social Relationships Are Special.** Strongest predictor of longevity (stronger than smoking, obesity, exercise). Loneliness is a public health crisis. Social trust underpins economic transactions.

**Measurement.**

- Social network size and quality
- Frequency of social contact
- Loneliness scales (UCLA Loneliness Scale)
- Trust surveys (World Values Survey)
- Civic participation rates
- Social support indices

**Example: Japan's Social Crisis.** Japan: High $U^F$, high $U^P$ (longevity), but increasingly low $U^S$. "Hikikomori" (social withdrawal), declining marriage, "kodokushi" (lonely deaths). Economic success doesn't guarantee social wellbeing.

### 10.3.5 Digital Dimension ($U^D$)

**What It Captures.** Participation in the digital world:

- Internet access and connectivity
- Digital literacy and skills
- Access to digital services (banking, government, healthcare)
- Data rights and privacy
- Protection from digital harms (cybercrime, misinformation)
- Ability to disconnect (digital wellbeing)

**Why a Separate Dimension?** Digital participation has become:

- Prerequisite for economic participation (job applications, banking)
- Primary mode of social interaction for many
- Major source of information and misinformation
- Site of new forms of harm (cyberbullying, addiction, surveillance)

The COVID-19 pandemic revealed: digital access is now as fundamental as electricity.

**Measurement.**

- Internet access (broadband, mobile)
- Digital literacy assessments
- E-government usage
- Screen time and digital wellbeing indicators
- Cybersecurity incident rates
- Data privacy indices

**Example: The Digital Divide.** Rural elderly: low $U^D$ (no internet, no skills). During pandemic: couldn't access telehealth, online shopping, video calls with family. Low $U^D$ cascaded into lower $U^P$ (health access), $U^S$ (isolation), $U^F$ (services).

### 10.3.6 Ecological Dimension ($U^{Eco}$)

**What It Captures.** Environmental quality and sustainability:

- Air and water quality
- Access to green spaces and nature
- Climate stability
- Biodiversity
- Resource sustainability
- Environmental justice (distribution of environmental harms)

**Why It's Increasingly Important.**

- Climate change threatens all other dimensions
- Environmental degradation affects health ($U^P$), livelihoods ($U^F$)
- Intergenerational equity: today's $U^F$ at cost of future $U^{Eco}$
- Nature exposure affects mental health ($U^E$)

**Theoretical Grounding.** Connects to:

- Environmental economics (externalities, sustainability)
- Ecological economics (natural capital, carrying capacity)
- Environmental psychology (nature and wellbeing)

**Measurement.**

- Air quality indices (PM2.5, ozone)
- Water quality measures
- Carbon footprint
- Green space access
- Ecological footprint
- Environmental Performance Index

**Example: China's Tradeoff.** 1980-2010: Rapid increase in $U^F$ (GDP growth). Simultaneously: Severe decline in $U^{Eco}$ (air pollution, water contamination). Health costs ($U^P$) partially offset financial gains. Now: Policy shift toward "ecological civilization."

## 10.4 The Three Utility Categories: INU, KNU, IDN

The six dimensions capture *what* matters. The three categories capture *for whom* it matters.

**Definition 10.2** (Utility Categories)**.**

$$INU : \textbf{\textit{Individual Utility}} \text{ — } \textit{What do I want for myself?}$$
$$KNU : \textbf{\textit{Collective Utility}} \text{ — } \textit{What do I want for us/my group?}$$
$$IDN : \textbf{\textit{Identity Utility}} \text{ — } \textit{Who am I / who do I want to be?}$$

### 10.4.1 Individual Utility (INU)

**What It Captures.** Standard self-interested preferences:

- My income, my health, my relationships
- Classical economic agent's utility function
- "What's in it for me?"

**Theoretical Grounding.** This is the utility that Arrow-Debreu models. It's real and important—but not the whole story.

### 10.4.2 Collective Utility (KNU)

**What It Captures.** Preferences over group outcomes:

- My family's welfare (not just mine)
- My community's prosperity
- My nation's success
- My company's performance

**Why It's Distinct from INU.** People sacrifice personal welfare for group welfare:

- Parents sacrifice for children
- Soldiers sacrifice for country
- Employees sacrifice for company

- Activists sacrifice for cause

This isn't reducible to individual utility—it's a separate category.

**Theoretical Grounding.** Connects to:

- Group selection in evolutionary theory
- Team reasoning in game theory
- Organizational identification in management
- Patriotism and nationalism in political science

### 10.4.3 Identity Utility (IDN)

**What It Captures.** Preferences over self-concept:

- Being a good person
- Being competent at valued activities
- Having a coherent life narrative
- Living according to one's values
- Being respected by those one respects

**Why It's Distinct.** People sacrifice both INU and KNU for identity:

- Refusing to lie even when profitable and harmless
- Maintaining principles even when group disagrees
- Choosing meaningful work over higher pay
- Accepting hardship to "be true to oneself"

**Theoretical Grounding.** Connects to:

- Akerlof and Kranton (2000) identity economics
- Bénabou and Tirole (2006) self-signaling
- Self-determination theory (authenticity)
- Narrative identity in psychology

## 10.5 The Time Structure: Four Horizons

Welfare is evaluated across time, not just at a moment.

**Definition 10.3** (Time Horizons)**.**

$$t = 0 : \textbf{\textit{Instant}} \text{ — } \textit{Right now, immediate experience}$$
$$t = 1 : \textbf{\textit{Short-term}} \text{ — } \textit{Days to weeks}$$
$$t = 2 : \textbf{\textit{Medium-term}} \text{ — } \textit{Months to years}$$
$$t = 3 : \textbf{\textit{Long-term}} \text{ — } \textit{Years to lifetime}$$

**Why Time Matters.** Many choices involve intertemporal tradeoffs:

- Education: Low $U_0^F$, high $U_3^F$
- Exercise: Low $U_0^E$ (effort), high $U_2^P$
- Saving: Low $U_0^F$, high $U_3^F$
- Addiction: High $U_0^E$, low $U_2^{P,E,S,F}$

**Discounting.** Future utility is weighted by discount factor $\delta_d(\Psi) \in (0, 1)$:

$$\text{Present value of } U_{d,t} = \delta_d^t \cdot U_{d,t} \tag{105}$$

Discount rates may vary by dimension:

- Financial: Standard exponential discounting
- Health: May be discounted less ("health is priceless")
- Ecological: Intergenerational discounting debates

## 10.6 Gains and Pains: The Prospect Theory Connection

**Why Separate Gains and Pains?** Kahneman and Tversky (1979) showed that losses loom larger than gains: people feel a \$100 loss more intensely than a \$100 gain.

This "loss aversion" applies across domains:

- Financial losses hurt more than equivalent gains help
- Health decline hurts more than equivalent improvement helps
- Social rejection hurts more than equivalent acceptance helps

**The Formal Structure.** For each dimension $d$, utility is:

$$U_d = G_d - \lambda_d \cdot P_d \tag{106}$$

where:

- $G_d \geq 0$: Gains in dimension $d$
- $P_d \geq 0$: Pains (losses) in dimension $d$
- $\lambda_d > 1$: Loss aversion coefficient for dimension $d$

**Empirical Estimates.**

- Financial: $\lambda_F \approx 2$ (Kahneman & Tversky)
- Health: $\lambda_P \approx 2 - 3$ (health state valuations)
- Social: $\lambda_S \approx 2 - 4$ (rejection sensitivity research)

**Policy Implication.** Preventing harm is more valuable than creating equivalent benefit. A policy that prevents \$100 loss creates more welfare than one that provides \$100 gain.

## 10.7 The Complete 144-Component Structure

**Counting Components.**

- 3 categories (INU, KNU, IDN)
- 2 valences (Gain, Pain)
- 6 dimensions (FEPSDE)
- 4 time horizons

Total: $3 \times 2 \times 6 \times 4 = 144$ components.

**Examples of Specific Components.**

| Component | Meaning | Example |
|---|---|---|
| $G_{INU,F,0}$ | My financial gain now | Bonus payment today |
| $P_{INU,P,2}$ | My physical pain medium-term | Chronic back pain |
| $G_{KNU,S,1}$ | Our group's social gain short-term | Team wins award |
| $P_{KNU,Eco,3}$ | Our collective ecological pain long-term | Climate change |
| $G_{IDN,E,2}$ | My identity emotional gain medium-term | Pride in achievement |
| $P_{IDN,S,0}$ | My identity social pain now | Public humiliation |

## 10.8 The Aggregate Welfare Function

**Definition 10.4** (Equilibrium Quality)**.** *The quality of an equilibrium $C^*$ is:*

$$Q(C^*) = \sum_{i \in \{INU, KNU, IDN\}} \sum_{d \in FEPSDE} \sum_{t=0}^{3} \omega_i \cdot \delta_d(\Psi)^t \left[G_{i,d,t} - \lambda_d(\Psi) \cdot P_{i,d,t}\right] \tag{107}$$

*where:*

- $\omega_i$*: Weight on utility category* $i$
- $\delta_d(\Psi)$*: Discount factor for dimension* $d$
- $\lambda_d(\Psi)$*: Loss aversion for dimension* $d$

**Parameter Values.** The weights $\omega_i$, discount factors $\delta_d$, and loss aversion $\lambda_d$ can be estimated from:

- Revealed preference (observed choices)
- Stated preference (surveys, experiments)
- Deliberative methods (citizens' assemblies)

Different societies may have different parameters—this is part of context $\Psi$.

## 10.9 FEPSDE vs. Alternative Welfare Measures

| Measure | Dimensions | Strengths | Weaknesses |
|---|---|---|---|
| GDP | Financial only | Widely available | Misses everything else |
| HDI | F, E, P (partial) | Simple, comparable | Only 3 dimensions |
| GNH | Multiple | Holistic | Hard to measure |
| OECD Better Life | 11 dimensions | Comprehensive | No aggregation |
| **FEPSDE** | $6 \times 3 \times 4$ | Structured, complete | Complex |

**Advantages of FEPSDE.**

1. **Structured:** Clear taxonomy enables systematic analysis
2. **Complete:** Covers material, psychological, social, digital, ecological
3. **Dynamic:** Explicit time structure captures sustainability
4. **Behavioral:** Gains/pains structure incorporates Prospect Theory
5. **Multi-level:** INU/KNU/IDN captures individual and social welfare

## 10.10 Applying FEPSDE: Country Comparison

**Example: USA vs. Denmark.**

| **Dimension** | **USA** | **Denmark** |
|---|---|---|
| Financial ($U^F$) | Higher (GDP/capita) | Lower (after taxes) |
| Emotional ($U^E$) | Lower (life satisfaction) | Higher |
| Physical ($U^P$) | Lower (life expectancy) | Higher |
| Social ($U^S$) | Lower (trust, community) | Higher |
| Digital ($U^D$) | Similar | Similar |
| Ecological ($U^{Eco}$) | Lower (footprint) | Higher |
| **Overall** $Q$ | ? | ? |

Which is higher depends on weights. If $\omega_F$ dominates: USA wins. If $\omega_{E,P,S,Eco}$ matter: Denmark wins. FEPSDE makes the tradeoffs explicit.

## 10.11 The K-Q Relationship

**Four Quadrants.**

| | **High K** | **Low K** |
|---|---|---|
| **High Q** | Ideal: Coherent + Good | Transition to better |
| **Low Q** | Trap: Coherent + Bad | Crisis: Incoherent + Bad |

**Examples.** **High K, High Q:** Nordic countries. Coherent configuration (institutions, norms, behavior aligned). High welfare across dimensions.

**High K, Low Q:** North Korea. Coherent configuration (everyone knows the rules, behaves accordingly). Low welfare by any reasonable measure.

**Low K, High Q potential:** Post-war Germany 1945-1955. Incoherent (old institutions destroyed, new ones forming). But moving toward high Q configuration.

**Low K, Low Q:** Failed states, civil wars. Incoherent (no stable expectations). Low welfare (violence, poverty, displacement).

**The Policy Implication.** Being in a "High K, Low Q" trap is dangerous. The system is stable but bad. Escaping requires accepting temporary low K (transition costs).

## 10.12 Summary: The Complete Welfare Framework

| **Element** | **Symbol** | **Question Answered** |
|---|---|---|
| Coherence | $K$ | Are we in equilibrium? |
| Quality | $Q$ | Is the equilibrium good? |
| Dimensions | FEPSDE | Good in what respects? |
| Categories | INU/KNU/IDN | Good for whom? |
| Time | $t = 0, 1, 2, 3$ | Good when? |
| Valence | G, P | Gains or avoided pains? |

The FEPSDE framework provides the vocabulary to discuss welfare beyond GDP, incorporating insights from psychology, health economics, environmental economics, and behavioral economics into a unified structure.

# 11 Novel Predictions

A unified framework must generate predictions that no constituent theory produces alone. This section develops predictions that emerge from the interaction of complementarity, context, and coherence—and applies the framework to concrete policy scenarios including trade wars, technological disruption, and institutional reform.

## 11.1 What Makes a Prediction "Novel"?

**Criteria for Novelty.** A prediction is novel if:

1. No single constituent theory generates it
2. It emerges from the *interaction* of framework elements
3. It is empirically testable
4. It differs from standard intuitions

**The Framework's Unique Features.** The $(C, \Psi, K, Q)$ framework uniquely combines:

- Cross-domain complementarities (financial $\leftrightarrow$ social $\leftrightarrow$ identity)
- Endogenous context dynamics
- Coherence as distinct from efficiency
- Multi-level analysis (individual to global)
- Multidimensional welfare (FEPSDE)

These combinations generate predictions unavailable to partial theories.

## 11.2 Prediction 1: Trade War Dynamics—The Trump Tariff Analysis

**The Policy Context.** In 2018-2019, the Trump administration imposed tariffs on approximately \$350 billion of Chinese imports, triggering retaliatory tariffs and ongoing trade tensions. In 2025, new rounds of tariffs have been announced.

**Standard Economic Prediction.** Classical trade theory predicts:

- Tariffs reduce welfare (deadweight loss)
- Both countries lose from trade war
- Effects are primarily financial ($U^F$)
- Policy is "irrational"—politicians acting against national interest

**Framework Prediction: Multi-Dimensional Effects.** The $(C, \Psi)$ framework predicts a more complex pattern:

**Phase 1: Immediate Effects (t = 0-1)**

| Dimension | USA | China | EU | Global |
|---|---|---|---|---|
| Financial ($U^F$) | $-$ | $-$ | $-$ | $-$ |
| Emotional ($U^E$) | $+/-$ | $-$ | $-$ | $-$ |
| Social ($U^S$) | + (in-group) | + (nationalism) | 0 | $-$ |
| Identity ($U^{IDN}$) | + ("winning") | + (resistance) | 0 | $-$ |

**Key insight:** While $U^F$ declines for all parties, $U^S$ and $U^{IDN}$ may *increase* for some groups. The tariffs serve identity and social functions beyond economics.

**Phase 2: Coherence Effects (t = 1-2)**

*Within USA:*

- Manufacturing regions: $K \uparrow$ (policy matches identity as "makers")
- Coastal/export regions: $K \downarrow$ (policy conflicts with globalist identity)
- Overall: Polarization—different regions at different $C^*$

*Within China:*

- Nationalism surge: $K \uparrow$ (external threat unifies)
- Export sector: $K \downarrow$ (business model disrupted)
- State response: Accelerate self-sufficiency (shift $\Psi$ to reduce dependence)

*Global trading system:*

- WTO rules: $K \downarrow$ (behavior diverges from institutional $C^*$)
- Supply chains: $K \downarrow$ (established complementarities disrupted)
- New blocs forming: Potential new $C^*$ equilibria

**Phase 3: Context Shift (t = 2-3)**

$$\Psi_{global,t+1} = \Psi_{global,t} + \eta \cdot (\text{tariff actions}) \tag{108}$$

The tariffs don't just affect prices—they shift context:

- $\Psi_{norm}$: "Free trade good" norm weakens
- $\Psi_{inst}$: WTO authority erodes
- $\Psi_{tech}$: Supply chains reconfigure
- $\Psi_{culture}$: Nationalist sentiment strengthens

**The Novel Prediction.**

> **Prediction 1:** Trade wars have persistent effects beyond their direct economic costs because they shift $\Psi$. Even if tariffs are removed, the context has changed: trust is lower, supply chains have restructured, nationalist identities have strengthened. The global trading system moves to a different $C^*$—possibly permanently.

**Formal Statement.** Let $\Psi^{FT}$ denote the free-trade context and $\Psi^{TW}$ the trade-war context.

$$\lim_{t\to\infty} \Psi_t \neq \Psi^{FT} \quad \text{even if tariffs removed at } t = T \tag{109}$$

**Hysteresis:** The path back is not the path forward. Context changes are partially irreversible.

**Welfare Analysis with FEPSDE.** Standard analysis: Tariffs reduce GDP $\Rightarrow$ bad.

FEPSDE analysis:

$$\Delta Q = \underbrace{\Delta U^F_{INU}}_{negative} + \underbrace{\Delta U^S_{KNU}}_{positive?} + \underbrace{\Delta U^{IDN}}_{positive?} + \text{time discounting} \tag{110}$$

For some voters, the identity and social gains ($U^{IDN}$, $U^S$) outweigh the financial losses ($U^F$)—especially if:

- $\omega_{IDN} > \omega_F$ (identity matters more than money)
- $\lambda_F$ is lower for losses framed as "sacrifice for the nation"
- $\delta_{IDN}$ is lower than $\delta_F$ (identity is less discounted)

**Why Standard Theory Misses This.**

- Trade theory: Only $U^F$, no $U^{IDN}$ or $U^S$
- Political economy: Explains special interests, not identity
- International relations: Explains power, not coherence

The framework integrates all three through the complementarity structure.

## 11.3 Prediction 2: Cross-Domain Coherence Spillovers

**Statement.** A negative shock to coherence in one domain predicts subsequent decline in coherence in other domains, even without direct causal links.

$$\Delta K_t^{financial} < 0 \Rightarrow \Delta K_{t+\tau}^{social} < 0 \Rightarrow \Delta K_{t+2\tau}^{political} < 0 \tag{111}$$

**Mechanism.** The complementarity matrix $C$ has cross-domain terms:

$$C = \begin{pmatrix} C^{FF} & C^{FS} & C^{FP} \\ C^{SF} & C^{SS} & C^{SP} \\ C^{PF} & C^{PS} & C^{PP} \end{pmatrix} \tag{112}$$

where F = financial, S = social, P = political.
When $C^{FS} \neq 0$, financial instability spills into social trust:

- Bank failures $\Rightarrow$ distrust of institutions
- Job losses $\Rightarrow$ community breakdown
- Inequality $\Rightarrow$ social resentment

**Example: 2008 Financial Crisis Cascade.**

1. 2008: $K^{financial}$ collapses (Lehman, correlations spike)
2. 2009-2011: $K^{social}$ declines (Occupy movement, Tea Party, polarization)
3. 2012-2016: $K^{political}$ declines (gridlock, populism, Brexit, Trump)
4. 2017-present: $K^{institutional}$ declines (norms erode, trust in democracy falls)

**Novel Prediction.**

> **Prediction 2:** Financial crises have political consequences with a lag of 5-10 years, mediated by social coherence. The 2008 crisis predicted the populist surge of 2016. Similarly, any future financial crisis will have political effects that manifest roughly a decade later.

**Why Constituent Theories Miss This.**

- Finance: Models markets in isolation
- Political science: Focuses on political variables
- Sociology: Studies social change without financial mechanisms

The framework connects all three through the coherence spillover mechanism.

## 11.4 Prediction 3: Asymmetric Recovery Dynamics

**Statement.** Recovery time from coherence loss is *convex* in the depth of the loss:

$$\tau_{recovery} = f(\Delta K), \quad f'' > 0 \tag{113}$$

Small shocks: Quick recovery. Large shocks: Disproportionately slow recovery.

**Mechanism.** For small deviations from $C^*$, the contraction property ensures quick return:

$$\|C_{t+1} - C^*\| \leq \gamma \|C_t - C^*\| \quad \text{for } \gamma < 1 \tag{114}$$

For large deviations, the system may exit the contraction region. Recovery then requires:

- Re-coordination of expectations (slow)
- Rebuilding of trust (very slow)
- Possible shift to different $C^*$ (path-dependent)

**Example: Post-Soviet Transition.**

- 1991: Coherence collapse ($K \to 0$) as Soviet institutions disappear
- Prediction (linear): Recovery in 5-10 years
- Reality: 30+ years later, many countries still haven't reached stable $C^*$
- Russia: Found a stable $C^*$ (authoritarian) but low $Q$
- Ukraine: Still searching for stable $C^*$
- Baltic states: Found high-$K$, high-$Q$ equilibrium (EU anchor helped)

**Novel Prediction.**

> **Prediction 3:** Policy responses to crises should be *non-linear* in crisis depth. A 10% decline in coherence requires more than twice the intervention needed for a 5% decline. Under-responding to large crises leads to persistent dysfunction.

## 11.5 Prediction 4: Innovation-Coherence Tradeoff

**Statement.** Innovation intensity $I_t$ predicts:

(a) Contemporaneous decline in coherence: $\partial K_t / \partial I_t < 0$

(b) Subsequent increase in coherence: $\partial K_{t+\tau} / \partial I_t > 0$ for $\tau > \bar{\tau}$

**Mechanism.** Innovation changes $\Psi_{tech}$, which shifts $C^*(\Psi)$. The old coherent configuration becomes incoherent under new technology. Agents must learn the new $C^*$ and re-coordinate.

**Example: AI and Labor Markets.**

- Pre-AI: Coherent labor market (skills match jobs, expectations stable)
- AI introduction: Skills mismatch, job uncertainty, anxiety
- Transition: Some workers adapt, others displaced, new roles emerge
- Post-transition: New coherent configuration (different jobs, different skills)

**The Adaptation Lag.**

$$\bar{\tau} = f(\eta_{norm}^{-1}, \eta_{inst}^{-1}) \tag{115}$$

The lag depends on how fast norms and institutions can adapt. Technology moves fast ($\eta_{tech}$ high); adaptation is slow ($\eta_{norm}$, $\eta_{inst}$ low).

**Novel Prediction.**

> **Prediction 4:** Periods of rapid innovation are also periods of high anxiety, political instability, and social conflict—not because innovation is bad, but because coherence is temporarily low. The "techlash" is a predictable coherence phenomenon, not irrational resistance to progress.

## 11.6 Prediction 5: The Identity-Economics Feedback

**Statement.** Economic policies that threaten group identity will face stronger resistance than economically equivalent policies that don't. Conversely, economically costly policies that affirm identity will receive more support than standard models predict.

**Mechanism.** Utility includes $U^{IDN}$. When policy threatens identity:

$$\Delta U^{total} = \Delta U^F + \omega_{IDN} \cdot \Delta U^{IDN} \tag{116}$$

If $\omega_{IDN}$ is large and $\Delta U^{IDN}$ is negative, even positive $\Delta U^F$ may not compensate.

**Example: Coal Communities and Climate Policy.**

- Standard analysis: Retraining + transfers can compensate job losses
- Reality: Fierce resistance even to generous compensation
- Reason: Coal mining is identity, not just income
- $\Delta U^F$ (compensation) $> 0$, but $\Delta U^{IDN}$ (identity loss) $\ll 0$

**Example: Brexit Economics.**

- Standard analysis: Brexit reduces GDP $\Rightarrow$ irrational
- Framework analysis: Brexit affirms national identity ($U^{IDN} \uparrow$)
- For voters with high $\omega_{IDN}$: Net welfare may be positive
- Not irrational—different utility function

**Novel Prediction.**

> **Prediction 5:** Policies should be evaluated not just on economic efficiency ($U^F$) but on identity coherence ($U^{IDN}$). Policies that are economically optimal but identity-threatening will fail politically. Successful reform requires either identity-neutral framing or explicit identity compensation.

## 11.7 Prediction 6: The Coherence Trap

**Statement.** High-coherence, low-quality equilibria are more stable than low-coherence, high-quality-potential configurations. Systems can be "trapped" in bad equilibria precisely because they are coherent.

**Formal Statement.** Let $C_A^*$ be a low-$Q$ equilibrium and $C_B^*$ a high-$Q$ equilibrium. If the system is at $C_A^*$ with $K \approx 1$, transition to $C_B^*$ requires passing through a valley where $K < 1$.

During transition:

- Current welfare: Falls (incoherence costs)
- Uncertainty: High (which equilibrium will emerge?)
- Political opposition: "Things were better before"

**Example: Middle-Income Trap.**

- Many countries reach middle income, then stagnate
- The middle-income configuration is coherent: export manufacturing, low wages, authoritarian stability
- Moving to high income requires: innovation, higher wages, institutional quality, political openness
- Transition is incoherent: old model breaks before new one works
- Many countries retreat to the old equilibrium

**Novel Prediction.**

> **Prediction 6:** Development is not continuous but involves discrete jumps between coherent configurations. The "middle-income trap" is a coherence trap. Escaping requires coordinated, simultaneous change across multiple dimensions—a "big push"—not gradual reform.

## 11.8 Prediction 7: Multi-Level Coherence Conflicts

**Statement.** Optimal coherence at one level may conflict with optimal coherence at another level. Individual rationality can produce collective irrationality, and vice versa.

**Example: Global Climate Coordination.**

- National level: Each country is coherent pursuing own interest
- Global level: Collective outcome is incoherent (tragedy of commons)
- $K^{national} \approx 1$ but $K^{global} \ll 1$

**Example: Corporate vs. Regional Coherence.**

- Company: Coherent strategy to offshore manufacturing
- Region: Loses jobs, tax base, community—incoherence
- $K^{org} \uparrow$ while $K^{region} \downarrow$

**Novel Prediction.**

> **Prediction 7:** Policies that optimize coherence at one level may destroy coherence at another level. Effective governance requires explicit analysis of cross-level coherence effects. The climate crisis (Weitzman 2009) is fundamentally a multi-level coherence problem: high $K$ at national level, low $K$ at global level.

## 11.9 Prediction 8: Mechanism Design for Context ($\Psi$-Design)

**The Intuition.** You want people to cooperate, but they have their own interests and private information. How do you set up the rules so they *want* to do what you need?

Consider kidney exchange: Alice needs a kidney, her husband wants to donate—but wrong blood type. Bob needs a kidney, his wife wants to donate—also wrong blood type. But Alice's husband matches Bob, and Bob's wife matches Alice. The solution exists, but how do you organize it? Who goes first? What if someone backs out?

Roth (2002) designed an algorithm that finds such chains and enables simultaneous operations. Thousands of lives saved—not through new medicine, but through better *rules*.

This is mechanism design: setting rules so that self-interest leads to good outcomes.

**The Deeper Problem.** Classical mechanism design assumes preferences are fixed. But what if the rules *change* the people?

Example: A community lacks trust. Classical solution: introduce contracts and courts. Short-run: Works! People cooperate because they fear punishment. Long-run problem: People learn "I don't need trust, I have contracts." Social trust erodes. When courts fail, cooperation collapses entirely.

This is the insight our framework adds: **mechanisms change the context in which they operate**. A rule that works today may destroy the social capital it needs tomorrow.

**The Core Insight.** Mechanism design (Hurwicz 1960; Myerson 1981; Maskin 1999) asks: given that agents have private information and strategic incentives, how can we design rules to achieve desired outcomes? Our framework extends this: **how do we design context $\Psi$ to achieve desired equilibria $C^*$?**

**Classical Mechanism Design.** The revelation principle (Myerson 1979) states that any outcome achievable by any mechanism can be achieved by a direct, incentive-compatible mechanism where agents truthfully reveal their types.

Key results:

- **Myerson (1981):** Optimal auction design—revenue maximization requires reserve prices and asymmetric treatment
- **Vickrey-Clarke-Groves:** Truthful revelation for public goods
- **Gibbard-Satterthwaite:** No perfect voting mechanism exists

**Market Design: Applied Mechanism Design.** Roth (2002, 2015) pioneered market design—engineering markets where standard mechanisms fail:

- **Kidney exchange:** Creating chains of compatible donors
- **School choice:** Deferred acceptance algorithms
- **Residency matching:** Stable matching that respects preferences

These are examples of $\Psi_I$-design: changing institutional rules to achieve better $C^*$.

**Framework Extension: Ψ-Design.** Classical mechanism design takes preferences as given and designs rules. Our framework goes further: Ψ **affects preferences themselves**.

| **Design Problem** | **Classical MD** | **Ψ-Design** |
|---|---|---|
| Given | Preferences, constraints | Initial $\Psi_0$ |
| Choose | Mechanism/rules | Ψ-intervention |
| To achieve | Outcome | Target $C^*$ with high $Q$ |
| Taking into account | Incentive compatibility | Ψ-dynamics, complementarities |

**Example: Designing Trust.** How do you create trust ($\Psi_S$) where none exists?

Classical answer: Design contracts and enforcement ($\Psi_I$) to make defection unprofitable.

Ψ-design answer: This works, but also changes $\Psi_S$ over time. Bohnet and Zeckhauser (2004) show that institutional trust can crowd out personal trust. The mechanism designer must consider:

- Short-run: Does the mechanism achieve desired behavior?
- Long-run: How does the mechanism change Ψ?
- Complementarities: How does $\Psi_I$ interact with $\Psi_S$, $\Psi_C$?

**Prediction 8.**

> **Prediction 8:** Mechanism design that ignores Ψ-dynamics will fail in the long run. Mechanisms that achieve short-run efficiency by substituting formal rules for social trust will erode the social capital ($\Psi_S$) on which future cooperation depends.
>
> Corollary: Sustainable mechanism design requires explicit modeling of $\dot{\Psi} = g(\Psi, \text{mechanism})$—how the mechanism itself changes context.

**Empirical Test.** Compare communities with:

1. Heavy reliance on formal contracts (high $\Psi_I$, low $\Psi_S$)
2. Reliance on informal trust (low $\Psi_I$, high $\Psi_S$)
3. Mixed systems (moderate both)

The framework predicts that Type 1 communities will show declining cooperation when contracts are incomplete or enforcement fails. Type 3 communities should show greatest resilience.

## 11.10 Summary: The Prediction Matrix

| Prediction | Novel Element | Policy Implication |
|---|---|---|
| 1. Trade wars | $\Psi$ hysteresis | Effects persist after tariffs end |
| 2. Cross-domain spillovers | $C^{FS} \neq 0$ | Financial crises cause political crises |
| 3. Asymmetric recovery | Convex $\tau(K)$ | Big crises need disproportionate response |
| 4. Innovation-coherence | $\Delta\Psi_{tech} \to \Delta K$ | Techlash is predictable |
| 5. Identity-economics | $U^{IDN}$ in welfare | Economic compensation insufficient |
| 6. Coherence trap | High-$K$, low-$Q$ stable | Escape requires big push |
| 7. Multi-level conflict | $K^{national} \neq K^{global}$ | Climate needs global coherence |
| 8. $\Psi$-Design | $\dot{\Psi} = g(\Psi, \text{mech})$ | Mechanism design must model context dynamics |

These predictions are testable, policy-relevant, and unavailable from any single constituent theory. They emerge from the integration of complementarity, context, and coherence that defines the framework.

# 12 Integration of Established Theories

A unified framework must demonstrate that established theories emerge as special cases. This section shows how classical, behavioral, institutional, organizational, and financial theories correspond to specific restrictions on the complementarity matrix $C$, the context variable $\Psi$, and their dynamics. Formal derivations are provided in Appendix **??**.

## 12.1 The Unification Logic

The general framework posits three equations:

$$U_i = U_i(x_i, x_{-i}, \Psi) \qquad \text{(Utility depends on others and context)} \tag{117}$$

$$C_{ij} = \frac{\partial^2 U_i}{\partial x_i \partial x_j} \qquad \text{(Complementarity structure)} \tag{118}$$

$$\Psi_{t+1} = f(\Psi_t, a_t) \qquad \text{(Context evolves with behavior)} \tag{119}$$

Each established theory imposes *restrictions* on this structure. Relaxing different restrictions yields different theories. The full framework relaxes all restrictions simultaneously.

## 12.2 Classical Economics: The Separability Restriction

**Arrow–Debreu General Equilibrium.** The foundational model of competitive markets (Arrow and Debreu 1954) assumes:

1. **Separable utilities:** $C_{ij} = 0$ for all $i \neq j$
2. **Exogenous context:** $\Psi_{t+1} = \Psi_t = \bar{\Psi}$
3. **No identity:** $U^{IDN} = 0$

Under these restrictions, each agent's optimization is independent. The coherence index $K = 1$ trivially—there are no cross-agent complementarities to align. The First Welfare Theorem holds: equilibrium is Pareto efficient.

**What Classical Economics Cannot Explain.** The separability assumption ($C_{ij} = 0$) rules out:

- Social preferences (fairness, reciprocity, altruism (Fehr and Fischbacher 2003))
- Strategic complementarity and coordination games
- Multiple equilibria from network effects
- Institutional change and path dependence

The framework shows *why* these phenomena require extensions: they all involve $C_{ij} \neq 0$.

## 12.3 Behavioral Economics: The Social Preference Extension

**Fehr–Schmidt Inequity Aversion.** Fehr and Schmidt (1999) introduce utility interdependence:

$$U_i = x_i - \alpha_i \sum_j \max(x_j - x_i, 0) - \beta_i \sum_j \max(x_i - x_j, 0) \tag{120}$$

This generates non-zero complementarities:

$$C_{ij} = \frac{\partial^2 U_i}{\partial x_i \partial x_j} \neq 0 \tag{121}$$

The key insight: $(\alpha_i, \beta_i)$ are not universal constants but depend on context. Laboratory experiments show they vary with framing, group identity, and institutional design. Thus: $\alpha_i = \alpha_i(\Psi)$, $\beta_i = \beta_i(\Psi)$.

**Bolton–Ockenfels ERC.** An alternative specification where utility depends on *share* of total payoff:

$$U_i = U_i\left(x_i, \frac{x_i}{\sum_j x_j}\right) \tag{122}$$

This also generates $C_{ij} \neq 0$, but the structure is state-dependent. Both models are special cases of the $(C, \Psi)$ framework with interdependent utilities.

**What Behavioral Economics Adds.** The move from $C_{ij} = 0$ to $C_{ij} \neq 0$ explains:

- Ultimatum game rejections (violate Arrow–Debreu predictions)
- Cooperation in public goods games
- Gift exchange in labor markets
- Context-dependent fairness norms

See Appendix **??** for formal derivations.

## 12.4 Identity Economics: The Temporal Extension

**Akerlof–Kranton Identity.** Akerlof and Kranton (2000) introduce identity as a utility dimension:

$$U_j = U_j(a_j, a_{-j}, c_j) \tag{123}$$

where $c_j$ is the agent's social category.

This adds two types of complementarity:

**Temporal complementarity:**

$$\frac{\partial^2 U}{\partial a_t \partial a_{t+1}} > 0 \tag{124}$$

Behaving consistently with identity today reinforces that identity tomorrow.

**Social complementarity:**

$$C_{ij} > 0 \text{ for } c_i = c_j \tag{125}$$

Members of the same category reinforce each other.

**Bénabou–Tirole Moral Identity.** Bénabou and Tirole (2006) model self-signaling:

$$U = v \cdot a - c(a) + \mu \cdot \mathbb{E}[\theta|a] \tag{126}$$

Agents care about what actions reveal about their character $\theta$. This creates strong intertemporal complementarity and explains taboos— actions that destroy self-image regardless of material payoff.

**Integration.** In the FEPSDE framework, identity maps to $U^{IDN}$: the utility from being who one believes oneself to be. The 144-component structure explicitly includes identity utility across all six dimensions and four time horizons.

## 12.5 Growth Theory: The Endogenous Context Extension

**Aghion–Howitt Endogenous Growth.** Aghion and Howitt (1998) model growth through innovation:

$$A_{t+1} = A_t \cdot (1 + g(n_t)) \tag{127}$$

Technology $A_t$ is part of context $\Psi$. Innovation changes context *endogenously*:

$$\Psi_{t+1} = f(\Psi_t, n_t) \neq \Psi_t \tag{128}$$

This violates the Arrow–Debreu assumption of fixed context.

**Creative Destruction as Coherence Dynamics.** Innovation disrupts existing complementarities:

1. Innovation occurs: $\Delta\Psi > 0$
2. Reference structure shifts: $C^*(\Psi_{old}) \to C^*(\Psi_{new})$
3. Coherence falls: $K_t \downarrow$ (old patterns no longer fit)

4. Adaptation: Agents learn new complementarities
5. Re-coherence: $K_t \uparrow$ at new equilibrium

**Growth as Repeated Cycles.**

- **Stagnation:** $K \approx 1$ (stable), but $Q$ doesn't grow
- **Growth:** $K$ fluctuates (creative destruction), but $Q$ rises over time

The framework reinterprets Schumpeterian dynamics as coherence dynamics.

## 12.6 Institutional Economics: The Multiple Equilibria Extension

**Acemoglu–Robinson Institutions.** Acemoglu and Robinson (2012) argue that institutions determine economic outcomes. In the framework:

$$\Psi = (\text{Property rights, Rule of law, Political openness}, \ldots) \tag{129}$$

Institutions determine the complementarity structure:

- **Inclusive institutions:** $C_{ij} > 0$ (cooperation pays)
- **Extractive institutions:** $C_{ij} \lesseqgtr 0$ (rent-seeking, zero-sum)

**Multiple Stable Equilibria.** Both configurations can be stable:

| **Equilibrium** | $C_{ij}$ | $K$ | $Q$ |
|---|---|---|---|
| Inclusive | $> 0$ | $\approx 1$ | High |
| Extractive | $\lesseqgtr 0$ | $\approx 1$ | Low |

**Key Insight.** High coherence ($K \approx 1$) does not imply high quality ($Q$). A society can be stably trapped in a bad equilibrium. This resolves the puzzle of persistent underdevelopment: extractive equilibria are self-reinforcing.

**Critical Junctures.** Major shocks can shift $\Psi$ between equilibria. The framework provides language for analyzing institutional transitions.

## 12.7 Organizational Economics: The Supermodularity Foundation

**Roberts Organizational Design.** Roberts (2004) models organizations as systems of complementary choices:

$$\Pi = \Pi(\text{Strategy, Structure, Processes, People, Culture, Incentives}) \tag{130}$$

with $\partial^2\Pi/\partial x_i \partial x_j > 0$ (supermodular complements).

**Direct Correspondence.** Roberts' complementarity matrix *is* $C$:

$$C_{ij}^{Roberts} = \frac{\partial^2 \Pi}{\partial x_i \partial x_j} \tag{131}$$

Roberts' "fit" *is* $K$:

$$K = 1 \text{ at coherent configuration}, \quad K < 1 \text{ at misfit} \tag{132}$$

**Coherent Configurations.** Due to complementarity, the profit function has multiple local maxima:

| **Configuration** | **Strategy** | **Structure** | **Culture** | **Incentives** |
|---|---|---|---|---|
| Cost Leader | Efficiency | Hierarchy | Discipline | Output-based |
| Innovator | Differentiation | Flat | Creative | Risk-reward |
| Customer Intimate | Service | Teams | Responsive | Satisfaction |

Each is a peak ($K \approx 1$). Mixtures are valleys ($K < 1$).

**Non-Concavity.** Complementarity generates non-concave payoff landscapes. This explains why partial organizational change often fails: it lands in the valley between peaks.

## 12.8 Financial Economics: The Market Coherence Extension

**CAPM.** The Capital Asset Pricing Model relates expected returns to covariance:

$$\mathbb{E}[r_i] = r_f + \beta_i(\mathbb{E}[r_m] - r_f) \tag{133}$$

**Covariance as Complementarity.** In financial markets, the complementarity matrix is the return covariance matrix:

$$C_{ij}^{market} = \text{Cov}(r_i, r_j) \tag{134}$$

**Market Coherence.**

$$K_{market} = 1 - \frac{\|\Sigma_t - \Sigma^*\|_F}{\|\Sigma^*\|_F} \tag{135}$$

- $K_{market} \approx 1$: Correlations stable, diversification works, EMH approximately holds
- $K_{market} < 1$: Correlations break down, diversification fails, crisis regime

**Financial Crises as Coherence Collapse.** The 2008 crisis: $K_{market}$ dropped from $\approx 0.95$ to $\approx 0.3$. "All assets fell together"—correlations spiked, destroying diversification.

**Jensen's Alpha.** Systematic $\alpha \neq 0$ indicates belief-payoff divergence: agents' expectations don't match realized complementarities. This is market incoherence.

## 12.9 International Trade Theory: The Factor Endowment Extension

**Heckscher–Ohlin Model (Heckscher 1919; Ohlin 1933).** The foundational model of international trade predicts that countries export goods that use their abundant factor intensively. A capital-rich country exports capital-intensive goods; a labor-rich country exports labor-intensive goods.

**Standard Assumptions.**

1. Identical technology across countries
2. Identical preferences across countries
3. Perfect factor mobility within, zero between countries
4. No transport costs, perfect competition

**Translation to $(C, \Psi)$ Framework.** Factor endowment is a component of context:

$$\Psi_{factor} = (K/L) \quad \text{(Capital-Labor ratio)} \tag{136}$$

Heckscher–Ohlin assumes all other context is identical:

$$\Psi^A_{tech} = \Psi^B_{tech} \quad \text{(Same technology)} \tag{137}$$
$$\Psi^A_{inst} = \Psi^B_{inst} \quad \text{(Same institutions)} \tag{138}$$
$$\Psi^A_{norm} = \Psi^B_{norm} \quad \text{(Same preferences)} \tag{139}$$

Only $\Psi^A_{factor} \neq \Psi^B_{factor}$ varies.

**Production Complementarity.** The production function $Y = F(K, L)$ implies factor complementarity:

$$C_{KL}^{prod} = \frac{\partial^2 F}{\partial K \partial L} > 0 \tag{140}$$

Capital and labor are complements in production.

**Trade as Coherent Configuration.** Given different factor endowments, there exists a coherent global specialization:

$$C^*(\Psi_{factor}^{high\ K/L}) = \text{“Produce capital-intensive goods”} \tag{141}$$

$$C^*(\Psi_{factor}^{low\ K/L}) = \text{“Produce labor-intensive goods”} \tag{142}$$

Free trade equilibrium is the reference structure $C^*$ for the global system:

- $K^{global} \approx 1$: Coherent international division of labor
- $Q^{trade} > Q^{autarky}$: Gains from trade

**Heckscher–Ohlin as Limiting Case.**

$$\text{H-O} = (C, \Psi)\text{-Framework}\Big|_{\substack{C_{ij}^{social}=0 \\ \Psi=\Psi_{factor}\text{ only} \\ U=U^F\text{ only} \\ \Psi_{t+1}=\Psi_t}} \tag{143}$$

The restrictions are severe:

| H-O Assumption | Framework Translation | What Is Excluded |
|---|---|---|
| Identical preferences | $U = U^F$ only | $U^S$, $U^{IDN}$ |
| Identical technology | $\Psi_{tech}$ constant | Innovation, tech transfer |
| Perfect competition | $C_{ij}^{market} = 0$ | Market power, networks |
| No institutions | $\Psi_{inst}$ irrelevant | Trade regimes, contracts |
| Static | $\Psi_{t+1} = \Psi_t$ | Path dependence, dynamics |

**What Heckscher–Ohlin Cannot Explain.** The restrictions rule out:

- **Protectionist politics:** If $U = U^F$ only, tariffs are irrational. But $U^{IDN}$ (“Made in America”) can outweigh $U^F$ losses.
- **Endogenous factor accumulation:** South Korea transformed from labor-abundant (1960) to capital/skill-abundant (2020). H-O explains the cross-section, not the transformation.
- **Institutional comparative advantage:** Hall–Soskice show that trade patterns depend on $\Psi_{inst}$, not just $\Psi_{factor}$.

- **Trade regime dynamics:** WTO erosion, bilateral deals, trade wars— these are $\Psi_{inst}^{global}$ dynamics that H-O ignores.

**The Full Framework on Trade.** Relaxing H-O restrictions yields richer predictions:

**1. Identity Utility in Trade Policy.**

$$\Delta U^{total} = \Delta U^F + \omega_{IDN} \cdot \Delta U^{IDN} \tag{144}$$

Tariffs may reduce $U^F$ but increase $U^{IDN}$ ("protecting our industries"). For voters with high $\omega_{IDN}$, protectionism is rational.

**2. Endogenous Factor Accumulation.**

$$\Psi_{factor,t+1} = f(\Psi_{factor,t}, \text{Investment}, \text{Education}, \text{Policy}) \tag{145}$$

Countries can *change* their comparative advantage through policy. Development strategy matters.

**3. Institutional Complementarity in Trade.** Hall and Soskice (2001) show:

- LMEs (USA, UK): Comparative advantage in radical innovation
- CMEs (Germany, Japan): Comparative advantage in incremental innovation

This reflects $\Psi_{inst}$, not $\Psi_{factor}$.

**4. Context Hysteresis in Trade Wars.** The Trump tariff analysis (Section 11) shows:

$$\lim_{t\to\infty} \Psi_t \neq \Psi^{pre-tariff} \quad \text{even if tariffs removed} \tag{146}$$

Tariffs shift $\Psi_{norm}$ (free trade norm weakens), $\Psi_{inst}$ (WTO erodes), and $\Psi_{tech}$ (supply chains reconfigure). The path back is not the path forward.

**Empirical Implications.** The framework generates testable predictions beyond H-O:

1. Trade policy support correlates with $U^{IDN}$ measures, not just $U^F$ exposure
2. Countries with institutional complementarity trade more within type (LME-LME, CME-CME)
3. Trade shocks have persistent effects even after policy reversal (hysteresis tests)
4. Factor accumulation responds to institutional context, not just prices

**Policy Implications.**

- Compensating losers with $U^F$ transfers may fail if $U^{IDN}$ losses dominate
- Trade agreements must address identity concerns, not just efficiency gains

- "Adjustment assistance" should target context change, not just retraining
- Global trade governance requires attention to $K^{global}$, not just $Q^{global}$

## 12.10 Agency Theory: The Alignment Extension

**Jensen–Meckling.** Jensen and Meckling (1976) analyze principal-agent conflicts:

$$U_P = f(e) - w \qquad \text{(Principal wants effort)} \tag{147}$$

$$U_A = u(w) - c(e) \qquad \text{(Agent dislikes effort)} \tag{148}$$

**Agency Cost as Incoherence.** Ideal alignment: $C^*_{PA} > 0$ (principal and agent interests complementary). Actual alignment: $C^{actual}_{PA} < C^*_{PA}$ (misalignment).

$$K_{PA} = 1 - \frac{\|C^{actual}_{PA} - C^*_{PA}\|}{\|C^*_{PA}\|} \tag{149}$$

Agency costs measure the deviation from coherence.

**Governance as Coherence Restoration.** Corporate governance mechanisms (performance pay, monitoring, ownership concentration) aim to increase $K_{PA}$ toward 1.

## 12.11 Prospect Theory: The Loss Aversion Extension

**Kahneman–Tversky.** Kahneman and Tversky (1979) showed that losses loom larger than gains:

$$V = \sum_i \pi(p_i) \cdot v(x_i - r) \tag{150}$$

with $|v(-x)| > v(x)$ for $x > 0$.

**Integration in FEPSDE.** The framework incorporates loss aversion directly:

$$U = \sum_d \sum_t \delta^t_d [G_{d,t} - \lambda_d \cdot P_{d,t}] \tag{151}$$

where $\lambda_d > 1$ captures loss aversion.

The separate treatment of Gains and Pains in the 144-component structure embodies the Prospect Theory insight that positive and negative outcomes are processed differently.

## 12.12 Summary: The Theory Space

Each theory occupies a position in "theory space" defined by restrictions:

| **Theory** | $C_{ij}$ | $\Psi$ dynamics | $U^{IDN}$ | $\lambda$ | Level |
|---|---|---|---|---|---|
| Arrow–Debreu | $= 0$ | Exogenous | No | $= 1$ | Individual |
| Heckscher–Ohlin | $= 0$ | Exogenous | No | $= 1$ | International |
| Fehr–Schmidt | $\neq 0$ | Slow | No | $= 1$ | Individual |
| Akerlof–Kranton | $\neq 0$ | Slow | **Yes** | $= 1$ | Individual |
| Bénabou–Tirole | $\neq 0$ | Slow | **Yes** | $= 1$ | Individual |
| Aghion–Howitt | $\neq 0$ | **Endogenous** | No | $= 1$ | Macro |
| Acemoglu–Robinson | $\neq 0$ | **Endogenous** | No | $= 1$ | Nation |
| Roberts | $> 0$ | Slow | No | $= 1$ | Organization |
| CAPM | $= \text{Cov}$ | Exogenous | No | $= 1$ | Market |
| Jensen–Meckling | $\lesseqgtr 0$ | Slow | No | $= 1$ | Organization |
| Prospect Theory | $\neq 0$ | Endogenous | No | $> \mathbf{1}$ | Individual |
| **Full Framework** | Any | Endogenous | Yes | $> 1$ | Multi-level |

**The Unification Claim.** The $(C, \Psi)$ framework is not merely another theory alongside others. It is a *meta-framework* that:

1. Shows which restrictions generate which theory
2. Explains why different theories apply in different contexts
3. Reveals what each theory cannot explain (by identifying active restrictions)
4. Provides a common language across economics subdisciplines

Detailed derivations for each theory are in Appendix **??**.

# 13 Empirical and Policy Implications

The framework generates specific empirical predictions and policy recommendations at each level of analysis. This section develops these implications systematically (see Appendix S for falsifiable predictions and Appendix R for evaluation protocols), from individual interventions to global governance.

## 13.1 General Principles

**The Dual Policy Objective.** Policy has two distinct objectives:

1. **Equilibrium selection:** Shape context $\Psi$ to reach a $C^*$ with high $Q$
2. **Stability maintenance:** Keep $K$ close to 1 to realize the welfare potential

These objectives can conflict: moving to a better equilibrium requires transitional instability. The framework makes this tradeoff explicit and quantifiable.

**The Coherence-Quality Distinction.** A key policy insight: interventions can target $K$ (stability) or $Q$ (quality) independently.

- High $K$, low $Q$: Stable but bad equilibrium $\rightarrow$ need disruption
- Low $K$, high $Q$ potential: Transition phase $\rightarrow$ need support
- Low $K$, low $Q$: Crisis $\rightarrow$ need both stabilization and reform

**Estimation Strategy.** The complementarity matrix $C_{ij}$ can be estimated from:

- Behavioral elasticities (how does $i$'s behavior respond to $j$'s?)
- Conditional covariances (how do outcomes co-move?)
- Experimental variation (how do treatments affect cross-responses?)

Time-variation in $K_t$ identifies phases of structural stability and disruption.

## 13.2 Level 1: Individual — Coaching and Therapy

**Diagnostic Framework.** Individual coherence can be assessed across FEPSDE dimensions:

| Dimension | Coherence Indicator | Incoherence Signal |
|---|---|---|
| Financial | Spending aligns with stated goals | Debt despite income |
| Emotional | Feelings match expressed values | Chronic dissonance |
| Physical | Behavior matches health goals | Know but don't do |
| Social | Relationships match priorities | Isolation despite wanting connection |
| Digital | Usage matches intentions | Compulsive behavior |
| Ecological | Footprint matches values | Green guilt |

**Intervention Strategies. For low $K$ (incoherence):**

- Identify misaligned elements (values vs. behavior, goals vs. habits)
- Choose direction: adjust values to match behavior, or behavior to match values
- Implement gradually to avoid overwhelming change
- Build supporting structures (habits, environments, relationships)

**For high $K$, low $Q$ (stable but unsatisfying):**

- Recognize that current stability is a trap
- Accept temporary incoherence as necessary for change
- Build resources before disrupting (financial buffer, social support)
- Set clear target configuration

**Empirical Measures.**

- Life satisfaction surveys with dimension-specific questions
- Time-use diaries vs. stated priorities
- Financial behavior vs. financial goals
- Health metrics vs. health values

**Applications.**

- Career counseling: Match job characteristics to life configuration
- Health coaching: Align habits with health goals
- Financial planning: Align spending with values
- Therapy: Resolve identity conflicts (IDN coherence)

**Behavioral Interventions: Nudges and Defaults.** Present-biased agents ($\beta < 1$) systematically deviate from their own long-run preferences (Laibson 1997; O'Donoghue and Rabin 1999). This creates scope for welfare-improving interventions that don't restrict choice:

**Nudges** (Thaler and Sunstein 2008) reshape choice architecture:

- Default enrollment in retirement savings: Madrian and Shea (2001) showed participation rises from 49% to 86% with automatic enrollment
- Chetty et al. (2014) found 85% of savers are "passive" (follow defaults)—policy should focus on default design, not incentives
- Commitment devices for present-biased agents (Ashraf et al. 2006; Bryan et al. 2010)

Framework interpretation: Nudges work by aligning $\Psi_I$ (institutional defaults) with agents' long-run preferences, overcoming $\Psi_T$ limitations (present bias).

## 13.3 Level 2: Household — Family Policy and Support

**Diagnostic Framework.** Household coherence requires alignment between members' expectations, resource allocation, and role definitions.

| Element | Coherence | Incoherence |
|---|---|---|
| Role expectations | Clear, shared, enacted | Conflicting, implicit, violated |
| Resource allocation | Matches priorities | Conflicts with stated values |
| Decision-making | Process matches beliefs | Egalitarian talk, unilateral action |
| Time allocation | Reflects priorities | Chronic time scarcity in priority areas |

**Policy Implications. Family support policy should:**

- Enable coherent configurations, not mandate specific ones
- Provide flexibility for both traditional and egalitarian households
- Reduce costs of role transitions (parental leave, childcare, eldercare)
- Support explicit negotiation of household arrangements

**Specific recommendations:**

- **Parental leave:** Flexible allocation between partners
- **Childcare:** Availability that enables both configurations
- **Tax policy:** Neutral between single and dual earner households
- **Work arrangements:** Part-time and remote options for coherence

**Empirical Measures.**

- Household satisfaction surveys (by member)
- Time-use studies by gender and role
- Conflict frequency and resolution patterns
- Role expectation surveys vs. actual allocation

## 13.4 Level 3: Organization — Management and Transformation

**Diagnostic Framework.** Organizational coherence ("fit") can be measured across Roberts' dimensions:

| Element | Coherent | Incoherent |
|---|---|---|
| Strategy–Structure | Innovation + flat / Efficiency + hierarchy | Innovation + hierarchy |
| Structure–Incentives | Flat + team rewards / Hierarchy + individual | Mixed signals |
| Incentives–Culture | Risk rewards + experimental / Safe + disciplined | Reward risk, punish failure |
| Culture–People | Creative + generalists / Efficient + specialists | Wrong people for culture |

**Policy Implications for Managers. Transformation strategy:**

1. **Assess current $K$:** How coherent is the current configuration?
2. **Choose target:** Which coherent configuration fits the market?
3. **Plan the valley:** Accept 2-4 years of $K < 1$
4. **Move everything together:** Partial change is worse than none
5. **Support the transition:** Extra resources, communication, patience

**Common mistakes:**

- Announce new strategy without changing structure
- Change incentives without changing culture
- Hire new people without adapting processes
- Expect quick results during transformation

**Empirical Measures.**

- Employee surveys on strategy-structure alignment
- Incentive analysis vs. stated strategy
- Culture audits vs. espoused values
- Performance metrics during transformation ($Q$ trajectory)

**Applications.**

- M&A integration: Assess configuration compatibility
- Digital transformation: Full-stack change required
- Turnarounds: Accept temporary instability
- Startup scaling: Maintain coherence while growing

## 13.5 Level 4: Region — Economic Development Policy

**Diagnostic Framework.** Regional coherence requires alignment between economic structure, labor market, education system, infrastructure, and culture.

| Configuration | Coherent Elements | Typical Regions |
|---|---|---|
| High-Tech Hub | STEM education, venture capital, risk culture | Silicon Valley, Shenzhen |
| Industrial | Vocational training, logistics, precision culture | Bavaria, Northern Italy |
| Financial Center | Business education, connectivity, trust | Zurich, Singapore |
| Tourism | Hospitality training, access, service culture | Alpine regions, islands |

**Policy Implications. Development strategy:**

1. **Diagnose current configuration:** What is the region good at?
2. **Assess feasibility of transformation:** How far is the target?
3. **If close:** Strengthen existing configuration
4. **If far:** Accept decade-long transformation or manage decline
5. **Avoid partial change:** High-tech strategy without STEM education fails

**The Rust Belt Lesson:**

- Old industrial regions often have $K \approx 1$ (coherent) but $Q$ low (obsolete)

- Transformation requires changing education, infrastructure, culture *simultaneously*
- Partial measures (tax breaks for tech companies) fail without ecosystem change
- Sometimes managing decline is more honest than promising transformation

**Empirical Measures.**

- Industry-education alignment (job vacancies vs. graduate skills)
- Infrastructure adequacy for target industries
- Cultural surveys (risk tolerance, entrepreneurship attitudes)
- Migration patterns (brain drain as coherence signal)

**Applications.**

- Regional development funds: Require coherent plans, not wish lists
- Education policy: Align with realistic economic trajectory
- Infrastructure investment: Match to target configuration
- Immigration policy: Attract skills that complement existing strengths

**Place-Based Policies: When and Why.** Kline and Moretti (2014) analyze when place-based policies (subsidizing locations rather than people) are justified:

- **Multiple equilibria:** If $C^*_{high}$ and $C^*_{low}$ both exist, policy can shift regions to better equilibrium
- **Agglomeration externalities:** Private location choices don't internalize spillovers (Krugman 1991; Glaeser 2008)
- **Mobility frictions:** When people can't move, help must come to them

**Spatial misallocation:** Hsieh and Moretti (2019) estimate US GDP is 36% lower due to housing constraints preventing workers from moving to productive cities. This is $\Psi_P$ (spatial context) constraining efficient allocation.

Framework interpretation: Place-based policy is $\Psi_P$ design—changing the spatial context to enable better regional $C^*$.

## 13.6 Level 5: Nation — Institutional Design and Reform

**Diagnostic Framework.** National coherence requires alignment between institutions, legal system, norms, political system, and social contract.

| Configuration | Key Elements | Examples |
|---|---|---|
| Liberal Market | Property rights, contract law, self-reliance | USA, UK |
| Coordinated Market | Corporatism, labor law, cooperation norms | Germany, Switzerland |
| State Capitalist | Party control, strategic direction, loyalty | China, Singapore |

Each can be coherent ($K \approx 1$). The question is which has higher $Q$ in context.

**Policy Implications. Institutional reform strategy:**

1. **Recognize path dependence:** History constrains options
2. **Assess current coherence:** Is the problem $K$ or $Q$?
3. **If low $K$:** Prioritize institutional consistency
4. **If low $Q$:** Accept that reform requires coherence costs
5. **Use external anchors:** EU accession, IMF programs, trade agreements

**The Estonia vs. Ukraine Lesson:**

- Both started with low $K$, low $Q$ (post-Soviet collapse)
- Estonia: Radical reform, EU anchor, accepted 5 years of low $K \to$ high $Q$
- Ukraine: Partial reform, no anchor, permanent low $K \to$ low $Q$
- Key insight: Partial reform can be worse than no reform

**Empirical Measures.**

- Institutional quality indices (World Bank Governance Indicators)
- Rule of law measures (contract enforcement, property rights)
- Trust surveys (generalized trust, institutional trust)
- Corruption perceptions (as coherence failure)
- FEPSDE welfare indicators across all dimensions

**Applications.**

- Constitutional design: Ensure internal consistency
- Anti-corruption: Address as coherence problem, not just enforcement
- EU accession: Use as external coherence anchor
- Development aid: Condition on coherent reform packages

## 13.7 Level 6: Global — International Cooperation and Governance

**Diagnostic Framework.** Global coherence requires alignment between trade, climate, security, migration, and financial regimes.

| Configuration | Key Elements | Historical Example |
|---|---|---|
| Multilateral | WTO, Paris, UN, open migration, IMF | Post-WWII order |
| Hegemonic | Single power sets rules | Pax Britannica, Pax Americana |
| Multipolar | Blocs, regional rules, competition | Pre-WWI, emerging? |

**The Climate Coordination Problem.** Climate change illustrates global incoherence:

- **Strategy** ($s \approx 0.6$): Rhetoric of cooperation, Paris commitments
- **Structure** ($\sigma \approx 0.3$): Reality of national competition, weak enforcement
- **Coherence** $K \ll 1$: Goals don't match institutions

**Policy Implications. For global coordination:**

1. **Accept structure constraint:** No world government $\Rightarrow$ limited $\Psi$ control
2. **Build coalitions:** Clubs of willing countries can create local coherence
3. **Use trade leverage:** Carbon border adjustments create incentives
4. **Technology solutions:** Reduce need for coordination by reducing costs
5. **Wait for crisis:** Major shocks can shift global $\Psi$

**Realistic expectations:**

- Full global coherence ($K \approx 1$) is unlikely without major crisis or hegemon
- Partial coherence (regional blocs, issue-specific regimes) is achievable
- National $K$ and global $K$ can conflict (nationally coherent, globally incoherent)

**Empirical Measures.**

- Treaty compliance rates
- Trade policy alignment vs. agreements
- Climate commitment vs. emissions trajectories
- Coordination indices for specific issue areas

**Applications.**

- Climate negotiations: Focus on structure, not just targets
- Trade agreements: Build coherent packages, not piecemeal deals
- Security architecture: Match commitments to capabilities
- Migration: Align rhetoric with actual policy capacity

## 13.8 Cross-Level Interactions

**Coherence Spillovers.** Incoherence at one level affects others:

- Incoherent individuals → family stress → organizational absenteeism
- Incoherent organizations → regional decline → national problems
- National incoherence → global coordination failure

**Policy Coordination.** Effective policy requires coherence *across* levels:

- Family policy must align with labor market institutions
- Education policy must align with regional economic strategy
- National institutions must align with international commitments

**The Multi-Level Welfare Function.** Total welfare aggregates across levels:

$$W_{total} = \sum_{l \in \{Ind, HH, Org, Reg, Nat, Glob\}} \omega_l \cdot W_l(K_l, Q_l) \tag{152}$$

Policy optimization requires considering:

- Level-specific coherence ($K_l$)
- Level-specific quality ($Q_l$)
- Cross-level complementarities
- Distributional effects within levels

## 13.9 Summary: The Policy Matrix

| Level | Key Actors | $K$ Tools | $Q$ Tools |
|---|---|---|---|
| Individual | Coach, therapist | Habit design, environment | Value clarification |
| Household | Family, counselor | Role negotiation, routines | Communication, support |
| Organization | Management | Structure, incentives | Strategy, culture |
| Region | Government, industry | Education, infrastructure | Cluster policy |
| Nation | Parliament, courts | Institutions, enforcement | Constitutional design |
| Global | UN, WTO, coalitions | Treaties, sanctions | Crisis, hegemony |

## 13.10 Implementation: The Technological Infrastructure

The policy framework outlined above is comprehensive but complex. Implementing it requires tools that didn't exist until recently.

**Why Previous Implementation Failed.** Multi-dimensional policy analysis was attempted before:

- OECD Better Life Index (2011): 11 dimensions, but no complementarity structure
- Human Development Index (1990): 3 dimensions, but no dynamics
- Happiness economics (2000s): Single outcome, ignores configuration

These efforts stalled because:

- Measuring multiple dimensions was expensive
- Analyzing interactions was computationally prohibitive
- Communicating complex tradeoffs to policymakers was impractical
- Real-time monitoring was impossible

**The 2024-2026 Infrastructure. 1. Real-Time Coherence Dashboards.** Policy makers can now access dashboards showing:

- Current $K$ estimates by level and dimension
- Trends in $\Psi$ components (trust, norms, institutional quality)
- Early warning signals of coherence breakdown
- Scenario simulations for proposed interventions

These dashboards aggregate diverse data streams (economic indicators, social media sentiment, institutional indices) and compute coherence metrics automatically.

**2. Policy Simulation Environments.** Before implementing a policy, analysts can simulate its effects:

> "Simulate the effect of a 10% tariff on Chinese goods over 5 years. Show effects on: (1) GDP ($U^F$), (2) manufacturing employment ($U^F$, regional), (3) consumer prices ($U^F$, short-term), (4) trade relationship trust ($\Psi_{inst}$), (5) nationalist sentiment ($U^{IDN}$). Include confidence intervals."

The simulation incorporates:

- Estimated complementarity matrices
- Context dynamics with realistic speeds
- Heterogeneous agent responses
- Feedback loops and second-order effects

**3. Natural Language Policy Interfaces.** Complex frameworks become accessible through conversation:

- "What is the coherence index for the German labor market?"
- "Why did Brexit reduce EU institutional coherence?"
- "What interventions would increase $K$ for rural communities?"
- "Compare the FEPSDE profiles of Nordic vs. Anglo-Saxon welfare states."

This democratizes access to sophisticated analysis.

**4. Automated Monitoring and Alerts.** Systems can monitor context continuously and alert when:

- $K$ falls below threshold (coherence crisis)
- $\Psi$ components diverge (cultural lag)
- Cross-level conflicts emerge
- Tipping points approach

**5. Multi-Stakeholder Scenario Planning.** Different groups can explore the same model with different weights:

- Business: Emphasize $U^F$, $U^D$
- Labor: Emphasize $U^S$, $U^P$, job security

- Environmentalists: Emphasize $U^{Eco}$
- Citizens: Deliberate on appropriate weights

The framework makes tradeoffs explicit without predetermining outcomes.

**The Policy Analysis Pipeline.** A complete policy analysis now follows this workflow:

1. **Data Integration:** Aggregate relevant data streams
2. **Context Estimation:** Use NLP to measure $\Psi$ components
3. **Complementarity Estimation:** Estimate $C$ from behavioral data
4. **Coherence Calculation:** Compute $K$ and $Q$
5. **Scenario Simulation:** Model intervention effects
6. **Stakeholder Communication:** Present findings in accessible format
7. **Monitoring:** Track outcomes against predictions

Each step is now automated or semi-automated. A policy analysis that would have taken months now takes days.

**Example: Climate Policy Coherence.** A climate ministry requests analysis:

> "Assess the coherence of our 2030 emissions targets with: current industrial structure, public opinion, technological readiness, international commitments, and political feasibility. Identify the binding constraints and recommend sequencing."

The system:

1. Extracts context from policy documents, surveys, and economic data
2. Estimates complementarities between climate action and other policy goals
3. Computes coherence indices for each dimension
4. Identifies misalignments (e.g., targets vs. public acceptance)
5. Simulates different sequencing strategies
6. Recommends: "Focus first on norm change ($\Psi_{norm}$) through communication campaign. Delay aggressive targets until $K_{public} > 0.7$."

This analysis is now feasible in a week, not a year.

**The Transformation.** The $(C, \Psi)$ framework is not an academic exercise—it is an *operational tool* for policy analysis, enabled by technological infrastructure that has matured only in the past 2-3 years.

For a comprehensive history of how computational capabilities evolved to make this framework feasible, see Appendix **??**.

Detailed applications and case studies are provided in Appendix **??**.

# 14 Limitations and Future Work

Every theoretical framework involves tradeoffs between generality, precision, and realism. This section provides an honest assessment of where the $(C, \Psi)$ framework falls short, what it cannot do, and where future work is needed.

## 14.1 Theoretical Limitations

### 14.1.1 The Contraction Assumption

**The Problem.** The existence and uniqueness of $C^*$ (Theorem 6.7) depends on the behavioral response operator $\Phi$ being a contraction:

$$\|\Phi(C_1, \Psi) - \Phi(C_2, \Psi)\|_F \leq \gamma \|C_1 - C_2\|_F, \quad \gamma < 1 \tag{153}$$

This assumption *fails* precisely when the framework is most needed:

- Financial panics (beliefs about others' beliefs spiral)
- Bank runs (withdrawal expectations self-fulfill explosively)
- Revolutions (coordination on regime change)
- Hyperinflation (price expectations accelerate)

**What This Means.** The framework describes stability and gradual change well. It has limited purchase on *explosive* dynamics— the moments when systems jump between equilibria.

**Possible Resolution.** Future work could:

- Develop regime-switching models where $\gamma < 1$ in normal times, $\gamma > 1$ during crises
- Use catastrophe theory to model discontinuous transitions
- Integrate agent-based modeling (Tesfatsion and Judd 2006) for micro-foundations of explosive dynamics

### 14.1.2 Linearity of Context Dynamics

**The Problem.** The context update equation

$$\Psi_{t+1} = \Psi_t + \eta \cdot \Delta a_t \tag{154}$$

is a first-order linear approximation. Real institutional change exhibits:

- **Thresholds:** Change only after critical mass (S-curve dynamics)

- **Hysteresis:** Path back differs from path forward
- **Irreversibility:** Some changes cannot be undone (extinctions, lost knowledge)
- **Discontinuities:** Sudden jumps (revolutions, technological breakthroughs)

**What This Means.** The linear model captures tendencies but misses the nonlinear phenomena that often matter most for policy.

**Possible Resolution.** Section 9 sketched nonlinear extensions (S-curve, punctuated equilibrium). Future work should:

- Specify functional forms for different context types
- Empirically estimate adjustment speeds and threshold parameters
- Develop simulation tools for nonlinear context dynamics

#### 14.1.3 The Aggregation Problem

**The Problem.** The welfare function aggregates across agents:

$$Q = \sum_i \sum_d \sum_t \omega_i \delta_d^t [G_{i,d,t} - \lambda_d P_{i,d,t}] \tag{155}$$

This utilitarian aggregation obscures:

- **Distribution:** Two equilibria with identical $Q$ may differ dramatically in how welfare is distributed
- **Power:** Who determines the weights $\omega_i$?
- **Capabilities:** Are people able to convert resources into welfare equally?

**What This Means.** The framework can compare equilibria by aggregate welfare but cannot resolve debates about justice, equality, or rights.

**Possible Resolution.** Future work could:

- Incorporate Rawlsian maximin (maximize welfare of worst-off)
- Add distributional constraints ($Q$ subject to Gini $< \bar{G}$)
- Develop capability-adjusted welfare measures following Sen
- Separate efficiency from equity analysis

### 14.1.4 No Theory of Equilibrium Selection

**The Problem.** The framework identifies that multiple equilibria exist ($C_A^*$, $C_B^*$, ...) but provides no theory of *which* equilibrium will be selected.

Selection depends on:

- History and initial conditions
- Focal points and coordination devices
- Power and political conflict
- Chance events and path dependence

None of these are modeled.

**What This Means.** The framework can say "these are the possible coherent configurations" but not "this is the configuration that will emerge."

**Possible Resolution.** Future work could:

- Integrate evolutionary game theory for selection dynamics
- Add political economy models for institutional choice
- Develop historical case studies of equilibrium selection
- Use agent-based models to simulate selection processes

## 14.2 Empirical Limitations

### 14.2.1 Identification of Complementarities

**The Problem.** The complementarity matrix $C$ consists of second derivatives of *unobserved* utility functions:

$$C_{ij} = \frac{\partial^2 U_i}{\partial x_i \partial x_j} \tag{156}$$

Utility is not directly observable. Identification requires:

- Strong functional form assumptions
- Exogenous variation in $x_j$ to estimate response of $x_i$
- Panel data to separate preference heterogeneity from complementarity

**What This Means.** Empirically estimating $C$ is extremely difficult. Most empirical work will rely on proxies and reduced-form evidence.

**Possible Resolution.**

- Use behavioral experiments to estimate $C$ in controlled settings
- Exploit natural experiments (policy changes, shocks) for identification
- Develop structural estimation methods with credible instruments
- Accept that some applications will be qualitative rather than quantitative

### 14.2.2 Parameter Proliferation

**The Problem.** The full FEPSDE model has:

- 144 utility components
- 6 dimension-specific discount rates $\delta_d$
- 6 dimension-specific loss aversion parameters $\lambda_d$
- 3 category weights $\omega_{INU}, \omega_{KNU}, \omega_{IDN}$
- Plus the full $144 \times 144$ complementarity matrix

This is an enormous parameter space.

**What This Means.**

- The full model cannot be estimated without heroic assumptions
- Risk of overfitting: with enough parameters, any data can be explained
- Different parameter choices can generate opposite predictions

**Possible Resolution.**

- Use simplified versions for specific applications (e.g., 2-dimension models)
- Impose theoretically motivated restrictions (e.g., symmetric $C$)
- Calibrate parameters from existing literature rather than estimating all
- Focus on qualitative predictions that are robust to parameter values
- Use Bayesian methods with informative priors

### 14.2.3 No Empirical Validation Yet

**The Problem.** This paper is theoretical. No empirical tests have been conducted. The predictions in Section 11 remain untested.

**What This Means.** The framework's empirical relevance is asserted, not demonstrated. It could be that real-world data reject the predictions.

**Possible Resolution.** Future work must:

- Test cross-domain spillover predictions using financial/political data
- Estimate asymmetric recovery dynamics from crisis data
- Validate innovation-coherence tradeoff with technology adoption data
- Conduct experimental tests of identity-economics predictions

#### 14.2.4 WEIRD Bias

**The Problem.** The empirical foundations (Section 4) draw primarily on:

- Laboratory experiments with university students
- Data from Western, Educated, Industrialized, Rich, Democratic populations
- English-language academic literature

Henrich et al. (2010) showed that WEIRD populations are often outliers, not representative of human psychology globally.

**What This Means.**

- Parameters like $\lambda$ (loss aversion) may vary cross-culturally
- The relative importance of FEPSDE dimensions may differ
- Collectivist societies may weight $KNU$ higher than individualist ones
- Some mechanisms may be culture-specific

**Possible Resolution.**

- Replicate key experiments in non-WEIRD populations
- Develop culture-specific parameterizations
- Test whether framework structure holds even if parameters differ
- Collaborate with researchers from diverse contexts

## 14.3 Conceptual Limitations

### 14.3.1 Normative Neutrality

**The Problem.** The framework ranks equilibria by aggregate FEPSDE welfare, but the weights are not determined by the framework itself.

- How much weight on Financial vs. Ecological?
- How much to discount the future?
- Whose utility counts (citizens, residents, future generations)?
- How to weight INU vs. KNU vs. IDN?

These are political and ethical choices, not scientific ones.

**What This Means.** The framework cannot resolve normative disagreements. It can clarify what is at stake but not determine the right answer.

**Our Position.** This is a feature, not a bug. A scientific framework should not dictate values. Its role is to:

- Make tradeoffs explicit
- Show consequences of different weight choices
- Provide vocabulary for democratic deliberation
- Let societies choose their own weights

### 14.3.2 Micro-Foundations

**The Problem.** The framework operates at the system level (complementarity structures, coherence). It does not provide micro-foundations for:

- How individuals form expectations
- How agents learn about $C^*$
- How coordination happens in practice
- What cognitive processes underlie coherence-seeking

**What This Means.** The framework describes equilibrium properties but not the process by which equilibria are reached. It is a "as if" model, not a literal description of individual cognition.

**Possible Resolution.**

- Integrate with learning models (Bayesian updating, reinforcement learning)
- Develop agent-based micro-foundations
- Connect to cognitive science and neuroscience of social coordination

### 14.3.3 Static vs. Dynamic Analysis

**The Problem.** Most of the formal analysis is comparative statics: comparing equilibria, not analyzing transitions between them.

But policy-relevant questions are often dynamic:

- How long does transition take?
- What is the path of $K$ during transformation?
- When do systems get stuck in transition?

**Possible Resolution.**

- Develop explicit dynamic models of transition
- Simulate adjustment paths with different policies
- Study historical transitions to calibrate adjustment speeds

## 14.4 What the Framework Cannot Do

To be explicit about scope:

1. **Cannot predict which equilibrium will be selected.** It identifies possible equilibria but not which will emerge.
2. **Cannot resolve normative disagreements.** It clarifies tradeoffs but cannot say what is right.
3. **Cannot model explosive dynamics.** It describes stability, not the moments when stability breaks down.
4. **Cannot explain individual behavior in detail.** It models system-level coherence, not individual psychology.
5. **Cannot be fully empirically validated with current data.** Many predictions require data that doesn't yet exist.
6. **Cannot capture everything.** 144 components is already complex; reality is more complex still.

## 14.5 Epistemic Honesty: Integration vs. Prediction

A critical distinction must be made explicit.

**The Honest Claim.** This framework is an *integrative schema*, not a *predictive theory* that anticipated the contributions it encompasses.

Appendix **??** shows how 56 years of Nobel Prize research maps onto the framework. But this mapping was constructed *ex post*—after the contributions were made. The framework did not predict Kahneman's loss aversion, Ostrom's polycentric governance, or Romer's endogenous growth. These were genuine scientific discoveries that *extended* economic understanding.

**What Would Be Dishonest.** It would be intellectually dishonest to claim that:

- The framework "contains" Prospect Theory (Kahneman 2002)
- The framework "anticipated" commons governance (Ostrom 2009)
- The framework "predicted" endogenous growth (Romer 2018)
- The framework "implied" institutional causality (Acemoglu 2024)

Each of these was a genuine discovery that required creativity, empirical work, and theoretical innovation. The framework was *built to accommodate them*, not the reverse.

**Genuine Extensions in the Nobel Record.** Several Nobel contributions represent *true extensions* that expanded economic understanding in ways no prior framework anticipated:

| **Laureate** | **Discovery** | **Framework Adaptation** |
|---|---|---|
| Kahneman (2002) | Loss aversion, reference points | Added $\lambda_d > 1$, $\Psi_{frame}$ |
| Simon (1978) | Bounded rationality | $C^{perceived} \neq C^{actual}$ |
| Ostrom (2009) | Polycentric governance | $\Psi_{inst}^{community}$ as third way |
| Thaler (2017) | Choice architecture | $\Psi_{architecture}$ as policy tool |
| Romer (2018) | Endogenous technology | $\Psi_{tech,t+1} = f(\Psi_{tech,t}, R\&D)$ |
| AJR (2024) | Causal identification | Empirical test of $\Psi_{inst} \to Q$ |

These were not "special cases" of a pre-existing theory. They were *expansions of knowledge* that the framework subsequently incorporated.

**What the Framework Actually Does.** The honest characterization is that the framework:

1. **Provides common language:** Different contributions can be compared using $(C, \Psi, C^*, K, Q)$
2. **Makes assumptions explicit:** Each theory's restrictions become visible
3. **Identifies connections:** Links between disparate literatures emerge
4. **Reveals gaps:** What has *not* been studied becomes clear
5. **Suggests extensions:** Where might the next discoveries lie?

**What the Framework Does NOT Do.**

1. **Does not replace creativity:** Genuine discoveries require insight beyond any schema
2. **Does not predict future Nobels:** We cannot say what will be discovered next
3. **Does not make prior work redundant:** Kahneman's experiments, Ostrom's fieldwork, Romer's mathematics remain essential
4. **Does not guarantee correctness:** The framework may need revision when future discoveries don't fit

**The Retrospective Fitting Problem.** Any sufficiently flexible framework can accommodate past findings. The test is whether it generates *novel, falsifiable predictions* that would not have been made without it.

Section 11 attempts this. But honesty requires acknowledging:

- The predictions have not yet been tested
- Some predictions may be too vague to falsify
- Future research may reveal predictions that fail

**Why This Matters.** Science progresses through genuine discovery, not retrospective categorization. The framework's value lies in facilitating communication, identifying gaps, and suggesting directions—not in claiming to have "known all along."

The 56 Nobel laureates in Appendix **??** made genuine contributions. The framework honors them by providing a structure for understanding how their insights connect—not by claiming credit for their creativity.

**A Constructive Stance.** Rather than asking "Does this framework contain all prior knowledge?" (to which the honest answer is "no—it was built to integrate it"), the productive question is:

> *"What does this framework suggest we should study next?"*

Some candidates:

- Cross-level coherence conflicts (individual vs. collective optimization)
- Digital context ($\Psi_{digital}$) and its rapid evolution
- Ecological-economic complementarities under climate change
- Identity formation and its interaction with institutional change

These are not "implied by" the framework in any predictive sense. They are *suggested by* the framework's structure as potentially important.

Whether they yield genuine insights is an empirical question that only future research can answer.

**Retrospective vs. Prospective Application.** A critical distinction clarifies the framework's epistemic status:

**Retrospective (1969–2025):** The framework was constructed *after* the Nobel contributions were made. Showing that Kahneman, Ostrom, or Romer "fit" the framework is *integration*, not *prediction*. This is valuable for understanding connections but does not demonstrate predictive power.

**Prospective (2026 onward):** The framework now exists as a complete structure. Any new paper published tomorrow can be immediately analyzed:

- Which $C_{ij}$ does it address?
- Which $\Psi$ components are relevant?
- What restrictions does it assume?
- What would the full framework add?
- How does it connect to existing literature?

This prospective capability is the framework's *operational value*. It provides a systematic protocol for integrating new research as it appears.

**The Difference:**

| | **Retrospective** | **Prospective** |
|---|---|---|
| Claim | "Framework integrates X" | "Framework classifies X" |
| Epistemic status | Post-hoc organization | Real-time analysis |
| Value | Understanding connections | Operational tool |
| Honesty required | "Built to fit" | "Applies to new cases" |

The 56 Nobel laureates in Appendix **??** demonstrate retrospective integration. The framework's prospective value will be demonstrated by its ability to usefully classify research that does not yet exist.

**A Testable Claim.** Here is a falsifiable prediction about the framework itself:

> *Any economics paper published in 2026 or later can be characterized in terms of* $(C, \Psi, C^*, K, Q)$*, with explicit identification of which restrictions it assumes and which components it addresses.*

If a paper appears that genuinely cannot be characterized this way— not because it extends the framework (which is valuable) but because it contradicts the framework's basic structure—then the framework requires revision.

This is the standard any unified framework must meet: not that it predicted everything, but that it can integrate anything— or honestly acknowledge when it cannot.

## 14.6 What the Framework Can Do

Despite these limitations:

1. **Provides unified language.** Economists, political scientists, psychologists, and organizational scholars can use $(C, \Psi, K, Q)$ to communicate across disciplines.

2. **Generates novel predictions.** Cross-domain spillovers, asymmetric recovery, identity-economics feedback— these predictions are unavailable from constituent theories.

3. **Clarifies policy tradeoffs.** The Trump tariff analysis shows how the framework illuminates tradeoffs that standard models miss.

4. **Explains puzzles.** Why do people support economically costly policies? Why do stable systems suddenly collapse? Why is reform so hard? The framework offers answers.

5. **Structures empirical research.** Even without full estimation, the framework suggests what to measure and what relationships to test.

6. **Enables better questions.** "What is the coherence index?" is more productive than "Is this policy efficient?" in many contexts.

## 14.7 Research Agenda

**Theoretical Extensions.**

- Nonlinear context dynamics with tipping points
- Equilibrium selection theory
- Dynamic transition analysis
- Multi-level coherence interactions

**Empirical Program.**

- Experimental estimation of complementarity parameters
- Cross-country validation of FEPSDE welfare measures
- Time-series tests of coherence spillover predictions
- Case studies of major transitions (EU integration, China's reform)

**Applied Work.**

- Trade policy analysis (beyond tariffs: supply chain resilience)
- Climate policy (global coherence problem)
- Digital transformation (innovation-coherence tradeoff)
- Institutional reform (coherence trap escape)

**Methodological Development.**

- Simulation tools for coherence dynamics
- Estimation methods for high-dimensional complementarity structures
- Survey instruments for FEPSDE measurement
- Case study protocols for qualitative application

The feasibility of this research agenda depends critically on the computational infrastructure described throughout this paper. The 60-year evolution from manual statistical analysis to AI-assisted economic modeling—documented in Appendix **??**—is what transforms these methodological goals from aspirations to achievable objectives.

## 14.8 Conclusion: The Value of Imperfect Frameworks

Every framework has limitations. The question is not whether limitations exist but whether the framework is useful despite them.

The $(C, \Psi)$ framework is useful because it:

- Asks questions that other frameworks don't ask
- Generates predictions that other frameworks don't generate
- Connects literatures that other frameworks don't connect
- Provides vocabulary that other frameworks don't provide

Its limitations are acknowledged. Its value lies in providing a coherent structure for asking better questions about stability, welfare, and institutional design— even when definitive answers remain elusive.

# 15 Conclusion

## 15.1 What We Have Done

This paper began with a puzzle: Why does economics remain fragmented into separate subdisciplines—classical, behavioral, institutional, organizational— each with its own assumptions, methods, and blind spots?

Our answer: The fragments are not competing theories but *limiting cases* of a more general structure. By identifying what each theory assumes away, we found what unifies them.

**The Core Innovation.** We introduced the complementarity-context framework $(C, \Psi)$:

- $C$: The complementarity matrix capturing how choices interact across agents and domains
- $\Psi \in \mathbb{R}^8$: The context vector decomposed into eight primitive dimensions
- $C^*(\Psi)$: The self-consistent reference structure—the configuration where beliefs and behavior mutually reinforce
- $K$: The coherence index measuring alignment between actual and reference structure
- $Q$: The quality index measuring welfare at equilibrium

**The Eight Context Dimensions.** A central methodological contribution is the operationalization of context through eight independently measurable dimensions:

| Symbol | Dimension | What It Captures | Dominant For |
|---|---|---|---|
| $\Psi_I$ | Institutional | Rules, enforcement, property rights | Democracy→Growth |
| $\Psi_S$ | Social | Trust, norms, social capital | Collective action |
| $\Psi_C$ | Cognitive | Information processing, biases | Individual decisions |
| $\Psi_K$ | Informational | Transparency, signal quality | Education→Earnings |
| $\Psi_E$ | Economic | Development, market thickness | Cash→Consumption |
| $\Psi_T$ | Temporal | Stability, time horizons | Patience→Growth |
| $\Psi_M$ | Market Scope | Geographic reach, openness | Trade effects |
| $\Psi_F$ | Factor Flexibility | Adjustment capacity, mobility | Management→Productivity |

These dimensions emerged not from armchair theorizing but from residual analysis: we asked what systematic variation remained after standard theories had been applied.

**Empirical Validation.** The framework is not merely theoretical. Across six canonical economic relationships, the eight $\Psi$ dimensions explain substantial heterogeneity:

| **Relationship** | $R^2$ | **Adj.** $R^2$ | **Dominant** $\Psi$ |
|---|---|---|---|
| Education $\rightarrow$ Earnings | 84.5% | 76.8% | $\Psi_K$, $\Psi_M$ |
| Management $\rightarrow$ Productivity | 78.9% | 68.3% | $\Psi_F$ |
| Democracy $\rightarrow$ Growth | 39.8% | 9.7% | $\Psi_I$, $\Psi_E$ |
| Patience $\rightarrow$ Growth | 67.3% | 51.0% | $\Psi_T$, $\Psi_K$ |
| Information $\rightarrow$ Behavior | 82.4% | 73.7% | $\Psi_K$ (–), $\Psi_F$ |
| Income $\rightarrow$ Health | 67.7% | 51.6% | $\Psi_T$, $\Psi_I$ |
| **Average** | **70.1%** | **55.2%** | — |

Three findings merit emphasis:

1. **Different dimensions dominate for different effects.** Factor flexibility ($\Psi_F$) matters most for management; information quality ($\Psi_K$) for education returns; temporal stability ($\Psi_T$) for patience effects. Context is not monolithic.

2. **Micro effects are better explained than macro effects.** Individual decisions show more systematic context-dependence ($R^2 \approx 75$–$85\%$) than aggregate outcomes ($R^2 \approx 40$–$70\%$). This is expected: aggregation introduces noise.

3. **No ninth dimension emerged.** We tested ethnic fractionalization, natural resources, and implementation capacity. None improved the model across all outcomes. Eight dimensions appear sufficient.

**The Framework as Metatheory.** Perhaps most importantly, the framework is proposed as a *metatheory*—not replacing existing theories but specifying *when* each applies:

- **Arrow-Debreu** applies when $C_{ij} \approx 0$ and $\Psi$ is stable
- **Behavioral economics** applies when $\Psi_C$ varies but institutions are fixed
- **Institutional economics** applies when $\Psi_I$ varies but individual behavior aggregates
- **The full framework** applies when multiple $\Psi$ dimensions vary with significant complementarity

Existing theories tell you *that* certain effects occur. Our framework tells you *when* they occur.

## 15.2 The Falsifiable Claim

We commit to a specific, testable proposition:

> **Core Claim:** *No economic effect is context-free.*

If any intervention produces identical effects across all $\Psi$ configurations—if education returns are the same in Switzerland and Nigeria, if information campaigns work equally in transparent and opaque environments, if management practices improve productivity identically with rigid and flexible labor markets—then our framework would face serious empirical challenges.

This is a strong hypothesis. Much of standard economics assumes context-independence (the "average treatment effect" literature). We hypothesize that context-dependence is universal and systematic.

The 70.1% average $R^2$ provides initial support. But the claim remains falsifiable: a single context-independent effect would refute it.

## 15.3 What This Changes

**For Economic Theory.** The division between "rational" and "behavioral" economics is artificial. All behavior is rational in the sense of coherence-seeking. The question is not whether agents maximize utility, but what enters their utility function and how choices interact.

Classical economics is not wrong—it is the special case where $C_{ij} = 0$. This case is useful for markets with minimal interdependence. But it cannot explain social preferences, coordination games, or institutional dynamics.

**For Empirical Research.** Treatment effect heterogeneity should not be treated as noise to be averaged away—it may contain signal to be explained. The eight $\Psi$ dimensions provide a systematic framework for understanding *why* effects vary across contexts.

The framework suggests a research program:

1. Measure $\Psi$ dimensions for the context of interest
2. Predict which dimensions will dominate for the effect being studied
3. Test whether heterogeneity follows the predicted pattern
4. Revise $\Psi$ measurement or framework structure based on results

This is more demanding than estimating average treatment effects, but more informative.

**For Policy Analysis.** Standard cost-benefit analysis considers only $U^F$ (financial utility). The FEPSDE framework shows this is radically incomplete. Policies affect all six dimensions, and tradeoffs between dimensions may be large.

More importantly, policy effects depend on context. An intervention that works in high-$\Psi_F$ environments may fail in low-$\Psi_F$ environments. Extrapolating from one context to another requires understanding *why* the original effect occurred.

## 15.4 Integration with Major Research Programs

The framework integrates with, rather than replaces, established research programs:

- **Heckman**: Selection and skill formation become context-dependent (Appendix Q)—the Heckman curve varies with $\Psi_I$ and $\Psi_{cultural}$ (Appendix Q)
- **Bloom**: Uncertainty effects depend on (Appendix R) $\Psi_F$ (flexibility) and $\Psi_T$ (stability)—firms respond to uncertainty differently in rigid vs. flexible environments (Appendix R)
- **Duflo**: RCT effects are conditional (Appendix S) on context—what works in one setting may not generalize, and $\Psi$ explains why (Appendix S)

Each research program provides evidence for context-dependence. The framework provides the structure to unify their findings.

## 15.5 Limitations Acknowledged

We have been explicit about what the framework cannot do:

- It cannot predict which equilibrium will be selected
- It cannot resolve normative disagreements about welfare weights
- It cannot model explosive dynamics when the contraction property fails
- Empirical validation, while promising (70.1% $R^2$), remains preliminary
- The eight dimensions may have WEIRD bias from their empirical foundations

These limitations are real. But every framework has limitations. The question is whether the framework is useful despite them— whether it asks better questions, generates new insights, and guides research productively.

The empirical validation suggests it does.

## 15.6 The Road Ahead

**Immediate Priorities.**

1. **Replication**: Test the 8-$\Psi$ model on additional outcomes beyond the six validated here
2. **Measurement refinement**: Develop better instruments for $\Psi_C$ (cognitive) and $\Psi_S$ (social)
3. **Causal identification**: Move from correlational evidence to causal tests using natural experiments in $\Psi$

**Theoretical Extension.**

- Nonlinear context dynamics with tipping points
- Explicit equilibrium selection mechanisms
- Dynamic transition analysis between $\Psi$ configurations
- Strategic context manipulation by large actors

**Policy Application.**

- Trade policy design that accounts for identity effects and $\Psi$-heterogeneity
- Climate governance as a global coherence problem across nations with different $\Psi$
- Development interventions calibrated to local $\Psi$ configurations
- Institutional reform that avoids coherence traps

## 15.7 Final Reflection

Economics began as moral philosophy—the study of how societies organize to meet human needs. Over time, it narrowed to the study of market efficiency. Behavioral economics recovered some of the lost terrain. Institutional economics recovered more.

This framework attempts to reunify the fragments. It says: Economics is the study of coherence— how beliefs, behaviors, and contexts align (or fail to align) to create stable patterns that determine human welfare.

The eight $\Psi$ dimensions operationalize "context" so it can be measured, tested, and used. The 70.1% explanatory power shows this operationalization has empirical bite. The falsifiable core claim—no effect is context-free—distinguishes the framework from unfalsifiable alternatives.

Efficiency is one aspect of coherence, the limiting case where choices don't interact. But most of economic life involves choices that do interact— creating complementarities, externalities, coordination problems, and social dilemmas.

Understanding these interactions requires a richer framework than "maximize utility subject to constraints." It requires understanding what enters utility (FEPSDE), how utilities interact ($C$), how context shapes interactions ($\Psi$), and how all three co-evolve.

The Complementarity-Context framework provides this richer structure. It is not complete. It is not final. But it is, we hope, a step toward a more unified and more useful economics— one that can address the great challenges of our time: climate change, inequality (Piketty 2014), institutional decay, technological disruption, and the search for coherent ways of living together on a finite planet.

*Rationality is not maximization. It is coherence.*
*Welfare is not wealth. It is flourishing across dimensions.*
*Context is not noise. It is the key to generalization.*

*And with eight measurable dimensions, we can finally put this vision to the test.*

# References


Acemoglu, D. and Robinson, J.A. (2012). *Why Nations Fail.* Crown Publishing.

Acemoglu, D. and Robinson, J.A. (2012). *Why Nations Fail: The Origins of Power, Prosperity, and Poverty.* Crown Business.

Aghion, P. and Howitt, P. (1998). *Endogenous Growth Theory.* MIT Press.

Akerlof, G.A. (1970). The Market for "Lemons": Quality Uncertainty and the Market Mechanism. *Quarterly Journal of Economics*, 84(3), 488–500.

Akerlof, G.A. and Kranton, R.E. (2000). Economics and Identity. *Quarterly Journal of Economics*, 115(3), 715–753.

Alesina, A. and La Ferrara, E. (2002). Who Trusts Others? *Journal of Public Economics*, 85(2), 207–234.

Almas, I., Cappelen, A.W., and Tungodden, B. (2020). Cutthroat Capitalism versus Cuddly Socialism: Are Americans More Meritocratic and Efficiency-seeking than Scandinavians? *Journal of Political Economy*, 128(5), 1753–1788.

Andreoni, J. (1988). Why Free Ride? Strategies and Learning in Public Goods Experiments. *Journal of Public Economics*, 37(3), 291–304.

Arrow, K.J. and Debreu, G. (1954). Existence of an Equilibrium for a Competitive Economy. *Econometrica*, 22(3), 265–290.

Arthur, W.B. (1997). *The Economy as an Evolving Complex System II.* Addison-Wesley.

Ashraf, N., Karlan, D., and Yin, W. (2006). Tying Odysseus to the Mast: Evidence from a Commitment Savings Product in the Philippines. *Quarterly Journal of Economics*, 121(2), 635–672.

Bak, P. (1996). *How Nature Works: The Science of Self-Organized Criticality.* Copernicus.

Bénabou, R. and Tirole, J. (2006). Incentives and Prosocial Behavior. *American Economic Review*, 96(5), 1652–1678.

Bénabou, R. and Tirole, J. (2006). Belief in a Just World and Redistributive Politics. *Quarterly Journal of Economics*, 121(2), 699–746.

Berg, J., Dickhaut, J., and McCabe, K. (1995). Trust, Reciprocity, and Social History. *Games and Economic Behavior*, 10(1), 122–142.

Bleakley, H. (2012). Health, Human Capital, and Development. *Annual Review of Economics*, 2(1), 283–310.

Bloom, N., Sadun, R., and Van Reenen, J. (2013). Does Management Matter? Evidence from India. *Quarterly Journal of Economics*, 128(1), 1–51.

Bohnet, I. and Zeckhauser, R. (2004). Trust, Risk and Betrayal. *Journal of Economic Behavior & Organization*, 55(4), 467–484.

Bolton, G.E. and Ockenfels, A. (2000). ERC: A Theory of Equity, Reciprocity, and Competition. *American Economic Review*, 90(1), 166–193.

Boyd, R. and Richerson, P.J. (2005). *The Origin and Evolution of Cultures*. Oxford University Press.

Bryan, G., Karlan, D., and Nelson, S. (2010). Commitment Devices. *Annual Review of Economics*, 2(1), 671–698.

Bulow, J.I., Geanakoplos, J.D., and Klemperer, P.D. (1985). Multimarket Oligopoly: Strategic Substitutes and Complements. *Journal of Political Economy*, 93(3), 488–511.

Camerer, C., Loewenstein, G., and Rabin, M. (Eds.) (2004). *Advances in Behavioral Economics*. Princeton University Press.

Cappelen, A.W., Hole, A.D., Sørensen, E.Ø., and Tungodden, B. (2007). The Pluralism of Fairness Ideals: An Experimental Approach. *American Economic Review*, 97(3), 818–827.

Charness, G. and Dufwenberg, M. (2006). Promises and Partnership. *Econometrica*, 74(6), 1579–1601.

Charness, G. and Rabin, M. (2002). Understanding Social Preferences with Simple Tests. *Quarterly Journal of Economics*, 117(3), 817–869.

Chetty, R., Hendren, N., Kline, P., and Saez, E. (2014). Where is the Land of Opportunity? The Geography of Intergenerational Mobility in the United States. *Quarterly Journal of Economics*, 129(4), 1553–1623.

Cohen, J.D., Ericson, K.M., Laibson, D., and White, J.M. (2020). Measuring Time Preferences. *Journal of Economic Literature*, 58(2), 299–347.

Combes, P.P. and Gobillon, L. (2015). The Empirics of Agglomeration Economies. *Handbook of Regional and Urban Economics*, 5, 247–348.

Davis, D.D. and Holt, C.A. (1993). *Experimental Economics*. Princeton University Press.

Duflo, E. (2012). Human Values and the Design of the Fight Against Poverty. Tanner Lectures on Human Values, Harvard University.

Dupas, P. (2011). Do Teenagers Respond to HIV Risk Information? Evidence from a Field Experiment in Kenya. *American Economic Journal: Applied Economics*, 3(1), 1–34.

Falk, A., Becker, A., Dohmen, T., Enke, B., Huffman, D., and Sunde, U. (2018). Global Evidence on Economic Preferences. *Quarterly Journal of Economics*, 133(4), 1645–1692.

Fama, E.F. (1970). Efficient Capital Markets: A Review of Theory and Empirical Work. *Journal of Finance*, 25(2), 383–417.

Fehr, E. and Charness, G. (2024). Social Preferences: Fundamental Characteristics and Economic Consequences. *Journal of Economic Literature* (forthcoming).

Fehr, E. and Fischbacher, U. (2004). Social Norms and Human Cooperation. *Trends in Cognitive Sciences*, 8(4), 185–190.

Fehr, E. and Gächter, S. (2000). Cooperation and Punishment in Public Goods Experiments. *American Economic Review*, 90(4), 980–994.

Fehr, E. and Schmidt, K.M. (1999). A Theory of Fairness, Competition, and Cooperation. *Quarterly Journal of Economics*, 114(3), 817–868.

Fehr, E. and Schmidt, K.M. (1999). A Theory of Fairness, Competition, and Cooperation. *Quarterly Journal of Economics*, 114(3), 817–868.

Frederick, S., Loewenstein, G., and O'Donoghue, T. (2002). Time Discounting and Time Preference: A Critical Review. *Journal of Economic Literature*, 40(2), 351–401.

Gintis, H. (2000). Strong Reciprocity and Human Sociality. *Journal of Theoretical Biology*, 206(2), 169–179.

Gintis, H. (2009). *The Bounds of Reason: Game Theory and the Unification of the Behavioral Sciences*. Princeton University Press.

Glaeser, E.L. (2008). *Cities, Agglomeration, and Spatial Equilibrium*. Oxford University Press.

Glaeser, E.L., Ponzetto, G.A.M., and Shleifer, A. (2009). Why Does Democracy Need Education? *Journal of Economic Growth*, 12(2), 77–99.

Greif, A. (2006). *Institutions and the Path to the Modern Economy*. Cambridge University Press.

Güth, W., Schmittberger, R., and Schwarze, B. (1982). An Experimental Analysis of Ultimatum Bargaining. *Journal of Economic Behavior & Organization*, 3(4), 367–388.

Hamilton, W.D. (1964). The Genetical Evolution of Social Behaviour I and II. *Journal of Theoretical Biology*, 7(1), 1–52.

Hanushek, E.A. and Woessmann, L. (2012). Do Better Schools Lead to More Growth? Cognitive Skills, Economic Outcomes, and Causation. *Journal of Economic Growth*, 17(4), 267–321.

Hofstede, G. (1980). *Culture's Consequences: International Differences in Work-Related Values*. Sage.

House, R.J., Hanges, P.J., Javidan, M., Dorfman, P.W., and Gupta, V. (2004). *Culture, Leadership, and Organizations: The GLOBE Study of 62 Societies*. Sage.

Hsieh, C.T. and Moretti, E. (2019). Housing Constraints and Spatial Misallocation. *American Economic Journal: Macroeconomics*, 11(2), 1–39.

Hurwicz, L. (1960). Optimality and Informational Efficiency in Resource Allocation Processes. In Arrow, Karlin, and Suppes (Eds.), *Mathematical Methods in the Social Sciences*. Stanford University Press.

Ichniowski, C., Shaw, K., and Prennushi, G. (1997). The Effects of Human Resource Management Practices on Productivity. *American Economic Review*, 87(3), 291–313.

Inglehart, R. (2014). *Cultural Evolution: People's Motivations are Changing, and Reshaping the World*. Cambridge University Press.

Jensen, R. (2007). The Digital Provide: Information (Technology), Market Performance, and Welfare in the South Indian Fisheries Sector. *Quarterly Journal of Economics*, 122(3), 879–924.

Kahneman, D. and Tversky, A. (1979). Prospect Theory: An Analysis of Decision under Risk. *Econometrica*, 47(2), 263–291.

Kline, P. and Moretti, E. (2014). People, Places, and Public Policy: Some Simple Welfare Economics of Local Economic Development Programs. *Annual Review of Economics*, 6(1), 629–662.

Kőszegi, B. and Rabin, M. (2006). A Model of Reference-Dependent Preferences. *Quarterly Journal of Economics*, 121(4), 1133–1165.

Laibson, D. (1997). Golden Eggs and Hyperbolic Discounting. *Quarterly Journal of Economics*, 112(2), 443–478.

Madrian, B.C. and Shea, D.F. (2001). The Power of Suggestion: Inertia in 401(k) Participation and Savings Behavior. *Quarterly Journal of Economics*, 116(4), 1149–1187.

Maskin, E. (1999). Nash Equilibrium and Welfare Optimality. *Review of Economic Studies*, 66(1), 23–38.

Maynard Smith, J. (1982). *Evolution and the Theory of Games*. Cambridge University Press.

McKelvey, R.D. and Palfrey, T.R. (1995). Quantal Response Equilibria for Normal Form Games. *Games and Economic Behavior*, 10(1), 6–38.

Milgrom, P. and Roberts, J. (1990). Rationalizability, Learning, and Equilibrium in Games with Strategic Complementarities. *Econometrica*, 58(6), 1255–1277.

Milgrom, P. and Roberts, J. (1990). The Economics of Modern Manufacturing: Technology, Strategy, and Organization. *American Economic Review*, 80(3), 511–528.

Milgrom, P. and Roberts, J. (1995). Complementarities and Fit: Strategy, Structure, and Organizational Change in Manufacturing. *Journal of Accounting and Economics*, 19(2-3), 179–208.

Myerson, R.B. (1979). Incentive Compatibility and the Bargaining Problem. *Econometrica*, 47(1), 61–73.

Myerson, R.B. (1981). Optimal Auction Design. *Mathematics of Operations Research*, 6(1), 58–73.

North, D.C. (1990). *Institutions, Institutional Change, and Economic Performance*. Cambridge University Press.

Nowak, M.A. and May, R.M. (1992). Evolutionary Games and Spatial Chaos. *Nature*, 359(6398), 826–829.

Nowak, M.A. (2006). Five Rules for the Evolution of Cooperation. *Science*, 314(5805), 1560–1563.

O'Donoghue, T. and Rabin, M. (1999). Doing It Now or Later. *American Economic Review*, 89(1), 103–124.

O'Donoghue, T. and Rabin, M. (2001). Choice and Procrastination. *Quarterly Journal of Economics*, 116(1), 121–160.

Piketty, T. (2014). *Capital in the Twenty-First Century*. Harvard University Press.

Preston, S.H. (1975). The Changing Relation between Mortality and Level of Economic Development. *Population Studies*, 29(2), 231–248.

Rabin, M. (1993). Incorporating Fairness into Game Theory and Economics. *American Economic Review*, 83(5), 1281–1302.

Rosenthal, S.S. and Strange, W.C. (2004). Evidence on the Nature and Sources of Agglomeration Economies. *Handbook of Regional and Urban Economics*, 4, 2119–2171.

Roth, A.E. (2002). The Economist as Engineer: Game Theory, Experimentation, and Computation as Tools for Design Economics. *Econometrica*, 70(4), 1341–1378.

Roth, A.E. (2015). *Who Gets What—and Why*. Houghton Mifflin Harcourt.

Samuelson, L. (1997). *Evolutionary Games and Equilibrium Selection*. MIT Press.

Schelling, T.C. (1971). Dynamic Models of Segregation. *Journal of Mathematical Sociology*, 1(2), 143–186.

Sen, A.K. (1999). *Development as Freedom*. Alfred A. Knopf.

Simon, H.A. (1962). The Architecture of Complexity. *Proceedings of the American Philosophical Society*, 106(6), 467–482.

Spence, M. (1973). Job Market Signaling. *Quarterly Journal of Economics*, 87(3), 355–374.

Stiglitz, J.E. (2000). The Contributions of the Economics of Information to Twentieth Century Economics. *Quarterly Journal of Economics*, 115(4), 1441–1478.

Strotz, R.H. (1956). Myopia and Inconsistency in Dynamic Utility Maximization. *Review of Economic Studies*, 23(3), 165–180.

Taylor, P.D. and Jonker, L.B. (1978). Evolutionary Stable Strategies and Game Dynamics. *Mathematical Biosciences*, 40(1-2), 145–156.

Topkis, D.M. (1998). *Supermodularity and Complementarity*. Princeton University Press.

Trivers, R.L. (1971). The Evolution of Reciprocal Altruism. *Quarterly Review of Biology*, 46(1), 35–57.

Tversky, A. and Kahneman, D. (1981). The Framing of Decisions and the Psychology of Choice. *Science*, 211(4481), 453–458.

Weibull, J.W. (1995). *Evolutionary Game Theory*. MIT Press.

Williamson, O.E. (1985). *The Economic Institutions of Capitalism*. Free Press.

Algan, Y. and Cahuc, P. (2010). Inherited Trust and Growth. *American Economic Review*, 100(5), 2060–2092.

Arrow, K.J., Cline, W.R., Mäler, K.G., Munasinghe, M., Squitieri, R., and Stiglitz, J.E. (1997). Intertemporal Equity, Discounting, and Economic Efficiency. *Climate Change 1995: Economic and Social Dimensions*. Cambridge University Press.

Arthur, W.B. (1989). Competing Technologies, Increasing Returns, and Lock-In by Historical Events. *Economic Journal*, 99(394), 116–131.

Arthur, W.B. (1994). *Increasing Returns and Path Dependence in the Economy.* University of Michigan Press.

Arthur, W.B. (2015). *Complexity and the Economy.* Oxford University Press.

Axelrod, R. (1984). *The Evolution of Cooperation.* Basic Books.

Berg, J., Dickhaut, J., and McCabe, K. (1995). Trust, Reciprocity, and Social History. *Games and Economic Behavior*, 10(1), 122–142.

Bolton, G.E. and Ockenfels, A. (2000). ERC: A Theory of Equity, Reciprocity, and Competition. *American Economic Review*, 90(1), 166–193.

Bowles, S. (2004). *Microeconomics: Behavior, Institutions, and Evolution.* Princeton University Press.

Boyd, R. and Richerson, P.J. (1985). *Culture and the Evolutionary Process.* University of Chicago Press.

Brown, T., Mann, B., Ryder, N., et al. (2020). Language Models are Few-Shot Learners. *Advances in Neural Information Processing Systems*, 33, 1877–1901.

Camerer, C.F. (2018). Artificial Intelligence and Behavioral Economics. *NBER Working Paper No. 24690.*

Camerer, C.F., Dreber, A., Holzmeister, F., et al. (2016). Evaluating Replicability of Laboratory Experiments in Economics. *Science*, 351(6280), 1433–1436.

Dana, J., Weber, R.A., and Kuang, J.X. (2007). Exploiting Moral Wiggle Room: Experiments Demonstrating an Illusory Preference for Fairness. *Economic Theory*, 33(1), 67–80.

Debreu, G. (1959). *Theory of Value: An Axiomatic Analysis of Economic Equilibrium.* Yale University Press.

de Quervain, D.J.F., Fischbacher, U., Treyer, V., et al. (2004). The Neural Basis of Altruistic Punishment. *Science*, 305(5688), 1254–1258.

Diamond, D.W. and Dybvig, P.H. (1983). Bank Runs, Deposit Insurance, and Liquidity. *Journal of Political Economy*, 91(3), 401–419.

Diener, E., Suh, E.M., Lucas, R.E., and Smith, H.L. (1999). Subjective Well-Being: Three Decades of Progress. *Psychological Bulletin*, 125(2), 276–302.

Edgeworth, F.Y. (1881). *Mathematical Psychics: An Essay on the Application of Mathematics to the Moral Sciences.* Kegan Paul.

Fehr, E. and Camerer, C.F. (2007). Social Neuroeconomics: The Neural Circuitry of Social Preferences. *Trends in Cognitive Sciences*, 11(10), 419–427.

Fehr, E., Bernhard, H., and Rockenbach, B. (2008). Egalitarianism in Young Children. *Nature*, 454(7208), 1079–1083.

Fehr, E. and Fischbacher, U. (2003). The Nature of Human Altruism. *Nature*, 425(6960), 785–791.

Fehr, E. and Gächter, S. (2000). Cooperation and Punishment in Public Goods Experiments. *American Economic Review*, 90(4), 980–994.

Gentzkow, M., Kelly, B., and Taddy, M. (2019). Text as Data. *Journal of Economic Literature*, 57(3), 535–574.

Gintis, H. (2007). A Framework for the Unification of the Behavioral Sciences. *Behavioral and Brain Sciences*, 30(1), 1–16.

Gneezy, U. and Rustichini, A. (2000). A Fine is a Price. *Journal of Legal Studies*, 29(1), 1–17.

Granovetter, M. (1985). Economic Action and Social Structure: The Problem of Embeddedness. *American Journal of Sociology*, 91(3), 481–510.

Hall, P.A. and Soskice, D. (2001). *Varieties of Capitalism.* Oxford University Press.

Heckscher, E.F. (1919). The Effect of Foreign Trade on the Distribution of Income. *Ekonomisk Tidskrift*, 21, 497–512.

Henrich, J., Boyd, R., Bowles, S., et al. (2000). In Search of Homo Economicus: Behavioral Experiments in 15 Small-Scale Societies. *American Economic Review*, 91(2), 73–78.

Henrich, J., Heine, S.J., and Norenzayan, A. (2010). The Weirdest People in the World? *Behavioral and Brain Sciences*, 33(2-3), 61–83.

Herrmann, B., Thöni, C., and Gächter, S. (2008). Antisocial Punishment Across Societies. *Science*, 319(5868), 1362–1367.

Jensen, M.C. and Meckling, W.H. (1976). Theory of the Firm: Managerial Behavior, Agency Costs and Ownership Structure. *Journal of Financial Economics*, 3(4), 305–360.

Johnson, S., McMillan, J., and Woodruff, C. (2002). Property Rights and Finance. *American Economic Review*, 92(5), 1335–1356.

Kahneman, D. (2011). *Thinking, Fast and Slow.* Farrar, Straus and Giroux.

Katz, M.L. and Shapiro, C. (1985). Network Externalities, Competition, and Compatibility. *American Economic Review*, 75(3), 424–440.

Keynes, J.M. (1936). *The General Theory of Employment, Interest, and Money.* Macmillan.

Knack, S. and Zak, P.J. (2003). Building Trust: Public Policy, Interpersonal Trust, and Economic Development. *Supreme Court Economic Review*, 10, 91–107.

Kahneman, D., Knetsch, J.L., and Thaler, R.H. (1990). Experimental Tests of the Endowment Effect and the Coase Theorem. *Journal of Political Economy*, 98(6), 1325–1348.

Kosfeld, M., Heinrichs, M., Zak, P.J., Fischbacher, U., and Fehr, E. (2005). Oxytocin Increases Trust in Humans. *Nature*, 435(7042), 673–676.

Krugman, P. (1991). Increasing Returns and Economic Geography. *Journal of Political Economy*, 99(3), 483–499.

Levitt, S.D. and List, J.A. (2007). What Do Laboratory Experiments Measuring Social Preferences Reveal About the Real World? *Journal of Economic Perspectives*, 21(2), 153–174.

List, J.A., Sadoff, S., and Wagner, M. (2011). So You Want to Run an Experiment, Now What? *Experimental Economics*, 14(4), 439–457.

Marshall, A. (1890). *Principles of Economics*. Macmillan.

Mas-Colell, A., Whinston, M.D., and Green, J.R. (1995). *Microeconomic Theory*. Oxford University Press.

Mill, J.S. (1848). *Principles of Political Economy*. John W. Parker.

Ogburn, W.F. (1922). *Social Change with Respect to Culture and Original Nature*. B.W. Huebsch.

Ohlin, B. (1933). *Interregional and International Trade*. Harvard University Press.

Ostrom, E. (1990). *Governing the Commons*. Cambridge University Press.

Putnam, R.D. (2000). *Bowling Alone: The Collapse and Revival of American Community*. Simon & Schuster.

Rangel, A., Camerer, C., and Montague, P.R. (2008). A Framework for Studying the Neurobiology of Value-Based Decision Making. *Nature Reviews Neuroscience*, 9(7), 545–556.

Roberts, J. (2004). *The Modern Firm*. Oxford University Press.

Schelling, T.C. (1978). *Micromotives and Macrobehavior*. W.W. Norton.

Slemrod, J. (1998). On Voluntary Compliance, Voluntary Taxes, and Social Capital. *National Tax Journal*, 51(3), 485–491.

Smith, A. (1759). *The Theory of Moral Sentiments*. A. Millar.

Smith, A. (1776). *An Inquiry into the Nature and Causes of the Wealth of Nations*. W. Strahan and T. Cadell.

Sonnenschein, H. (1972). Market Excess Demand Functions. *Econometrica*, 40(3), 549–563.

Stigler, G.J. and Becker, G.S. (1977). De Gustibus Non Est Disputandum. *American Economic Review*, 67(2), 76–90.

Tesfatsion, L. and Judd, K.L. (Eds.) (2006). *Handbook of Computational Economics, Vol. 2: Agent-Based Computational Economics*. North-Holland.

Thaler, R.H. and Sunstein, C.R. (2008). *Nudge: Improving Decisions About Health, Wealth, and Happiness*. Yale University Press.

Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention Is All You Need. *Advances in Neural Information Processing Systems*, 30, 5998–6008.

Vives, X. (2005). Complementarities and Games: New Developments. *Journal of Economic Literature*, 43(2), 437–479.

Weitzman, M.L. (2009). On Modeling and Interpreting the Economics of Catastrophic Climate Change. *Review of Economics and Statistics*, 91(1), 1–19.